Foundations of Reinforcement Learning

Mihir Rao

Contents

1 Introduction and MDP Basics **4**
1.1 Bandits . . . 4
1.2 Markov decision processes . . . 4
1.3 Environment, policy, and value . . . 6

2 Bellman Equations **7**
2.1 Markov policies suffice . . . 7
2.2 Notation . . . 8
2.3 Policy evaluation . . . 8
2.4 Optimal policies and the Bellman optimality equation . . . 9

3 Planning **11**
3.1 Value iteration . . . 11
3.2 Policy iteration . . . 11
3.3 The infinite-horizon discounted setting . . . 12
3.4 From planning to learning . . . 15

4 Concentration Inequalities **16**
4.1 Hoeffding's inequality . . . 16
4.2 Bernstein's inequality . . . 17
4.3 Concentration without independence . . . 18

5 Martingale Concentrations **20**
5.1 Martingales . . . 20
5.2 Azuma-Hoeffding and Azuma-Bernstein . . . 20
5.3 Union bound and uniform concentration . . . 21
5.4 Learning settings for MDPs . . . 22

6 Generative Models **24**
6.1 Coarse analysis . . . 24
6.2 Refined analysis . . . 25

7 Introduction to Exploration **27**
7.1 Q-learning . . . 27
7.2 Summary of the simulator setting . . . 29
7.3 Online reinforcement learning . . . 30
7.4 Multi-armed bandits . . . 31

8 Exploration in Multi-Arm Bandits **32**
8.1 Regret versus best-arm identification . . . 32
8.2 Explore-then-commit and ϵ-greedy . . . 32
8.3 The UCB algorithm . . . 33
8.4 Contextual bandits . . . 34

9 Exploration in RL **36**
9.1 The UCB-VI algorithm . . . 36
9.2 Q-learning with UCB . . . 39
9.3 Summary . . . 39

10 Lower Bounds for Multi-Arm Bandits **41**
10.1 Intuition with two arms . . . 41
10.2 Reduction to deterministic algorithms . . . 41
10.3 Proof of the lower bound . . . 42
10.4 Lower bound for best-arm identification . . . 43

11 Lower Bounds for MDPs **44**

12 Offline RL **46**
12.1 Setting . . . 46
12.2 What causes suboptimality . . . 46
12.3 Pessimistic value iteration . . . 47
12.4 Linear and tabular MDPs . . . 49

13 RL in Large State Space **50**
13.1 Why function approximation . . . 50
13.2 Linear function approximation . . . 50

14 Least-Squares Value Iteration **52**
14.1 Least squares as an approximate Bellman backup . . . 52
14.2 Core sets . . . 52
14.3 Analysis . . . 53

15 Exploration in Large State Space **55**
15.1 LSVI-UCB . . . 55
15.2 Global optimism . . . 57

16 General Function Approximation **58**
16.1 Assumptions and the Bellman error . . . 58
16.2 Fitted Q-iteration . . . 59

17 Exploration in General Function Approximation **62**
17.1 Bellman rank . . . 62
17.2 Bellman Eluder dimension . . . 63
17.3 GOLF . . . 64

18 Multiagent Reinforcement Learning **66**
18.1 Matrix games . . . 66
18.2 Markov games . . . 67
18.3 Solution concepts . . . 68
18.4 Direct combinations of game theory and single-agent RL . . . 69

19 Two-Player Zero-Sum Games **70**
19.1 Bellman equations and planning . . . 70
19.2 Exploration: optimistic Nash value iteration . . . 71
19.3 Decentralized learning . . . 72
19.4 V-learning . . . 73

20 Multiplayer General-Sum Games **74**
20.1 General-sum Markov games . . . 74
20.2 Equilibria . . . 74

20.3 Optimistic Nash value iteration and the curse of multiagents . . . 75
20.4 V-learning for general-sum games . . . 76

21 Partially Observable Reinforcement Learning I **78**
21.1 Partially observable MDPs . . . 78
21.2 Belief states . . . 79
21.3 Predictive state representations . . . 80

22 Partially Observable Reinforcement Learning II **82**
22.1 Setting . . . 82
22.2 Why POMDPs are hard . . . 82
22.3 The weakly revealing condition . . . 83
22.4 Optimistic maximum likelihood estimation . . . 84

Lecture 1: Introduction and MDP Basics

Reinforcement learning (RL) is a framework for sequential decision making. An agent (the learner or decision maker) takes actions that affect an environment, and the environment sends feedback to the agent, typically a state and a reward. The goal is to find a good policy, one that maximizes the cumulative reward.

The course studies the mathematics of RL, with proofs: models and learning settings (simulator, exploration, function approximation); the challenges of exploration versus exploitation, large state spaces, multiple agents, and partial observability; algorithms such as value iteration, Q-learning, and fitted Q-iteration, built on *optimism* for strategic exploration and *pessimism* for offline data; and their sample and computational complexity. Practical algorithms often need more than 10^6 samples and long training, so the central questions are what the sample complexity of a problem is and how to improve it, and how to design faster algorithms.

1.1 Bandits

Definition 1.1 [Multi-armed bandit]. A multi-armed bandit MAB($\mathcal{A}, r$) has a set $\mathcal{A} = [K]$ of K arms (where $[K] = \{1, \dots, K\}$). Each pull of arm i returns a reward, either a deterministic r_i or a random variable with mean r_i, for example $\mathcal{N}(r_i, \sigma^2)$; r records the reward distributions. The goal is to identify the best arm, the arm with the highest mean reward.

A contextual multi-armed bandit C-MAB($\mathcal{S}, \mathcal{A}, r$) adds a context or state: there are several K-armed machines, indexed by a row number $s \in \mathcal{S}$.

Algorithm 1.1: C-MAB interaction

```
for k = 1, ..., K do
    the environment selects a context (row) s
    the learner takes an action a
    the environment reveals the reward r
```

The goal is to identify the best arm for each row, that is, an optimal policy. A policy is either deterministic, $\pi : \mathcal{S} \to \mathcal{A}$, or stochastic, $\pi : \mathcal{S} \to \Delta_{\mathcal{A}}$, where $\Delta_{\mathcal{A}}$ is the probability simplex over $\mathcal{A}$; we write $\pi(a \mid s)$ for the probability that π takes action a in state s.

1.2 Markov decision processes

In a contextual bandit the action affects only the reward, not the next state. A Markov decision process also models how actions drive the state, and hence the temporal correlation between states

at different steps.

Algorithm 1.2: MDP interaction

```
for episode k = 1, ..., K do
    the environment samples the initial state s_1
    for step h = 1, ..., H do
        the learner takes an action a_h
        the environment transitions to s_{h+1} and reveals the reward r_h
```

Definition 1.2 [Episodic MDP]. An episodic Markov decision process MDP$(\mathcal{S}, \mathcal{A}, r, P, H)$ consists of

- a set of states $\mathcal{S}$ and a set of actions $\mathcal{A}$;
- reward functions $r_h(s, a)$: the reward for taking action a in state s at step h;
- transition probabilities $P_h(s' \mid s, a)$: the probability of moving to s' at step $h+1$ after taking a in s at step h;
- the episode length (horizon) H.

In each episode (for example, one game of chess) the environment draws an initial state s_1. At each step $h \in [H]$ the agent observes $s_h \in \mathcal{S}$, picks $a_h \in \mathcal{A}$, receives $r_h(s_h, a_h)$, and the environment moves to $s_{h+1} \sim P_h(\cdot \mid s_h, a_h)$. The episode ends when s_{H+1} is reached (Figure 1).

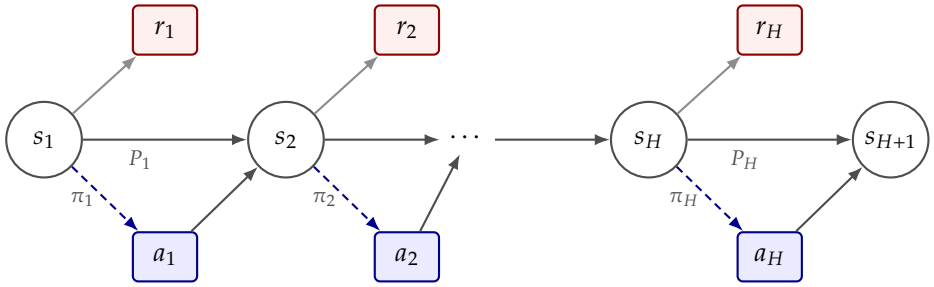


Figure 1: One episode of an MDP. At step h the policy π_h chooses a_h from s_h (dashed), the agent receives the reward $r_h = r_h(s_h, a_h)$, and the next state s_{h+1} is drawn from $P_h(\cdot \mid s_h, a_h)$, which depends on both the state and the action.

Three remarks on the model:

1. Rewards are deterministic for simplicity. This is essentially without loss of generality: the transitions already carry a similar level of noise, and the transitions are the harder part to learn.
2. The next-state distribution depends only on the current state and action, not on earlier ones. This is the key Markov property: for a fixed action sequence, $(s_1, \dots, s_H)$ is a Markov chain, so conditioned on s_h the past $(s_1, \dots, s_{h-1})$ is independent of the future $(s_{h+1}, \dots, s_H)$. Partially observable models (Lectures 21 and 22) give up this property for what the agent observes.

3. The course mostly uses the episodic formulation, which is closely related to the infinite-horizon discounted formulation (Lecture 3).

1.3 Environment, policy, and value

The ENVIRONMENT is the pair (P, r). Problems differ in how much is known about it:

1. the environment is known (PLANNING);
2. the environment is unknown but can be simulated: a SIMULATOR returns the reward and a next state for any queried state-action pair;
3. the environment is unknown and the agent must learn either by interacting with it from the initial state (the ONLINE setting) or from pre-collected data (the OFFLINE setting).

Definition 1.3 [MARKOV AND GENERAL POLICIES]. A MARKOV POLICY is a collection $\pi = \{\pi_h\}_{h=1}^H$ with $\pi_h : \mathcal{S} \to \mathcal{A}$ (deterministic) or $\pi_h : \mathcal{S} \to \Delta_{\mathcal{A}}$ (stochastic); the action depends only on the current state. A GENERAL POLICY may use the whole history: $\pi_h : (\mathcal{S} \times \mathcal{A})^{h-1} \times \mathcal{S} \to \mathcal{A}$ or $\Delta_{\mathcal{A}}$, a map from histories $(s_1, a_1, \ldots, s_h)$ to actions.

In an MDP it suffices to consider Markov policies (Lecture 2). In multiagent and partially observable RL, Markov policies are no longer powerful enough, and general policies return.

Definition 1.4 [VALUE FUNCTIONS]. For a Markov policy π, the V-VALUE and Q-VALUE at step h are the expected cumulative rewards to the end of the episode,

$$V_h^\pi(s) = \mathbb{E}_\pi\Big[\sum_{h'=h}^{H} r_{h'}(s_{h'}, a_{h'}) \,\Big|\, s_h = s\Big], \qquad Q_h^\pi(s,a) = \mathbb{E}_\pi\Big[\sum_{h'=h}^{H} r_{h'}(s_{h'}, a_{h'}) \,\Big|\, s_h = s, a_h = a\Big].$$

In $V_h^\pi(s)$ the expectation is over the future sequence $(a_h, s_{h+1}, a_{h+1}, \ldots)$, that is,

$$\mathbb{E}_\pi = \mathbb{E}_{a_h \sim \pi_h(\cdot|s_h)} \mathbb{E}_{s_{h+1} \sim P_h(\cdot|s_h,a_h)} \mathbb{E}_{a_{h+1} \sim \pi_{h+1}(\cdot|s_{h+1})} \cdots,$$

and similarly for Q_h^π. For $h > 1$ these definitions make sense only for Markov policies, since a general policy's future reward depends on the past history as well. At $h = 1$ there is no history, so $V_1^\pi(s)$ and $Q_1^\pi(s,a)$ are defined for general policies too.

Definition: The goal of RL is to find an OPTIMAL POLICY $\pi^\star$ that maximizes $V_1^\pi(s_1)$ when the initial state s_1 is fixed, or $\mathbb{E}_{s_1 \sim \mu} V_1^\pi(s_1)$ when s_1 is drawn from a distribution μ. Throughout the course s_1 is fixed for simplicity; most arguments extend directly to random initial states.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 1 (Spring 2024); written from the Spring 2024 Lecture 1 notes and slides and the Spring 2020 Lecture 1 notes.

Lecture 2: Bellman Equations

This unit works with the episodic MDP $\mathrm{MDP}(\mathcal{S}, \mathcal{A}, r, P, H)$ of Lecture 1 and a fixed initial state s_1. It shows that Markov policies suffice, derives the Bellman equations for evaluating a policy, and proves that the greedy policy of the Bellman optimality equation is optimal.

2.1 Markov policies suffice

General policies can depend on the whole history and are hard to handle. In an MDP nothing is lost by restricting to Markov policies.

Proposition 2.1 [Sufficiency of Markov policies]. *For any general policy π there is a Markov policy μ with $V_1^\pi(s_1) = V_1^\mu(s_1)$.*

In particular, if an optimal policy is a general policy, some Markov policy achieves the same value, so every MDP has an optimal policy that is Markov.

Proof. Let $P_h^\pi(s, a)$, the visitation measure, be the probability of visiting (s, a) at step h when starting from s_1 and following π (write $P_h^\pi(s)$ for its state marginal). Then

$$V_1^\pi(s_1) = \sum_{h=1}^{H} \mathbb{E}_\pi[r_h(s_h, a_h)] = \sum_{h=1}^{H} \sum_{s,a} P_h^\pi(s, a)\, r_h(s, a),$$

and only the visitation measure depends on the policy. It therefore suffices to construct μ with $P_h^\pi(s, a) = P_h^\mu(s, a)$ for all (s, a, h). Let $\tau_h = (s_1, a_1, \ldots, s_h)$ be the trajectory up to step h and let μ average π over the past history:

$$\mu_h(a \mid s) = \mathbb{E}_\pi[\pi_h(a \mid \tau_h) \mid s_h = s] = \frac{\sum_{\tau_h : s_h = s} P_h^\pi(\tau_h)\, \pi_h(a \mid \tau_h)}{P_h^\pi(s)} \quad \text{if } P_h^\pi(s) \neq 0,$$

and arbitrary otherwise. We prove equality of the visitation measures by induction on h. At $h = 1$, $P_1^\pi(s) = P_1^\mu(s)$ because the initial state does not depend on the policy. For a fixed h:
States to state-action pairs. If $P_h^\pi(s) = P_h^\mu(s)$ for all s, then for every (s, a) (both sides vanish when $P_h^\pi(s) = 0$)

$$\begin{aligned} P_h^\mu(s, a) &= P_h^\mu(s)\mu_h(a \mid s) = P_h^\pi(s)\mu_h(a \mid s) \\ &= \sum_{\tau_h : s_h = s} P_h^\pi(\tau_h)\pi_h(a \mid \tau_h) = \sum_{\tau_h : s_h = s} P_h^\pi(\tau_h, a) = P_h^\pi(s, a). \end{aligned}$$

State-action pairs to next states. If $P_h^\pi(s,a) = P_h^\mu(s,a)$ for all (s,a), then

$$P_{h+1}^\mu(s') = \sum_{s,a} P_h(s' \mid s,a) P_h^\mu(s,a) = \sum_{s,a} P_h(s' \mid s,a) P_h^\pi(s,a) = P_{h+1}^\pi(s').$$

Both outer equalities use the Markov property: given (s_h, a_h) the next state has law $P_h(\cdot \mid s_h, a_h)$ whatever the policy and the earlier history. □

From now on, all policies in MDPs are Markov.

2.2 Notation

In the TABULAR setting the state and action sets are finite, with $S = |\mathcal{S}|$ states and $A = |\mathcal{A}|$ actions. A value function V_h^π is then a vector in $\mathbb{R}^S$ and Q_h^π a vector in $\mathbb{R}^{SA}$. For a function $V : \mathcal{S} \to \mathbb{R}$ write

$$(P_h V)(s,a) = \mathbb{E}_{s' \sim P_h(\cdot \mid s,a)} V(s'),$$

a matrix-vector product with the transition matrix $P_h \in \mathbb{R}^{SA \times S}$ whose $((s,a), s')$ entry is $P_h(s' \mid s,a)$. Without loss of generality $r_h(s,a) \in [0,1]$, so that $V_h^\pi \in [0,H]^S$ and $Q_h^\pi \in [0,H]^{SA}$.

2.3 Policy evaluation

POLICY EVALUATION asks: given P, r, and a Markov policy π, compute $V_1^\pi(s_1)$. Expanding the expectation over whole trajectories $\tau_H = (s_1, a_1, \ldots, s_H, a_H)$ gives

$$V_1^\pi(s_1) = \sum_{\tau_H} P^\pi(\tau_H \mid s_1) \sum_{h=1}^{H} r_h(s_h, a_h), \qquad P^\pi(\tau_H \mid s_1) = \pi_1(a_1 \mid s_1) P_1(s_2 \mid s_1, a_1) \pi_2(a_2 \mid s_2) \cdots,$$

but the sum has $(SA)^H$ terms, exponential in the horizon. Dynamic programming is much faster, because the value functions at consecutive steps satisfy the BELLMAN EQUATIONS

$$V_h^\pi(s) = \sum_{a \in \mathcal{A}} \pi_h(a \mid s)\, Q_h^\pi(s,a), \qquad Q_h^\pi(s,a) = r_h(s,a) + (P_h V_{h+1}^\pi)(s,a), \qquad V_{H+1}^\pi = 0, \tag{2.1}$$

which follow directly from the definitions by conditioning on the first action and the next state.

Algorithm 2.1: Policy evaluation by the Bellman equations

Input: transitions P, rewards r, Markov policy π

$V_{H+1}^\pi(s) \leftarrow 0$ for all s

for $h = H, H-1, \ldots, 1$ **do**

  $Q_h^\pi(s,a) \leftarrow r_h(s,a) + (P_h V_{h+1}^\pi)(s,a)$ for all (s,a)

  $V_h^\pi(s) \leftarrow \sum_{a \in \mathcal{A}} Q_h^\pi(s,a) \pi_h(a \mid s)$ for all s

Starting from the zero vector $V^\pi_{H+1} \in \mathbb{R}^S$, the algorithm computes $Q^\pi_H \in \mathbb{R}^{SA}$, then $V^\pi_H \in \mathbb{R}^S$, then Q^π_{H-1}, and so on. Each step costs $O(S^2A)$, so the total cost is polynomial in S, A, H and only linear in H.

2.4 Optimal policies and the Bellman optimality equation

Only the first equation in (2.1) involves the policy, by averaging $Q^\pi_h(s,\cdot)$ over actions. An optimal policy should instead pick the action that maximizes the Q-value, which suggests the Bellman optimality equation

$$V^\star_h(s) = \max_{a\in\mathcal{A}} Q^\star_h(s,a), \qquad Q^\star_h(s,a) = r_h(s,a) + (P_h V^\star_{h+1})(s,a), \qquad V^\star_{H+1} = 0, \tag{2.2}$$

together with the greedy policy

$$\pi^\star_h(a \mid s) = \mathbf{1}\Big[a = \arg\max_{a'\in\mathcal{A}} Q^\star_h(s,a')\Big]. \tag{2.3}$$

The Bellman equations of $\pi^\star$ coincide with (2.2), so $V^\star$ and $Q^\star$ are the value functions of $\pi^\star$: $V^\star_h = V^{\pi^\star}_h$ and $Q^\star_h = Q^{\pi^\star}_h$.

Proposition 2.2 [Optimality of the greedy policy]. *The deterministic Markov policy $\pi^\star$ of (2.3) is optimal. More strongly, for all (s,a,h),*

$$V^\star_h(s) = \max_\pi V^\pi_h(s), \qquad Q^\star_h(s,a) = \max_\pi Q^\pi_h(s,a),$$

where the maxima are over Markov policies. Thus $\pi^\star$ attains the highest value from every state (and state-action pair) at every step simultaneously; by Proposition 2.1 it is also optimal among general policies.

Proof. Backward induction on h. At $h = H+1$ there is no reward left, so $V^\star_{H+1} = \max_\pi V^\pi_{H+1} = 0$. For a fixed $h \le H$ there are two steps.
From V_{h+1} to Q_h. Assume $V^\star_{h+1}(s') = \max_\pi V^\pi_{h+1}(s')$ for all s'. By the Bellman equation,

$$\begin{aligned}\max_\pi Q^\pi_h(s,a) &= r_h(s,a) + \max_\pi \sum_{s'} P_h(s' \mid s,a) V^\pi_{h+1}(s') \\ &= r_h(s,a) + \sum_{s'} P_h(s' \mid s,a) \max_\pi V^\pi_{h+1}(s') = Q^\star_h(s,a).\end{aligned}$$

The middle step is the key one. In general a maximum cannot be moved inside a sum, since $\max_x (f(x)+g(x)) \le \max_x f(x) + \max_x g(x)$, but equality holds when the individual maxima are attained at the same point. Here the induction hypothesis says that $\pi^\star$ attains $\max_\pi V^\pi_{h+1}(s')$ for every s' simultaneously.
From Q_h to V_h. Assume $Q^\star_h(s,a) = \max_\pi Q^\pi_h(s,a)$ for all (s,a). Since $Q^\pi_h(s,a)$ fixes the action at

step h, it depends only on $\pi_{h+1}, \dots, \pi_H$, and its maximum over these is attained at $\pi^\star$ for all (s, a) simultaneously. Hence

$$\begin{aligned}\max_{\pi} V_h^{\pi}(s) &= \max_{\pi_h} \max_{\pi_{h+1:H}} \sum_a Q_h^{\pi}(s,a)\pi_h(a \mid s) = \max_{\pi_h} \sum_a \Big(\max_{\pi_{h+1:H}} Q_h^{\pi}(s,a)\Big)\pi_h(a \mid s) \\ &= \max_{\pi_h} \sum_a Q_h^{\star}(s,a)\pi_h(a \mid s) = \max_a Q_h^{\star}(s,a) = V_h^{\star}(s),\end{aligned}$$

where the next-to-last step puts probability one on an action with the highest Q-value. □

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 2 (Spring 2024); written from the Spring 2024 Lecture 2 notes and the Spring 2020 Lecture 3 and Lecture 4 notes.

Lecture 3: Planning

Planning is the task of computing an optimal policy $\pi^\star$ when the environment P and r are known. Lecture 2 showed that $\pi^\star$ is the greedy policy of the Bellman optimality equation, $\pi_h^\star(s) = \arg\max_a Q_h^\star(s,a)$. This unit gives two planning algorithms for episodic MDPs, relates the episodic and the discounted settings, and starts the move from planning to learning.

3.1 Value iteration

Dynamic programming on the Bellman optimality equation gives value iteration.

Algorithm 3.1: Value iteration

Input: transitions P, rewards r

$V_{H+1}^\star(s) \leftarrow 0$ for all s

for $h = H, H-1, \dots, 1$ **do**

  $Q_h^\star(s,a) \leftarrow r_h(s,a) + (P_h V_{h+1}^\star)(s,a)$ for all (s,a)

  $V_h^\star(s) \leftarrow \max_{a\in\mathcal{A}} Q_h^\star(s,a)$ for all s

return $\pi_h^\star(s) = \arg\max_{a\in\mathcal{A}} Q_h^\star(s,a)$ for all (s,h)

As in policy evaluation, the algorithm computes $Q_H^\star \in \mathbb{R}^{SA}$, $V_H^\star \in \mathbb{R}^S$, $Q_{H-1}^\star$, and so on backward from the zero vector, at a total cost polynomial in S, A, H and linear in H.

3.2 Policy iteration

Policy iteration computes $\pi^\star$ using only the Bellman equation of a fixed policy, not the optimality equation.

Algorithm 3.2: Policy iteration

Input: transitions P, rewards r

initialize $\pi^0 = \{\pi_h^0\}_{h=1}^H$ to an arbitrary policy

for $t = 1, 2, \dots$ **do**

  evaluate $Q_h^{\pi^{t-1}}(s,a)$ for all (s,a,h) *// policy evaluation*

  $\pi_h^t(s) \leftarrow \arg\max_{a\in\mathcal{A}} Q_h^{\pi^{t-1}}(s,a)$ for all (s,h) *// policy improvement*

Each iteration evaluates $Q^{\pi^{t-1}}$ by dynamic programming (Lecture 2) and then improves the policy by acting greedily with respect to it.

Theorem 3.1 [Policy iteration]. *In policy iteration, π^t is optimal for every $t \geq H$.*

Proof. We prove by induction on t the stronger claim that $Q_h^{\pi^t} = Q_h^\star$ for all $h \geq H - t$. In the table below a $\star$ in row t and column h marks a step where the claim holds: at $t = 0$ only the last step is correct, and each iteration extends correctness one step backward.

	$h = 1$	$\cdots$	$h = H-2$	$h = H-1$	$h = H$
$t = 0$					$\star$
$t = 1$				$\star$	$\star$
$t = 2$			$\star$	$\star$	$\star$
$\vdots$					
$t = H-1$	$\star$	$\cdots$	$\star$	$\star$	$\star$

At $t = 0$, $Q_H^{\pi^0}(s,a) = Q_H^\star(s,a) = r_H(s,a)$, since no steps follow. Suppose the claim holds at iteration t. For every $h \geq H - t$, the improvement step and the hypothesis give

$$\pi_h^{t+1}(s) = \arg\max_a Q_h^{\pi^t}(s,a) = \arg\max_a Q_h^\star(s,a) = \pi_h^\star(s).$$

Now $Q_{h-1}^{\pi^{t+1}}(s,a)$ fixes the action at step $h-1$, so it depends only on $\pi_h^{t+1}, \ldots, \pi_H^{t+1}$, which we just showed are optimal. Hence $Q_{h-1}^{\pi^{t+1}} = Q_{h-1}^\star$ for every $h \geq H - t$, that is, $Q_h^{\pi^{t+1}} = Q_h^\star$ for all $h \geq H - (t+1)$. Finally, for $t \geq H$ the claim at iteration $t - 1 \geq H - 1$ covers every step, so π^t is greedy with respect to $Q^\star$ at every step and equals $\pi^\star$. □

3.3 The infinite-horizon discounted setting

Instead of a finite horizon, the INFINITE-HORIZON DISCOUNTED setting MDP$(\mathcal{S}, \mathcal{A}, r, P, \gamma)$ allows infinitely many steps but discounts future rewards by a DISCOUNT FACTOR $\gamma \in [0, 1)$. Its main benefit is that the setting can be *stationary*: r and P do not depend on the step, and neither do the policy and the values. For a stationary policy π,

$$V^\pi(s) = \mathbb{E}_\pi\Big[\sum_{h=1}^{\infty} \gamma^{h-1} r(s_h, a_h) \;\Big|\; s_1 = s\Big], \qquad Q^\pi(s,a) = \mathbb{E}_\pi\Big[\sum_{h=1}^{\infty} \gamma^{h-1} r(s_h, a_h) \;\Big|\; s_1 = s, a_1 = a\Big].$$

With $r(s,a) \in [0,1]$ both are geometric series with values in $[0, \beta]$, where we write $\beta = 1/(1-\gamma)$; typically γ is close to 1.

Bellman equations and policy evaluation. The values satisfy

$$V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s,a), \qquad Q^\pi(s,a) = r(s,a) + \gamma (P V^\pi)(s,a),$$

or in vector form

$$Q^\pi = r + \gamma P^\pi Q^\pi, \qquad P^\pi \in [0,1]^{SA \times SA}, \quad P^\pi_{(s,a),(s',a')} = P(s' \mid s,a)\pi(a' \mid s').$$

There is no last step to start a backward recursion from, so evaluation is a fixed-point iteration $Q_{t+1} = \mathcal{T}^\pi Q_t$ with the operator $\mathcal{T}^\pi Q = r + \gamma P^\pi Q$, started from any $Q_0 \in [0,\beta]^{SA}$.

Lemma 3.2 [Policy evaluation converges]. *$\|Q_t - Q^\pi\|_\infty \le \gamma^t \|Q_0 - Q^\pi\|_\infty \le \gamma^t \beta$. In particular $\|Q_t - Q^\pi\|_\infty \le \epsilon$ once $t \ge \log(\epsilon/\beta)/\log\gamma \approx \beta \log(\beta/\epsilon)$.*

Proof. Since $Q^\pi = \mathcal{T}^\pi Q^\pi$,

$$\begin{aligned}\|Q_{t+1} - Q^\pi\|_\infty = \|\mathcal{T}^\pi Q_t - \mathcal{T}^\pi Q^\pi\|_\infty &= \gamma \|P^\pi (Q_t - Q^\pi)\|_\infty \\ &\le \gamma \, \|P^\pi\|_\infty \, \|Q_t - Q^\pi\|_\infty = \gamma \, \|Q_t - Q^\pi\|_\infty,\end{aligned}$$

because the induced ℓ_∞ norm of a matrix is its largest absolute row sum, which is 1 for the stochastic matrix P^π. Unroll the recursion, and use $\log(1/\gamma) = -\log(1 - 1/\beta) \approx 1/\beta$. □

Value iteration. The Bellman optimality equation

$$V^\star(s) = \max_a Q^\star(s,a), \qquad Q^\star(s,a) = r(s,a) + \gamma (P V^\star)(s,a)$$

reads $Q^\star = \mathcal{T}^\star Q^\star$ with $\mathcal{T}^\star Q = r + \gamma P \mathcal{M} Q$, where $(\mathcal{M}Q)(s) = \max_a Q(s,a)$ is the (nonlinear) maximum operator. Value iteration is the fixed-point iteration $Q_{t+1} = \mathcal{T}^\star Q_t$ from $Q_0 \in [0,\beta]^{SA}$.

Lemma 3.3 [Value iteration converges]. *$\|Q_t - Q^\star\|_\infty \le \gamma^t \|Q_0 - Q^\star\|_\infty$, so $\|Q_t - Q^\star\|_\infty \le \epsilon$ once $t \ge \beta \log(\beta/\epsilon)$.*

Proof. As before, $\|Q_{t+1} - Q^\star\|_\infty = \gamma \|P(\mathcal{M}Q_t - \mathcal{M}Q^\star)\|_\infty \le \gamma \|\mathcal{M}Q_t - \mathcal{M}Q^\star\|_\infty \le \gamma \|Q_t - Q^\star\|_\infty$, where the last step uses $|\max_a Q_t(s,a) - \max_a Q^\star(s,a)| \le \max_a |Q_t(s,a) - Q^\star(s,a)|$ for each s. □

An approximate Q is turned into a policy by acting greedily, and the loss is at most a factor 2β.

Lemma 3.4 [Greedy policies]. *For any Q, the greedy policy $\pi_Q(s) = \arg\max_a Q(s,a)$ satisfies $\|V^{\pi_Q} - V^\star\|_\infty \le 2\beta \|Q - Q^\star\|_\infty$.*

Proof. Since $Q(s, \pi_Q(s)) \geq Q(s, \pi^\star(s))$,

$$\begin{aligned} 0 \leq V^\star(s) - V^{\pi_Q}(s) &= Q^\star(s, \pi^\star(s)) - Q^{\pi_Q}(s, \pi_Q(s)) \\ &\leq \left[Q^\star(s, \pi^\star(s)) - Q(s, \pi^\star(s))\right] + \left[Q(s, \pi_Q(s)) - Q^\star(s, \pi_Q(s))\right] \\ &\quad + \left[Q^\star(s, \pi_Q(s)) - Q^{\pi_Q}(s, \pi_Q(s))\right] \\ &\leq 2\|Q^\star - Q\|_\infty + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi_Q(s))}\left[V^\star(s') - V^{\pi_Q}(s')\right] \\ &\leq 2\|Q^\star - Q\|_\infty + \gamma \|V^\star - V^{\pi_Q}\|_\infty. \end{aligned}$$

Take the maximum over s and rearrange. □

Linear programming. The optimal values can also be computed by a linear program. For any weights $\alpha(s) > 0$, consider

$$\min_{V \in [0,\beta]^S} \sum_{s \in \mathcal{S}} \alpha(s) V(s) \quad \text{subject to} \quad V(s) \geq r(s,a) + \gamma (PV)(s,a) \ \text{ for all } (s,a).$$

Lemma 3.5 [Linear programming]. *The minimizer of this linear program is $V^\star$.*

Proof. Let V be a minimizer, and suppose that for some state s, $V(s) \geq \max_a \big(r(s,a) + \gamma (PV)(s,a)\big) + \Delta$ with $\Delta > 0$. Let V' equal V except $V'(s) = V(s) - \Delta$. Since $V' \leq V$ and $P \geq 0$, the constraints at s still hold, $V'(s) \geq r(s,a) + \gamma (PV)(s,a) \geq r(s,a) + \gamma (PV')(s,a)$, and so do those at every other state. But V' has a strictly smaller objective, a contradiction. Hence $V(s) = \max_a \big(r(s,a) + \gamma (PV)(s,a)\big)$ for all s, which is the Bellman optimality equation, whose unique solution is $V^\star$ by Lemma 3.3. □

Episodic versus discounted. Bellman equations, policy evaluation, and planning all carry over to the discounted setting, but the algorithms become fixed-point iterations: morally similar to dynamic programming, yet less intuitive and more involved to analyze. The course therefore works in the episodic setting, and results transfer to the discounted setting by truncation. The most reward that can be collected after the first h steps is $f(h) = \sum_{h'=h}^{\infty} \gamma^{h'} = \gamma^h/(1-\gamma)$, and $f(\tilde{H}) \leq \epsilon$ whenever $\tilde{H} \geq c \cdot \beta \log(\beta/\epsilon)$ for a constant c. Truncating the discounted problem at this effective horizon $\tilde{H} = \widetilde{O}(\beta)$, that is, comparing it with the episodic MDP with horizon $\tilde{H}$ and rewards $\tilde{r}_h = \gamma^{h-1} r$, changes the value of every policy by at most ϵ, so the optimal values change by at most ϵ as well. Moreover, $\tilde{H}$ steps of value iteration on the discounted MDP, started from $Q_0 = 0$, are equivalent to value iteration on the truncated episodic MDP (the iterates satisfy $\tilde{Q}_h = \gamma^{h-1} Q_{\tilde{H}-h+1}$), so the finite-horizon analysis can replace the fixed-point argument.

3.4 From planning to learning

Planning and policy evaluation require knowing P and r. In the LEARNING setting the agent must perform well by interacting with an unknown environment, and a key ingredient is estimating transitions from random samples. Estimating the transition probability from a fixed (s, a) to a next state with two possible values is a coin-toss problem: observe independent $X_1, \dots, X_n$ with $X_i = 1$ (heads) with probability p and 0 (tails) otherwise, and estimate the unknown p by the sample average. The law of large numbers gives $\frac{1}{n}\sum_{i=1}^{n} X_i \to \mathbb{E}X = p$ as $n \to \infty$, but only in the limit. With finitely many samples we want a bound

$$\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}X\right| \le f(n)$$

with f decreasing in n, describing how fast the average concentrates. Such bounds are CONCENTRATION INEQUALITIES.

Asymptotic notation. To keep only leading-order behavior, let $f, g : \mathbb{R}^d \to \mathbb{R}$ take vector arguments x.

- $g(x) \le O(f(x))$ if there are $C > 0$ and $N > 0$ such that $g(x) \le Cf(x)$ for every x with $\min_{i\in[d]} x_i \ge N$: for large x, g is at most f up to a constant factor.
- $g(x) \le o(f(x))$ if for every $c > 0$ there is $N > 0$ such that $g(x) \le cf(x)$ for every x with $\min_i x_i \ge N$: g is asymptotically strictly smaller.
- $f(x) \ge \Omega(g(x))$ means $g(x) \le O(f(x))$, $f(x) \ge \omega(g(x))$ means $g(x) \le o(f(x))$, and $f(x) = \Theta(g(x))$ means both $f \ge \Omega(g)$ and $f \le O(g)$.
- The tilde versions also hide polylogarithmic factors: $g(x) \le \widetilde{O}(f(x))$ if there are $C > 0$, a polynomial p, and $N > 0$ with $g(x) \le Cf(x)\,p(\log x)$ for every x with $\min_i x_i \ge N$, where $\log x = (\log x_1, \dots, \log x_d)$. The notations $\tilde{o}$, $\tilde{\Theta}$, $\tilde{\Omega}$, $\tilde{\omega}$ are defined similarly.

The first concentration inequality bounds sums of independent bounded random variables; it is proved in Lecture 4.

Theorem 3.6 [HOEFFDING'S INEQUALITY]. *Let $X_1, \dots, X_n$ be independent random variables with $X_i \in [a_i, b_i]$ almost surely, and let $R_i = b_i - a_i$. Then for all $t \ge 0$,*

$$\mathbb{P}\Big(\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \ge t\Big) \le \exp\Big(-\frac{2t^2}{\sum_{i=1}^{n} R_i^2}\Big).$$

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 3 (Spring 2024); written from the Spring 2024 Lecture 3 notes and the Spring 2020 Lecture 4 and Lecture 5 notes.

Lecture 4: Concentration Inequalities

This unit proves Hoeffding's inequality (Theorem 3.6), states the sharper Bernstein inequality, and explains why reinforcement learning needs concentration without independence.

4.1 Hoeffding's inequality

Hoeffding's inequality bounds the upper tail of $\sum_{i=1}^{n}(X_i - \mathbb{E}X_i)$ for independent $X_i \in [a_i, b_i]$ with ranges $R_i = b_i - a_i$. Applying it to $-X_i$ bounds the lower tail, so with high probability the centered sum stays close to 0. Setting the tail probability to $\delta = \exp(-2t^2/\sum_i R_i^2)$ and solving for t gives the form used most often.

Corollary 4.1 [Hoeffding, high-probability form]. *Under the conditions of Theorem 3.6, for any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \le \sqrt{\frac{\sum_{i=1}^{n} R_i^2 \log(1/\delta)}{2}}.$$

If moreover the X_i are identically distributed with range R, then with probability at least $1-\delta$,

$$\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}X \le \frac{R}{\sqrt{n}}\sqrt{\frac{\log(1/\delta)}{2}} = \widetilde{O}\Big(\frac{R}{\sqrt{n}}\Big).$$

So the sample average estimates the mean with error $\widetilde{O}(R/\sqrt{n})$, which vanishes as $n \to \infty$. This $1/\sqrt{n}$ rate is the standard statistical error rate and appears throughout the course. For the coin toss of Lecture 3, $|\frac{1}{n}\sum_i X_i - p| \le c\sqrt{\log(1/\delta)/n}$ with probability at least $1-\delta$ for an absolute constant c.

The proof has three ingredients: Markov's inequality, a tail bound for sums of sub-Gaussian variables, and the fact that bounded variables are sub-Gaussian.

Lemma 4.2 [Markov's inequality]. *If $X \ge 0$, then $\mathbb{P}(X \ge a) \le \mathbb{E}[X]/a$ for every $a > 0$.*

Proof. $\mathbb{E}[X] \ge \mathbb{P}(X < a)\cdot 0 + \mathbb{P}(X \ge a)\cdot a$. □

Definition 4.3 [Sub-Gaussian]. A random variable X is sub-Gaussian with parameter σ if

$$\mathbb{E}\big[e^{\lambda(X-\mathbb{E}X)}\big] \le \exp(\lambda^2\sigma^2/2) \qquad \text{for all } \lambda \in \mathbb{R}.$$

Lemma 4.4 [Sums of sub-Gaussian variables]. *If $X_1, \dots, X_n$ are independent and X_i is sub-Gaussian*

with parameter σ_i, then for all $t \geq 0$,

$$\mathbb{P}\Big(\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \geq t\Big) \leq \exp\Big(-\frac{t^2}{2\sum_{i=1}^{n}\sigma_i^2}\Big).$$

Proof. For any $\lambda > 0$, the map $x \mapsto e^{\lambda x}$ is increasing, so by Markov's inequality and independence,

$$\mathbb{P}\Big(\sum_{i}(X_i - \mathbb{E}X_i) \geq t\Big) = \mathbb{P}\big(e^{\lambda\sum_i(X_i-\mathbb{E}X_i)} \geq e^{\lambda t}\big) \leq \frac{\prod_{i=1}^{n}\mathbb{E}\big[e^{\lambda(X_i-\mathbb{E}X_i)}\big]}{e^{\lambda t}} \leq \exp\Big(-\lambda t + \frac{\lambda^2\sum_i\sigma_i^2}{2}\Big).$$

The exponent is minimized at $\lambda^\star = t/\sum_i\sigma_i^2$, where it equals $-t^2/(2\sum_i\sigma_i^2)$. □

Lemma 4.5 [Hoeffding's lemma]. *If $X \in [a,b]$ almost surely and $R = b - a$, then $\mathbb{E}\big[e^{\lambda(X-\mathbb{E}X)}\big] \leq e^{\lambda^2R^2/8}$ for all λ; that is, X is sub-Gaussian with parameter $R/2$.*

Proof. Let $Z = X - \mathbb{E}X$ and $\psi(\lambda) = \log\mathbb{E}e^{\lambda Z}$. By Taylor's theorem, $\psi(\lambda) = \psi(0) + \lambda\psi'(0) + \frac{\lambda^2}{2}\psi''(\lambda')$ for some λ' between 0 and λ. Differentiating,

$$\psi'(\lambda) = \mathbb{E}\Big[\frac{e^{\lambda Z}}{\mathbb{E}e^{\lambda Z}}Z\Big], \qquad \psi''(\lambda) = \mathbb{E}\Big[\frac{e^{\lambda Z}}{\mathbb{E}e^{\lambda Z}}Z^2\Big] - \Big(\mathbb{E}\Big[\frac{e^{\lambda Z}}{\mathbb{E}e^{\lambda Z}}Z\Big]\Big)^2,$$

so $\psi(0) = 0$ and $\psi'(0) = \mathbb{E}Z = 0$. Moreover $\psi''(\lambda)$ is the variance of a random variable Y whose law is the tilted law $dQ = \frac{e^{\lambda Z}}{\mathbb{E}e^{\lambda Z}}\,dP$ of Z. The variable Y has the same support as Z, an interval of length R, so $\psi''(\lambda) = \mathbb{E}_Q(Y - \mathbb{E}_QY)^2 \leq \mathbb{E}_Q(Y-m)^2 \leq R^2/4$, where m is the midpoint of the interval. (The cruder bound R^2, from $|Y - \mathbb{E}_QY| \leq R$, already suffices up to constants.) Hence $\psi(\lambda) \leq \lambda^2R^2/8$. □

Proof of Theorem 3.6. By Lemma 4.5 each X_i is sub-Gaussian with $\sigma_i = R_i/2$, and Lemma 4.4 gives $\exp\big(-t^2/(2\sum_i R_i^2/4)\big) = \exp\big(-2t^2/\sum_i R_i^2\big)$. □

The proof used boundedness only through sub-Gaussianity, so Lemma 4.4 applies to any sub-Gaussian variables. A Gaussian $X \sim \mathcal{N}(\mu,\sigma^2)$ satisfies $\mathbb{E}e^{\lambda(X-\mu)} = e^{\lambda^2\sigma^2/2}$, so sums of independent Gaussians also concentrate, with error $\widetilde{O}(\sigma/\sqrt{n})$ for the average, even though they are unbounded.

4.2 Bernstein's inequality

Bernstein's inequality uses the variance as well as the range, and gives a sharper bound when the variance is small.

Theorem 4.6 [Bernstein's inequality]. *Let $X_1,\dots,X_n$ be independent with $|X_i - \mathbb{E}X_i| \leq R$ almost surely*

and $\mathrm{Var}(X_i) \le \sigma_i^2$. *Then for all* $t \ge 0$,

$$\mathbb{P}\Big(\sum_{i=1}^n (X_i - \mathbb{E}X_i) \ge t\Big) \le \exp\Big(-\frac{t^2}{2\sum_{i=1}^n \sigma_i^2 + \frac{2}{3}Rt}\Big).$$

The proof is omitted. To compare with Hoeffding, set the right side equal to δ. Since $u + v \le 2\max\{u, v\}$,

$$\log\frac{1}{\delta} = \frac{t^2}{2\sum_i \sigma_i^2 + \frac{2}{3}Rt} \ge \frac{t^2}{\max\{4\sum_i \sigma_i^2, \frac{4}{3}Rt\}} = \min\Big\{\frac{t^2}{4\sum_i \sigma_i^2}, \frac{3t}{4R}\Big\},$$

so $t \le \max\big\{2\sqrt{\sum_i \sigma_i^2 \log(1/\delta)},\ \frac{4}{3}R\log(1/\delta)\big\}$.

Corollary 4.7 [Bernstein, high-probability form]. *Under the conditions of Theorem 4.6, for any* $\delta \in (0,1)$, *with probability at least* $1-\delta$,

$$\sum_{i=1}^n (X_i - \mathbb{E}X_i) \le C\Big(\sqrt{\textstyle\sum_{i=1}^n \sigma_i^2 \log(1/\delta)} + R\log(1/\delta)\Big)$$

for an absolute constant C. *For identically distributed* X_i *with variance at most* σ^2,

$$\frac{1}{n}\sum_{i=1}^n X_i - \mathbb{E}X \le \widetilde{O}\Big(\frac{\sigma}{\sqrt{n}} + \frac{R}{n}\Big).$$

The term $\sigma/\sqrt{n}$ is the leading-order term and R/n the lower-order term, which vanishes much faster as $n \to \infty$. Compared with Hoeffding's $\widetilde{O}(R/\sqrt{n})$, the leading term improves from $R/\sqrt{n}$ to $\sigma/\sqrt{n}$. Since $\sigma \le R$, Bernstein is never worse, and it is used when $\sigma \ll R$.

Example: *Biased coin.* For Bernoulli(p) samples, $\mathrm{Var}(X) = p(1-p) \le p$, so with probability at least $1-\delta$,

$$\Big|\frac{1}{n}\sum_i X_i - p\Big| \le c\Big(\sqrt{\frac{p}{n}\log\frac{1}{\delta}} + \frac{1}{n}\log\frac{1}{\delta}\Big) = \widetilde{O}\Big(\sqrt{\frac{p}{n}} + \frac{1}{n}\Big),$$

much sharper than Hoeffding's $\widetilde{O}(1/\sqrt{n})$ when $p \ll 1$.

4.3 Concentration without independence

Independence is generally false in reinforcement learning, because the learner's choices depend on what it has seen.

Example: *Adaptive arm choices.* In a multi-armed bandit with Bernoulli arms, the reward $R_j^{(i)}$ of arm j at round i is 1 with probability p_j and 0 otherwise, independently over i and j. Let a_i be the arm pulled at round i and $Z_i = R_{a_i}^{(i)} - p_{a_i}$. The arm is chosen by a possibly randomized policy using

the history, $a_i = f(\rho_i, z_{i-1}, a_{i-1}, \rho_{i-1}, \ldots, z_1, a_1, \rho_1)$, where ρ_i is the policy's internal randomness at round i. So Z_i depends on a_i, which depends on $Z_1, \ldots, Z_{i-1}$: the Z_i are not independent, and the results above do not apply. However, let $\mathcal{F}_{i-1}$ collect everything before the reward of round i, namely $(\rho_i, z_{i-1}, a_{i-1}, \rho_{i-1}, \ldots, z_1, a_1, \rho_1)$. Given $\mathcal{F}_{i-1}$ the arm a_i is determined, and

$$\mathbb{E}[Z_i \mid \mathcal{F}_{i-1}] = \mathbb{E}\left[R_{a_i}^{(i)} - p_{a_i} \;\middle|\; a_i\right] = 0.$$

This property is enough for $\sum_{i=1}^{n} Z_i$ to concentrate even though the terms are dependent (Lecture 5).

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 4 (Spring 2024); written from the Fall 2025 concentration notes and the Spring 2020 Lecture 2 notes.

Lecture 5: Martingale Concentrations

Lecture 4 ended with sums whose terms are dependent but have conditional mean zero given the past. This unit shows that such sums concentrate as well as independent ones, adds the union bound and uniform concentration, and then sets up the problem of learning an MDP from samples.

5.1 Martingales

Let $Z_1, \dots, Z_n$ be random variables and $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}_n$ a filtration, where $\mathcal{F}_{i-1}$ contains all the randomness before Z_i is sampled (including, but not limited to, $Z_1, \dots, Z_{i-1}$), and Z_i is determined by $\mathcal{F}_i$.

Definition 5.1 [Martingale difference sequence]. $\{Z_i\}_{i=1}^n$ is a martingale difference sequence with respect to $\{\mathcal{F}_i\}$ if $\mathbb{E}[Z_i \mid \mathcal{F}_{i-1}] = 0$ for all i. A sequence $\{S_i\}$ is a martingale if $S_i = \sum_{j=1}^{i} Z_j$ for a martingale difference sequence $\{Z_j\}$.

Since $\mathbb{E}[S_{t+1} - S_t \mid \mathcal{F}_t] = \mathbb{E}[Z_{t+1} \mid \mathcal{F}_t] = 0$, the expected change of a martingale is zero whatever history we condition on: the path may wander, but there is no cumulative drift. The adaptive bandit rewards $Z_i = R_{a_i}^{(i)} - p_{a_i}$ of Lecture 4 form a martingale difference sequence.

Example: *Dependent but conditionally centered.* Let $Z_1 = \pm 1$ with probability $\frac{1}{2}$ each, and let Z_2 depend on Z_1:

	$Z_2 = -1$	$Z_2 = 0$	$Z_2 = 1$
given $Z_1 = -1$	0.5	0	0.5
given $Z_1 = 1$	0	1	0

Z_1 and Z_2 are clearly dependent, yet $\mathbb{E}Z_1 = 0$ and $\mathbb{E}[Z_2 \mid Z_1 = -1] = \mathbb{E}[Z_2 \mid Z_1 = 1] = 0$, so (Z_1, Z_2) is a martingale difference sequence.

5.2 Azuma-Hoeffding and Azuma-Bernstein

Theorem 5.2 [Azuma-Hoeffding inequality]. *Let $\{X_i - \mathbb{E}X_i\}_{i=1}^n$ be a martingale difference sequence with respect to $\{\mathcal{F}_i\}$, and suppose that, conditionally on $\mathcal{F}_{i-1}$, X_i lies in an interval of length R_i. Then for all $t \geq 0$,*

$$\mathbb{P}\Big(\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \geq t\Big) \leq \exp\Big(-\frac{2t^2}{\sum_{i=1}^{n} R_i^2}\Big).$$

In particular, if $|X_i - \mathbb{E}X_i| \leq R_i$, the bound is $\exp\big(-t^2/(2\sum_i R_i^2)\big)$.

Proof. The proof of Hoeffding's inequality used independence in exactly one step,

$$\mathbb{E}\big[e^{\lambda \sum_i (X_i - \mathbb{E}X_i)}\big] = \prod_i \mathbb{E}\big[e^{\lambda(X_i - \mathbb{E}X_i)}\big],$$

and the martingale condition replaces it. For any function f, the tower property gives $\mathbb{E}[f(X_1, \dots, X_n)] = \mathbb{E}\big[\mathbb{E}[f(X_1, \dots, X_n) \mid \mathcal{F}_{n-1}]\big]$, where the inner expectation is over X_n only. Since $\sum_{i<n}(X_i - \mathbb{E}X_i)$ is determined by $\mathcal{F}_{n-1}$,

$$\begin{aligned}\mathbb{E}\Big[e^{\lambda \sum_{i=1}^{n}(X_i - \mathbb{E}X_i)}\Big] &= \mathbb{E}\Big[e^{\lambda \sum_{i=1}^{n-1}(X_i - \mathbb{E}X_i)}\, \mathbb{E}\big[e^{\lambda(X_n - \mathbb{E}X_n)} \,\big|\, \mathcal{F}_{n-1}\big]\Big] \\ &\le e^{\lambda^2 R_n^2/8}\, \mathbb{E}\Big[e^{\lambda \sum_{i=1}^{n-1}(X_i - \mathbb{E}X_i)}\Big] \le \cdots \le \exp\Big(\frac{\lambda^2 \sum_{i=1}^{n} R_i^2}{8}\Big).\end{aligned}$$

The first inequality is Hoeffding's lemma (Lemma 4.5) applied conditionally: whatever history $\mathcal{F}_{n-1}$ we condition on, $X_n - \mathbb{E}X_n$ has mean zero and lies in an interval of length R_n. The rest peels off $j = n-1, n-2, \dots$ in the same way. The Chernoff argument of Lemma 4.4 with $\sigma_i = R_i/2$ finishes the proof. □

Theorem 5.3 [Azuma-Bernstein inequality]. *Let $\{X_i - \mathbb{E}X_i\}_{i=1}^n$ be a martingale difference sequence with respect to $\{\mathcal{F}_i\}$ such that $\mathrm{Var}(X_i \mid \mathcal{F}_{i-1}) \le \sigma_i^2$ and $X_i - \mathbb{E}X_i \le R$ for all i. Then for all $t \ge 0$,*

$$\mathbb{P}\Big(\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \ge t\Big) \le \exp\Big(-\frac{t^2}{2\sum_{i=1}^{n}\sigma_i^2 + \frac{2}{3}Rt}\Big).$$

So the rates for independent variables survive without independence: with probability at least $1 - \delta$,

Inequality	Bound on $\frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}X_i)$
Hoeffding, Azuma-Hoeffding	$O\big(R\sqrt{\log(1/\delta)/n}\big)$
Bernstein, Azuma-Bernstein	$O\big(\sigma\sqrt{\log(1/\delta)/n} + R\log(1/\delta)/n\big)$

5.3 Union bound and uniform concentration

Lemma 5.4 [Union bound]. *For any events A and B, $\mathbb{P}(A \cup B) + \mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B)$, so $\mathbb{P}(A \cup B) \le \mathbb{P}(A) + \mathbb{P}(B)$.*

For example, if $Z = \sum_i (X_i - \mathbb{E}X_i)$ satisfies $\mathbb{P}(Z \ge t) \le \delta$ and $\mathbb{P}(Z \le -t) \le \delta$, then $\mathbb{P}(|Z| \ge t) \le 2\delta$: with probability at least $1 - 2\delta$, $|Z| \le t$.

The union bound also controls quantities that depend on the data. Let $x_1, \dots, x_n$ be i.i.d. on a finite

set $\mathcal{S}$ with $|\mathcal{S}| = S$, and consider functions $f : \mathcal{S} \to [0,1]$. For a *fixed* f, Hoeffding gives

$$\mathbb{P}\Big(\Big|\sum_{i=1}^{n}\big(f(x_i) - \mathbb{E}f(x_i)\big)\Big| \geq t\Big) \leq 2\exp\Big(-\frac{2t^2}{n}\Big).$$

If instead $\hat{f}$ is chosen using the data, for example $\hat{f} = \arg\min_{f\in\mathcal{F}} \ell(f, \{x_i\}_{i=1}^n)$ for a function class $\mathcal{F}$ and a loss ℓ, this does not apply directly. The relaxation

$$\mathbb{P}\Big(\Big|\sum_{i}\big(\hat{f}(x_i) - \mathbb{E}\hat{f}(x_i)\big)\Big| \geq t\Big) \leq \mathbb{P}\Big(\exists f \in \mathcal{F} : \Big|\sum_{i}\big(f(x_i) - \mathbb{E}f(x_i)\big)\Big| \geq t\Big)$$

reduces the question to UNIFORM CONCENTRATION over $\mathcal{F}$.

Finite classes. By the union bound the right side is at most $2|\mathcal{F}|e^{-2t^2/n}$, so with probability at least $1-\delta$,

$$\sup_{f\in\mathcal{F}}\Big|\frac{1}{n}\sum_{i=1}^{n}\big(f(x_i) - \mathbb{E}f(x_i)\big)\Big| \leq \sqrt{\frac{1}{2n}\log\frac{2|\mathcal{F}|}{\delta}} = O\Big(\sqrt{\frac{1}{n}\log\frac{|\mathcal{F}|}{\delta}}\Big).$$

Since $|\mathcal{F}|$ enters only through its logarithm, exponentially many functions are affordable.

Infinite classes. Let $\mathcal{F}$ be all functions $\mathcal{S} \to [0,1]$, a subset of $[0,1]^S$. Discretize with the ϵ-grid $G_\epsilon = \{0, \epsilon, 2\epsilon, \ldots, \lfloor 1/\epsilon \rfloor \epsilon\}$ and the ϵ-NET $\mathcal{F}_\epsilon = G_\epsilon^S$. Then $|\mathcal{F}_\epsilon| \approx (1/\epsilon)^S$, and every $f \in \mathcal{F}$ has some $f_\epsilon \in \mathcal{F}_\epsilon$ with $\sup_x |f(x) - f_\epsilon(x)| \leq \epsilon$. Applying the finite-class bound to $\mathcal{F}_\epsilon$ and paying at most 2ϵ for the discretization, with probability at least $1-\delta$, for all $f \in \mathcal{F}$,

$$\Big|\frac{1}{n}\sum_{i=1}^{n}\big(f(x_i) - \mathbb{E}f(x_i)\big)\Big| \leq \Big|\frac{1}{n}\sum_{i=1}^{n}\big(f_\epsilon(x_i) - \mathbb{E}f_\epsilon(x_i)\big)\Big| + 2\epsilon \leq O\Big(\sqrt{\frac{1}{n}\Big(S\log\frac{1}{\epsilon} + \log\frac{1}{\delta}\Big)}\Big) + 2\epsilon.$$

Choosing $\epsilon = 1/\sqrt{n}$ gives $O\big(\sqrt{(S\log n + \log(1/\delta))/n}\big) = \widetilde{O}\big(\sqrt{S/n}\big)$.

5.4 Learning settings for MDPs

When P and r are unknown they must be learned from samples. The settings of Lecture 1 are the simulator (generative model), the online setting, and the offline setting.

Definition 5.5 [SIMULATOR]. A SIMULATOR (GENERATIVE MODEL) is an oracle that, queried with any triple (s, a, h), returns a next state $s' \sim P_h(\cdot \mid s, a)$ and the reward $r = r_h(s, a)$.

Rewards are assumed noise-free: for fixed (s, a), $P_h(\cdot \mid s, a)$ is a multinomial distribution and much harder to learn than the scalar reward, so little is lost. The strong assumption is that *any* state can be queried, which is often impossible in practice. In a maze, querying a position and a move returns the next position (unchanged on hitting a wall) and reward 1 at the exit; the simulator can teleport anywhere, whereas in practice reaching a given position is itself a hard problem. In the ONLINE setting each episode starts from the initial state, and the agent can only choose actions.

Definition: The goal is to output, with high probability, an ϵ-optimal policy $\hat{\pi}$, one with $V_1^{\star}(s_1) - V_1^{\hat{\pi}}(s_1) \le \epsilon$. The sample complexity is the number of samples (simulator queries) needed to do so.

The simplest algorithm with a simulator sweeps over all triples (s, a, h), estimates the transitions by empirical frequencies, and plans in the estimated model. Such a sweep is possible only because the simulator can query any state.

Algorithm 5.1: Model-based value iteration with a simulator

Input: number of samples n per triple
for each $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$ **do**
  query (s, a, h) n times to obtain next states $s'_1, \dots, s'_n$
  $\hat{P}_h(s' \mid s, a) \leftarrow \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[s'_i = s']$ for all $s' \in \mathcal{S}$
$\hat{V}_{H+1} \leftarrow 0$
for $h = H, \dots, 1$ **do**
  $\hat{Q}_h(s, a) \leftarrow r_h(s, a) + (\hat{P}_h \hat{V}_{h+1})(s, a)$ for all (s, a)
  $\hat{V}_h(s) \leftarrow \max_a \hat{Q}_h(s, a)$ for all s
return $\hat{\pi}_h(s) = \arg\max_a \hat{Q}_h(s, a)$

The algorithm runs value iteration (Lecture 3) in the empirical MDP $\mathrm{MDP}(\hat{P}, r)$, so $\hat{\pi}$ is optimal for that MDP. Lecture 6 bounds how large n must be.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 5 (Spring 2024); written from the Fall 2025 concentration and simulator notes and the Spring 2020 Lecture 3 and Lecture 5 notes.

Lecture 6: Generative Models

This unit bounds the sample complexity of model-based value iteration with a simulator (Algorithm 5.1): n queries of every triple (s, a, h), the empirical transitions $\hat{P}_h$, and the output $\hat{\pi}$, which is optimal for the empirical MDP. Write V^π, Q^π for the value functions of policy π in the true MDP $M = \mathrm{MDP}(P, r)$ and $\hat{V}^\pi, \hat{Q}^\pi$ for its value functions in $\hat{M} = \mathrm{MDP}(\hat{P}, r)$. Throughout, $\iota = \log(HSA/\delta)$.

6.1 Coarse analysis

Theorem 6.1 [Value iteration, coarse bound]. *There is an absolute constant c such that if $n \geq cH^4S\iota/\epsilon^2$, then with probability at least $1-\delta$ the policy $\hat{\pi}$ is ϵ-optimal.*

The total number of samples is $N = nHSA = cH^5S^2A\iota/\epsilon^2$. The factor $1/\epsilon^2$ comes from concentration, $\sqrt{1/n} \approx \epsilon$, and S^2A is the number of parameters of each transition matrix. The refined analysis below removes one factor of S.

The first ingredient, the simulation lemma, expresses a difference of values under the same policy through the difference of the transitions.

Lemma 6.2 [Simulation lemma]. *For any policy π and any $(s, h) \in \mathcal{S} \times [H]$,*

$$V_h^\pi(s) - \hat{V}_h^\pi(s) = \mathbb{E}_{\hat{M},\pi}\Big[\sum_{i=h}^{H} \big((P_i - \hat{P}_i)V_{i+1}^\pi\big)(s_i, a_i) \;\Big|\; s_h = s\Big],$$

where $\mathbb{E}_{\hat{M},\pi}$ is over future states and actions with actions drawn from π and states drawn from $\hat{P}$.

Proof. By the Bellman equations, adding and subtracting $\hat{P}_h V_{h+1}^\pi$,

$$\begin{aligned} Q_h^\pi(s,a) - \hat{Q}_h^\pi(s,a) &= (P_h V_{h+1}^\pi)(s,a) - (\hat{P}_h \hat{V}_{h+1}^\pi)(s,a) \\ &= \big((P_h - \hat{P}_h)V_{h+1}^\pi\big)(s,a) + \big(\hat{P}_h(V_{h+1}^\pi - \hat{V}_{h+1}^\pi)\big)(s,a). \end{aligned}$$

Averaging over $a_h \sim \pi_h(\cdot \mid s)$,

$$V_h^\pi(s) - \hat{V}_h^\pi(s) = \mathbb{E}_\pi\big[\big((P_h - \hat{P}_h)V_{h+1}^\pi\big)(s_h, a_h) \;\big|\; s_h = s\big] + \mathbb{E}_{\hat{M},\pi}\big[V_{h+1}^\pi(s_{h+1}) - \hat{V}_{h+1}^\pi(s_{h+1}) \;\big|\; s_h = s\big].$$

This expresses the difference at step h through the difference at step $h+1$ plus one additive term; expanding until step H gives the claim. □

Lemma 6.3 [Uniform deviation]. *There is an absolute constant c such that with probability at least $1-\delta$, for all $(s,a,h) \in \mathcal{S} \times \mathcal{A} \times [H]$ and all $V \in [0,H]^S$,*

$$\left|\left((P_h - \hat{P}_h)V\right)(s,a)\right| \le cH\sqrt{\frac{S\iota}{n}}.$$

Proof. Fix (s,a,h) and write $p = P_h(\cdot \mid s,a)$, $\hat{p} = \hat{P}_h(\cdot \mid s,a)$. First, $\|\hat{p} - p\|_1 \le c\sqrt{S\iota/n}$ with probability at least $1 - \delta/(HSA)$. Indeed, $\|\hat{p} - p\|_1 = \max_{u \in \{\pm 1\}^S} \langle \hat{p} - p, u \rangle$, and for a fixed sign vector u, $\langle \hat{p} - p, u \rangle = \frac{1}{n}\sum_{i=1}^{n} \left(u(s_i') - \mathbb{E}u(s_i')\right)$ is an average of i.i.d. terms with range 2, so Hoeffding bounds it by t except with probability $e^{-nt^2/2}$. A union bound over the 2^S sign vectors with $t = \sqrt{2(S\log 2 + \log(HSA/\delta))/n}$ gives the claim. Then $|\langle p - \hat{p}, V \rangle| \le \|p - \hat{p}\|_1 \|V\|_\infty \le cH\sqrt{S\iota/n}$ for every $V \in [0,H]^S$ simultaneously. A union bound over the HSA triples finishes the proof. □

Proof of Theorem 6.1. On the event of Lemma 6.3, for every policy π, the simulation lemma gives

$$|V_1^\pi(s_1) - \hat{V}_1^\pi(s_1)| \le \mathbb{E}_{\hat{M},\pi}\left[\sum_{h=1}^{H} \left|\left((P_h - \hat{P}_h)V_{h+1}^\pi\right)(s_h,a_h)\right|\right] \le H \cdot cH\sqrt{\frac{S\iota}{n}} \le \frac{\epsilon}{2}$$

once $n \ge c'H^4 S\iota/\epsilon^2$. Then

$$V_1^{\pi^\star}(s_1) - V_1^{\hat{\pi}}(s_1) = \left[V_1^{\pi^\star}(s_1) - \hat{V}_1^{\pi^\star}(s_1)\right] + \left[\hat{V}_1^{\pi^\star}(s_1) - \hat{V}_1^{\hat{\pi}}(s_1)\right] + \left[\hat{V}_1^{\hat{\pi}}(s_1) - V_1^{\hat{\pi}}(s_1)\right] \le \frac{\epsilon}{2} + 0 + \frac{\epsilon}{2},$$

since $\hat{\pi}$ is optimal for $\hat{M}$, so the middle term is nonpositive. The bound for $\hat{\pi}$ is legitimate even though $\hat{\pi}$ depends on the data, because Lemma 6.3 holds for all V simultaneously. □

6.2 Refined analysis

The S^2A scaling is loose: the correct scaling is SA, linear in S (lower bounds show that linear scaling is necessary), and value iteration already achieves it. Only the analysis changes.

Theorem 6.4 [Value iteration, refined bound]. *There is an absolute constant c such that if $n \ge cH^4\iota/\epsilon^2$, then with probability at least $1-\delta$ the policy $\hat{\pi}$ is ϵ-optimal. The total number of samples is $nHSA = cH^5SA\iota/\epsilon^2$.*

Although P has S^2A parameters, a near-optimal policy does not require learning P accurately along every direction V; it suffices to learn P along a few value functions. The key is a bound for a single V.

Lemma 6.5 [Fixed value function]. *For a fixed (s,a,h) and a fixed $V \in [0,H]^S$, with probability at least*

$1-\delta/(HSA)$,

$$\left|\left((P_h-\hat{P}_h)V\right)(s,a)\right| \le cH\sqrt{\frac{\iota}{n}}$$

for an absolute constant c. The same holds for a random V that is independent of the samples $s'_1,\dots,s'_n$ drawn for (s,a,h).

Proof. For fixed V, $(\hat{P}_hV)(s,a)=\frac{1}{n}\sum_{i=1}^n V(s'_i)$ is an average of i.i.d. terms in $[0,H]$ with mean $(P_hV)(s,a)$, so Hoeffding applies. For a random V independent of the samples, let $E(V)$ be the failure event; then $\mathbb{P}(E(V))=\int \mathbb{P}(E(V)\mid V=v)\,d\mu(v)\le \delta/(HSA)$, because the conditional law of the samples given $V=v$ is unchanged and the bound holds for each fixed v. □

The coarse analysis pays an extra $\sqrt{S}$ to cover value functions that adapt to the randomness in $\hat{P}_h$. Indeed, since $\hat{p}-p$ sums to zero, the largest deviation over $V\in[0,H]^S$ is attained by the adaptive choice $V(s')=H\cdot\mathbf{1}[\hat{p}(s')>p(s')]$, which gives $\langle\hat{p}-p,V\rangle$ equal to H times the total variation distance $\frac{1}{2}\|\hat{p}-p\|_1$; a fixed V is not correlated with the fluctuation $\hat{p}-p$ and sees only the $1/\sqrt{n}$ error of an average.

Proof of Theorem 6.4. Decompose

$$V_1^{\pi^\star}(s_1)-V_1^{\hat{\pi}}(s_1)=\underbrace{V_1^{\pi^\star}(s_1)-\hat{V}_1^{\pi^\star}(s_1)}_{A_1}+\underbrace{\hat{V}_1^{\pi^\star}(s_1)-\hat{V}_1^{\hat{\pi}}(s_1)}_{A_2}+\underbrace{\hat{V}_1^{\hat{\pi}}(s_1)-V_1^{\hat{\pi}}(s_1)}_{A_3}.$$

As before $A_2\le 0$. For A_1, the simulation lemma and Lemma 6.5 applied to $V=V_{h+1}^{\pi^\star}$, which is fixed because $\pi^\star$ does not depend on the simulator's outputs, give (after a union bound over (s,a,h))

$$|A_1|\le \mathbb{E}_{\hat{M},\pi^\star}\Big[\sum_{h=1}^H\left|\left((P_h-\hat{P}_h)V_{h+1}^{\pi^\star}\right)(s_h,a_h)\right|\Big]\le cH^2\sqrt{\frac{\iota}{n}}\le\frac{\epsilon}{2}$$

once $n\ge c'H^4\iota/\epsilon^2$. For A_3, the simulation lemma gives

$$|A_3|\le \mathbb{E}_{\hat{M},\hat{\pi}}\Big[\sum_{h=1}^H\left|\left((P_h-\hat{P}_h)V_{h+1}^{\hat{\pi}}\right)(s_h,a_h)\right|\Big].$$

Value iteration computes $\hat{\pi}_{h+1},\dots,\hat{\pi}_H$ from $\hat{P}_{h+1},\dots,\hat{P}_H$ only, so the true value $V_{h+1}^{\hat{\pi}}$ depends only on the samples drawn at steps $h+1,\dots,H$ and is independent of the samples behind $\hat{P}_h$. Lemma 6.5 therefore applies again and $|A_3|\le cH^2\sqrt{\iota/n}\le\epsilon/2$. Combining the three terms proves the theorem. □

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 6 (Spring 2024); written from the Fall 2025 simulator notes and the Spring 2020 Lecture 5 and Lecture 6 notes.

Lecture 7: Introduction to Exploration

This unit finishes the simulator setting with Q-learning, a model-free alternative to value iteration, and summarizes what is known there. It then turns to the online setting, where the agent must explore, shows that random exploration can fail exponentially, and introduces the multi-armed bandit as the simplest exploration problem.

7.1 Q-learning

Value iteration with a simulator performs a batch update: with $\hat{P}_h(s' \mid s,a) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[s_i' = s']$,

$$\hat{Q}_h(s,a) = r_h(s,a) + (\hat{P}_h \hat{V}_{h+1})(s,a) = r_h(s,a) + \sum_{i=1}^{n} \frac{1}{n} \hat{V}_{h+1}(s_i'), \tag{7.1}$$

so all the data must be collected and stored first. Q-learning instead updates incrementally as soon as a sample arrives,

$$Q_h^t(s,a) = (1-\alpha_t) Q_h^{t-1}(s,a) + \alpha_t \left[r_h(s,a) + V_{h+1}^{t-1}(s_t') \right],$$

keeping a $1-\alpha_t$ fraction of the previous estimate and adding an α_t fraction of a one-sample Bellman update, where α_t is the learning rate.

Algorithm 7.1: Q-learning with a simulator

Input: learning rates $\{\alpha_t\}_{t \geq 1}$

$Q_h^0(s,a) \leftarrow H$ for all (s,a,h), and $V_h^0(s) \leftarrow \max_a Q_h^0(s,a)$; $V_{H+1}^t \leftarrow 0$

for $t = 1, 2, \ldots$ **do**

  for $h = 1, \ldots, H$ **do**

    for each $(s,a) \in \mathcal{S} \times \mathcal{A}$ **do**

      sample $s_t' \sim P_h(\cdot \mid s,a)$

      $Q_h^t(s,a) \leftarrow (1-\alpha_t) Q_h^{t-1}(s,a) + \alpha_t \left[r_h(s,a) + V_{h+1}^{t-1}(s_t') \right]$

    $V_h^t(s) \leftarrow \max_{a \in \mathcal{A}} Q_h^t(s,a)$ for all s

Two features distinguish Q-learning from value iteration.

1. It is model-free: it never forms an estimate $\hat{P}$ of the model and learns directly how to act.
2. It replaces the exact Bellman backup by an incremental update with one noisy sample. It can discard old data and needs only $O(HSA)$ space, while value iteration stores all its data, $\widetilde{O}(SA\,\mathrm{poly}(H)/\epsilon^2)$ space. This is the key advantage of Q-learning.

Learning rates. With $\alpha_1 = 1$, unrolling the update writes Q-learning as a weighted average of past one-sample backups,

$$Q_h^t(s,a) = \sum_{i=1}^{t} \gamma_t^i \left[r_h(s,a) + V_{h+1}^{i-1}(s_i') \right] = r_h(s,a) + \sum_{i=1}^{t} \gamma_t^i V_{h+1}^{i-1}(s_i'), \tag{7.2}$$

where s_i' is the ith sample and the weights $\gamma_t^i = \alpha_i \prod_{j=i+1}^{t}(1-\alpha_j)$ satisfy $\sum_{i=1}^{t} \gamma_t^i = 1$. Value iteration (7.1) uses uniform weights. Q-learning recovers uniform weights $\gamma_t^i = 1/t$ when $\alpha_t = 1/t$, favors later samples when $\alpha_t > 1/t$, and favors earlier samples when $\alpha_t < 1/t$ (Figure 2). Another difference is that value iteration always backs up the same estimate $\hat{V}_{h+1}$, while Q-learning uses the sequence $V_{h+1}^0, \dots, V_{h+1}^{t-1}$, whose early members can be very inaccurate. The algorithm should therefore favor later samples, and in theory the best choice is $\alpha_t = \frac{H+1}{H+t} \approx \frac{H}{t} > \frac{1}{t}$. The rate must not be too large either: at the extreme $\alpha_t = 1$ only the last, noisy sample is used.

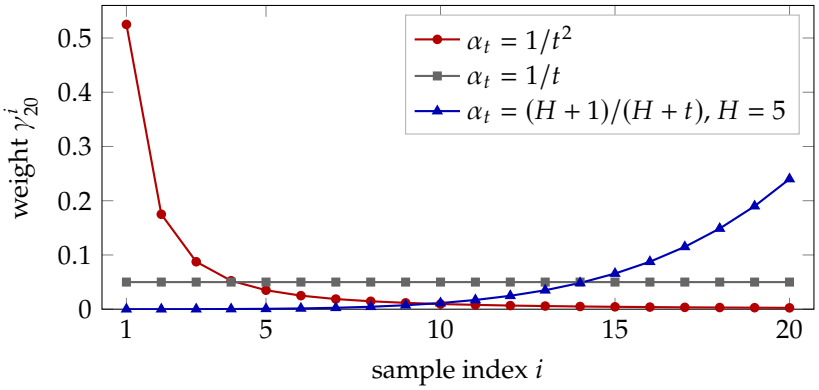


Figure 2: The weights γ_t^i of (7.2) at $t = 20$, computed exactly. The rate $\alpha_t = 1/t^2 < 1/t$ favors early samples, $\alpha_t = 1/t$ gives the uniform weights of value iteration, and $\alpha_t = (H+1)/(H+t) > 1/t$ favors late samples.

Lemma 7.1 [Properties of the weights]. *For $\alpha_t = \frac{H+1}{H+t}$: (i) $\sum_{i=1}^{t}(\gamma_t^i)^2 \le \frac{2H}{t}$ for all t; (ii) $\sum_{t=i}^{\infty} \gamma_t^i = 1 + \frac{1}{H}$ for all i.*

Proof. (i) The ratio $\gamma_t^{i+1}/\gamma_t^i = \frac{\alpha_{i+1}}{\alpha_i(1-\alpha_{i+1})} = \frac{H+i}{i} \ge 1$, so the weights increase in i and $\max_i \gamma_t^i = \gamma_t^t = \alpha_t \le \frac{H+1}{t} \le \frac{2H}{t}$. Hence $\sum_i (\gamma_t^i)^2 \le \max_i \gamma_t^i \sum_i \gamma_t^i \le \frac{2H}{t}$. (ii) Write $\gamma_t^i = \alpha_i x_t$ with $x_i = 1$ and $x_t = x_{t-1}(1-\alpha_t) = x_{t-1}\frac{t-1}{H+t}$. Then $Hx_t = (t-1)x_{t-1} - tx_t$, and summing over $t > i$ telescopes to $H\sum_{t>i} x_t = ix_i = i$, since $tx_t \to 0$. So $\sum_{t \ge i} \gamma_t^i = \alpha_i\left(1 + \frac{i}{H}\right) = \frac{H+1}{H+i} \cdot \frac{H+i}{H} = 1 + \frac{1}{H}$. □

Theorem 7.2 [Q-learning with a simulator]. *Let $\alpha_t = \frac{H+1}{H+t}$ and $\iota = \log(HSAT/\delta)$. With probability at least $1-\delta$, for all $t \le T$ and all $h \in [H]$,*

$$\left\| \frac{1}{t} \sum_{i=1}^{t} Q_h^i - Q_h^\star \right\|_\infty \le c\sqrt{\frac{H^5 \iota}{t}}.$$

To make the right side at most ϵ takes $t = cH^5\iota/\epsilon^2$ iterations of HSA samples each, $\widetilde{O}(H^6SA/\epsilon^2)$ samples in total: one factor of H worse than value iteration, because of the uneven weighting.

Proof. Since $\sum_i \gamma_t^i = 1$ and $Q_h^\star = r_h + P_h V_{h+1}^\star$, (7.2) gives

$$Q_h^t(s,a) - Q_h^\star(s,a) = \underbrace{\sum_{i=1}^t \gamma_t^i \left[V_{h+1}^{i-1}(s_i') - V_{h+1}^\star(s_i')\right]}_{\beta_1} + \underbrace{\sum_{i=1}^t \gamma_t^i \left[V_{h+1}^\star(s_i') - (P_h V_{h+1}^\star)(s,a)\right]}_{\beta_2}.$$

The terms of β_2 are independent, mean zero, and bounded by H, so by Hoeffding and Lemma 7.1(i), with probability at least $1 - \delta/(HSAT)$, $|\beta_2| \le c\sqrt{\sum_i (\gamma_t^i)^2 H^2 \iota} \le c\sqrt{H^3\iota/t}$. A union bound over (s,a,h) and over the iterations $t \le T$ makes this hold everywhere. Using $\|V_{h+1}^{i-1} - V_{h+1}^\star\|_\infty \le \|Q_{h+1}^{i-1} - Q_{h+1}^\star\|_\infty$ (the maximum is 1-Lipschitz in ℓ_∞),

$$\|Q_h^t - Q_h^\star\|_\infty \le \sum_{i=1}^t \gamma_t^i \|Q_{h+1}^{i-1} - Q_{h+1}^\star\|_\infty + c\sqrt{\frac{H^3\iota}{t}}.$$

Sum over t, exchange the order of summation, and apply Lemma 7.1(ii) and $\sum_{j=1}^t 1/\sqrt{j} \le 2\sqrt{t}$:

$$\begin{aligned}\sum_{j=1}^t \|Q_h^j - Q_h^\star\|_\infty &\le \sum_{i=1}^t \Big(\sum_{j=i}^t \gamma_j^i\Big) \|Q_{h+1}^{i-1} - Q_{h+1}^\star\|_\infty + 2c\sqrt{H^3 t\iota} \\ &\le \left(1 + \frac{1}{H}\right) \sum_{i=0}^{t-1} \|Q_{h+1}^i - Q_{h+1}^\star\|_\infty + 2c\sqrt{H^3 t\iota}.\end{aligned}$$

Add the $j = 0$ term $\|Q_h^0 - Q_h^\star\|_\infty \le H \le \sqrt{H^3 t\iota}$ to the left and the nonnegative $i = t$ term to the right, so both sums run from 0 to t. Unrolling from h to $H+1$, where $Q_{H+1}^i = Q_{H+1}^\star = 0$,

$$\sum_{i=0}^t \|Q_h^i - Q_h^\star\|_\infty \le \left[1 + \left(1 + \frac{1}{H}\right) + \cdots + \left(1 + \frac{1}{H}\right)^{H-h}\right](2c+1)\sqrt{H^3 t\iota} \le e(2c+1)\sqrt{H^5 t\iota},$$

since each of the at most H terms is at most $(1 + 1/H)^H \le e$. Finally, by convexity of the norm, $\|\frac{1}{t}\sum_{i=1}^t Q_h^i - Q_h^\star\|_\infty \le \frac{1}{t}\sum_{i=1}^t \|Q_h^i - Q_h^\star\|_\infty \le e(2c+1)\sqrt{H^5\iota/t}$. □

7.2 Summary of the simulator setting

Algorithm	Sample complexity	Space
Value iteration	$\widetilde{O}(H^5SA/\epsilon^2)$	$\widetilde{O}\big(\min\{H^5SA/\epsilon^2, HS^2A\}\big)$
Q-learning	$\widetilde{O}(H^6SA/\epsilon^2)$	$O(HSA)$
Lower bound	$\Omega(H^4SA/\epsilon^2)$	

Value iteration either stores all samples or, more cheaply when possible, the transition matrices $\hat{P}_h \in [0,1]^{S \times SA}$; Q-learning stores only the value tables. Two comments:

- A sharper analysis that replaces Hoeffding's inequality by a careful use of Bernstein's improves both sample complexities by a factor of H, so value iteration is minimax optimal: it matches the lower bound.
- These bounds are for non-stationary MDPs, where $P_1, \ldots, P_H$ may all differ. For stationary MDPs ($P_h = P$ for all h) the minimax optimal rate is $\widetilde{O}(H^3SA/\epsilon^2)$.

7.3 Online reinforcement learning

Without a simulator, every episode starts from the initial state, as in a video game that always starts at the first level. A simulator makes reaching any state trivial; online, the agent must work out how to get there. Balancing the effort to reach and learn about states against playing the best policy found so far is the trade-off between EXPLORATION and EXPLOITATION.

The simplest exploration strategy, taking uniformly random actions, can fail badly.

Example: *Combination lock.* Consider two states and two actions over horizon H (Figure 3). The episode starts in state g. At each step h, one action $a_h^\star$ keeps the agent in g and the other moves it to b, which is absorbing under both actions; which action is the right one may change from step to step and is unknown to the learner. All rewards are zero except reward 1 for being in g at step H. The optimal policy plays $a_h^\star$ in g (anything in b), with $V_1^\star(\mathsf{g}) = 1$, and the only way to see a nonzero reward is the exact action sequence $(a_1^\star, \ldots, a_{H-1}^\star)$. Uniformly random actions produce it with probability $2^{-(H-1)}$ per episode, so random exploration needs $\Omega(2^H)$ episodes in expectation to discover the optimal policy, even though the MDP has only two states and two actions.

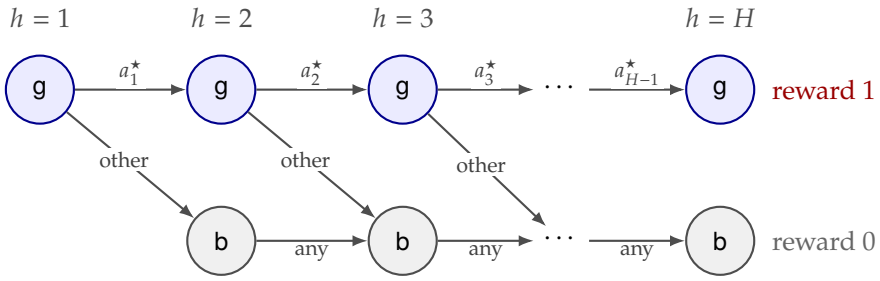


Figure 3: The combination lock. Only the unknown sequence $a_1^\star, \ldots, a_{H-1}^\star$ keeps the agent in g until step H and earns the reward.

The popular ϵ-GREEDY strategy takes a uniformly random action with probability ϵ and the greedy action with respect to the current estimates otherwise. Its exploration comes only from the random actions, so in the worst case, as in the combination lock, it also needs exponentially many episodes. Strategic exploration is developed first in the simplest setting, the bandit, and then extended to MDPs (Lectures 8 and 9).

7.4 Multi-armed bandits

Algorithm 7.2: Multi-armed bandit

```
for t = 1, ..., T do
    the learner picks an arm i_t ∈ [A]
    the learner observes a stochastic reward R_{i_t}
```

The rewards satisfy $R_i \in [0,1]$ almost surely with $\mathbb{E}R_i = r_i$, and the best arm is $i^\star = \arg\max_{i\in[A]} r_i$. There are two goals.

1. *Find an ϵ-optimal arm $\hat{i}$*, with $r_{\hat{i}} \geq r_{i^\star} - \epsilon$. This depends only on the final output, not on how the learner played along the way.
2. *Achieve low regret*, where the REGRET

$$\mathrm{Regret}(T) = T \cdot r_{i^\star} - \sum_{t=1}^{T} r_{i_t} \in [0, T]$$

compares the expected reward of always playing the best arm with that of the arms actually played. Low regret requires both finding out which arms are good (exploration) and playing well in most rounds (exploitation).

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 7 (Spring 2024); written from the Fall 2025 simulator notes and the Spring 2020 Lecture 6, Lecture 7, and Lecture 8 notes.

Lecture 8: Exploration in Multi-Arm Bandits

This unit studies exploration in the multi-armed bandit of Lecture 7: A arms with rewards $R_i \in [0,1]$ of mean r_i, best arm $i^\star$, and T rounds. Simple strategies that explore at random achieve regret of order $T^{2/3}$; the optimism principle behind UCB achieves the optimal $\sqrt{T}$.

8.1 Regret versus best-arm identification

The two goals of Lecture 7 are related.

Lemma 8.1 [Conversion]. *Let $c > 0$.*

1. *If an algorithm has regret at most $cT^{1-\alpha}$ for some $\alpha \in (0,1]$, then the uniform distribution $\hat{D}$ over the arms it played, $\hat{D} = \text{Uniform}\{i_1, \ldots, i_T\}$, satisfies $\mathbb{E}_{i\sim\hat{D}} r_i \geq r_{i^\star} - \epsilon$ after $T = (c/\epsilon)^{1/\alpha}$ rounds.*
2. *If an algorithm finds an ϵ-optimal arm with $c\epsilon^{-\beta}$ samples, then running it and committing to its output has regret at most $2c^{1/(1+\beta)}T^{\beta/(1+\beta)}$.*

Proof. (1) Dividing $T\, r_{i^\star} - \sum_t r_{i_t} \leq cT^{1-\alpha}$ by T gives $\frac{1}{T}\sum_t r_{i_t} \geq r_{i^\star} - c/T^\alpha$, and $c/T^\alpha = \epsilon$ at $T = (c/\epsilon)^{1/\alpha}$. (2) The first $c\epsilon^{-\beta}$ rounds cost regret at most 1 each, and the remaining rounds at most ϵ each, so $\text{Regret}(T) \leq c\epsilon^{-\beta} + T\epsilon$. Choosing $\epsilon = (c/T)^{1/(1+\beta)}$ balances the two terms. □

For example, regret $\sqrt{T}$ gives an ϵ-optimal arm (distribution) with $1/\epsilon^2$ samples, and $1/\epsilon^2$ samples give regret $T^{2/3}$.

8.2 Explore-then-commit and ϵ-greedy

The ϵ-first algorithm (explore-then-commit) pulls each arm n times, computes the empirical means $\hat{r}_i = \frac{1}{n}\sum_{j=1}^n R_i^{(j)}$, and plays $\hat{i} = \arg\max_i \hat{r}_i$ in all remaining rounds.

Theorem 8.2 [Explore-then-commit]. *Let $\iota = \log(A/\delta)$ and $n = c\iota/\epsilon^2$. With probability at least $1-\delta$, $\max_i |\hat{r}_i - r_i| \leq \epsilon/2$, and therefore $r_{\hat{i}} \geq r_{i^\star} - \epsilon$.*

Proof. For a fixed arm, Hoeffding gives $|\hat{r}_i - r_i| \leq c'\sqrt{\iota/n} \leq \epsilon/2$ with probability at least $1 - \delta/A$; take a union bound over the arms. Then $r_{\hat{i}} \geq \hat{r}_{\hat{i}} - \frac{\epsilon}{2} \geq \hat{r}_{i^\star} - \frac{\epsilon}{2} \geq r_{i^\star} - \epsilon$. □

Corollary 8.3. *With probability at least $1-\delta$, explore-then-commit has regret at most $c(A\iota T^2)^{1/3}$.*

Proof. It uses $nA = c' A\iota/\epsilon^2$ samples to find an ϵ-optimal arm; apply Lemma 8.1(2) with $\beta = 2$. □

Algorithm 8.1: ϵ-greedy

maintain the empirical means $\hat{r}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} R_i^{(j)}$
for $t = 1, \dots, T$ **do**
  with probability ϵ, pull a uniformly random arm i_t
  otherwise, pull $i_t = \arg\max_i \hat{r}_i$
  observe R_{i_t} and update $\hat{r}_{i_t}$

With a suitable choice of ϵ, ϵ-greedy also has regret $\widetilde{O}(A^{1/3}T^{2/3})$ with high probability.

8.3 The UCB algorithm

Both strategies above achieve only $T^{2/3}$ regret. The upper confidence bound (UCB) algorithm achieves the optimal $\widetilde{O}(\sqrt{T})$ through optimism in the face of uncertainty. Instead of the unbiased empirical mean, it uses an optimistic estimate that, with high probability, is at least the true mean:

$$\tilde{r}_i^t = \begin{cases} \frac{1}{n_i^t} \sum_{j=1}^{n_i^t} R_i^{(j)} + \mathrm{bonus}(n_i^t), & n_i^t \geq 1, \\ 1, & n_i^t = 0, \end{cases} \qquad \mathrm{bonus}(n) = c\sqrt{\frac{\log(AT/\delta)}{n}},$$

where $n_i^t = \sum_{\tau=1}^{t-1} \mathbf{1}[i_\tau = i]$ is the number of pulls of arm i before round t. A smaller count gives a larger bonus, which encourages exploring rarely pulled arms.

Algorithm 8.2: UCB for multi-armed bandits

for $t = 1, \dots, T$ **do**
  pull $i_t = \arg\max_i \tilde{r}_i^t$
  observe R_{i_t} and update $\tilde{r}_{i_t}^{t+1}$

Theorem 8.4 [Regret of UCB]. *Let $\iota = \log(AT/\delta)$. With probability at least $1 - \delta$, UCB has regret at most $c\sqrt{AT\iota}$.*

Lemma 8.5 [Confidence bounds]. *With probability at least $1 - \delta$, for all $t \in [T]$ and $i \in [A]$, $|\tilde{r}_i^t - r_i| \leq c\sqrt{\iota/\max\{n_i^t, 1\}}$, and in particular $\tilde{r}_i^t \geq r_i$.*

This is Hoeffding's inequality applied to the average of the n_i^t rewards of arm i, together with the bonus. Because the count n_i^t is itself random and depends on past rewards, a rigorous proof uses martingale concentration (Lecture 5) with a union bound over arms and counts. Optimism then follows once the constant in the bonus is large enough.

Proof of Theorem 8.4. On the event of Lemma 8.5,

$$\text{Regret}(T) = \sum_{t=1}^{T} (r_{i^\star} - r_{i_t}) \le \sum_{t=1}^{T} (\tilde{r}^t_{i^\star} - r_{i_t}) \le \sum_{t=1}^{T} (\tilde{r}^t_{i_t} - r_{i_t}) \le \sum_{t=1}^{T} c \sqrt{\frac{\iota}{\max\{n^t_{i_t}, 1\}}} \le c' \sum_{t=1}^{T} \sqrt{\frac{\iota}{n^t_{i_t} + 1}},$$

using optimism, then the greedy choice of i_t with respect to $\tilde{r}^t$, then the confidence bound. Let $N_i = n_i^{T+1}$ be the total number of pulls of arm i. Grouping the rounds by arm and using Cauchy-Schwarz with $\sum_i N_i = T$,

$$\sum_{t=1}^{T} \frac{1}{\sqrt{n^t_{i_t} + 1}} = \sum_{i=1}^{A} \sum_{m=0}^{N_i - 1} \frac{1}{\sqrt{m+1}} \le \sum_{i=1}^{A} 2\sqrt{N_i} \le 2\sqrt{A \sum_i N_i} = 2\sqrt{AT}. \qquad \square$$

Algorithm	Regret
ϵ-first	$\widetilde{O}(A^{1/3}T^{2/3})$
ϵ-greedy	$\widetilde{O}(A^{1/3}T^{2/3})$
UCB	$\widetilde{O}(\sqrt{AT})$
Lower bound (Lecture 10)	$\Omega(\sqrt{AT})$

Since $\text{Regret}(T) \le T$ always, a nontrivial bound must be sublinear in T; we consider $T > A$, since otherwise no algorithm can do better than linear regret. UCB matches the lower bound up to logarithmic factors, so it is minimax optimal.

8.4 Contextual bandits

Algorithm 8.3: Contextual bandit

```
for t = 1, ..., T do
    the environment reveals a context s_t
    the learner picks an arm i_t
    the learner receives a stochastic reward R_{i_t}(s_t)
```

Here $R_i(s) \in [0,1]$ almost surely with $\mathbb{E}R_i(s) = r_i(s)$ for each arm $i \in [A]$ and context $s \in [S]$, and the best policy is $i^\star(s) = \arg\max_i r_i(s)$. The regret decomposes by context:

$$\text{Regret}(T) = \sum_{t=1}^{T} \left[r_{i^\star(s_t)}(s_t) - r_{i_t}(s_t) \right] = \sum_{s=1}^{S} \sum_{t=1}^{T_s} \left[r_{i^\star(s)}(s) - r_{i_t(s)}(s) \right],$$

where T_s is the number of rounds with context s and $i_t(s)$ is the arm pulled the t-th time s appears. So a contextual bandit with S contexts is S separate bandits. Running UCB for each context and

taking a union bound over contexts ($\iota = \log(SAT/\delta)$), with probability at least $1-\delta$,

$$\text{Regret}(T) \le c \sum_{s=1}^{S} \sqrt{AT_s \iota} \le c\sqrt{S \sum_s AT_s \iota} = c\sqrt{SAT\iota},$$

again by Cauchy-Schwarz. Going from bandits to contextual bandits costs a factor of $\sqrt{S}$.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 8 (Spring 2024); written from the Spring 2020 Lecture 7 and Lecture 8 notes.

Lecture 9: Exploration in RL

In bandits, optimism solved exploration: add a bonus to each estimate and act greedily. A contextual bandit is an MDP without transitions. In a general MDP an action also determines which states are reached, and random exploration can need exponentially many episodes (Lecture 7). This unit combines optimism with value iteration, so that bonuses propagate through the Bellman backup, and obtains regret polynomial in S, A, H.

9.1 The UCB-VI algorithm

The agent plays K episodes from a fixed initial state s_1, using policy π_k in episode k, and suffers

$$\mathrm{Regret}(K) = \sum_{k=1}^{K} \left[V_1^\star(s_1) - V_1^{\pi_k}(s_1) \right], \qquad T = KH \text{ samples in total.}$$

Let $N_h(s,a)$ and $N_h(s,a,s')$ count the visits to (s,a) and the transitions $(s,a) \to s'$ at step h in the data collected so far.

Algorithm 9.1: UCB-VI

Input: bonus $b(N) = c\sqrt{H^2\iota/N}$ with $\iota = \log(HSAT/\delta)$

$\mathcal{D} \leftarrow \emptyset$; $Q_h(s,a) \leftarrow H$ for all (s,a,h); $V_{H+1} \leftarrow 0$
for episode $k = 1, \dots, K$ **do**
  for $h = H, \dots, 1$ **do** *// value iteration with bonus*
    for each (s,a) with $N_h(s,a) \geq 1$ **do**
      $\hat{P}_h(s' \mid s,a) \leftarrow N_h(s,a,s')/N_h(s,a)$ for all s'
      $Q_h(s,a) \leftarrow \min\{H,\ r_h(s,a) + (\hat{P}_h V_{h+1})(s,a) + b(N_h(s,a))\}$
    $V_h(s) \leftarrow \max_a Q_h(s,a)$ for all s
  for $h = 1, \dots, H$ **do** *// execute the greedy policy*
    take $a_h = \arg\max_a Q_h(s_h,a)$, observe $r_h(s_h,a_h)$ and s_{h+1}
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(h, s_h, a_h, s_{h+1})\}$

UCB-VI runs value iteration in every episode on the empirical model, adding an exploration bonus based on Hoeffding's inequality; unvisited pairs keep the maximal value H.

Theorem 9.1 [Regret of UCB-VI]. *With probability at least* $1-\delta$,

$$\mathrm{Regret}(K) \leq c\Big(\sqrt{H^3SAT\iota} + \underbrace{H^3S^2A\iota^2}_{\text{lower order}}\Big), \qquad \iota = \log(HSAT/\delta).$$

Write a superscript k for the quantities used in episode k, which are computed from the first $k-1$ episodes: counts N_h^k, the empirical model $\hat{P}_h^k$, and values Q_h^k, V_h^k. The trajectory of episode k is

$(s_h^k, a_h^k)_{h=1}^H$, and $b_h^k = b(N_h^k(s_h^k, a_h^k))$.

Lemma 9.2 [Optimism]. *With probability at least* $1-\delta$, $Q_h^k(s,a) \geq Q_h^\star(s,a)$ *and* $V_h^k(s) \geq V_h^\star(s)$ *for all* (k,h,s,a).

Proof. Fix k and induct backward over h. At $h = H+1$, $V_{H+1}^k = 0 = V_{H+1}^\star$. Suppose $V_{h+1}^k \geq V_{h+1}^\star$. If (s,a) has not been visited, or the minimum in the update is H, then $Q_h^k(s,a) = H \geq Q_h^\star(s,a)$. Otherwise

$$Q_h^k(s,a) - Q_h^\star(s,a) = \big(\hat{P}_h^k(V_{h+1}^k - V_{h+1}^\star)\big)(s,a) + \big((\hat{P}_h^k - P_h)V_{h+1}^\star\big)(s,a) + b(N_h^k(s,a)).$$

The first term is nonnegative by the induction hypothesis, and Hoeffding's inequality (in martingale form, with a union bound over (s,a,h) and counts) gives $|((\hat{P}_h^k - P_h)V_{h+1}^\star)(s,a)| \leq b(N_h^k(s,a))$, because $V_{h+1}^\star$ is a fixed function with values in $[0,H]$. Hence $Q_h^k \geq Q_h^\star$, and $V_h^k = \max_a Q_h^k \geq \max_a Q_h^\star = V_h^\star$. □

Proof of Theorem 9.1. Optimism. By Lemma 9.2, $\mathrm{Regret}(K) \leq \sum_k \big(V_1^k - V_1^{\pi_k}\big)(s_1)$, the gap between the value maintained by the algorithm and the true value of its greedy policy.
One-step recursion. Let $\delta_h^k = (V_h^k - V_h^{\pi_k})(s_h^k)$. Because π_k takes $a_h^k = \arg\max_a Q_h^k(s_h^k, a)$, $\delta_h^k = (Q_h^k - Q_h^{\pi_k})(s_h^k, a_h^k)$, and

$$\begin{aligned}(Q_h^k - Q_h^{\pi_k})(s_h^k, a_h^k) &\leq \big(\hat{P}_h^k V_{h+1}^k - P_h V_{h+1}^{\pi_k}\big)(s_h^k, a_h^k) + b_h^k \\ &= \big((\hat{P}_h^k - P_h)V_{h+1}^\star\big) + \big((\hat{P}_h^k - P_h)(V_{h+1}^k - V_{h+1}^\star)\big) + \big(P_h(V_{h+1}^k - V_{h+1}^{\pi_k})\big) + b_h^k,\end{aligned}$$

all evaluated at (s_h^k, a_h^k). The first term is at most b_h^k as in the optimism lemma. For the second term, call it $\mathcal{A}$, let $G = V_{h+1}^k - V_{h+1}^\star \in [0,H]^S$, which is not a fixed function, so we bound each entry of $\hat{P}_h^k - P_h$ separately. Writing $P(s') = P_h(s' \mid s_h^k, a_h^k)$ and $N = N_h^k(s_h^k, a_h^k)$, Bernstein's inequality (with a union bound over s') and $c\sqrt{P(s')\iota/N} \leq P(s')/H + c^2 H\iota/(4N)$ give

$$\mathcal{A} \leq \sum_{s'} c\Big(\sqrt{\frac{P(s')\iota}{N}} + \frac{\iota}{N}\Big)G(s') \leq \sum_{s'}\Big(\frac{P(s')}{H} + \frac{c'H\iota}{N}\Big)G(s') \leq \frac{1}{H}\big(P_h(V_{h+1}^k - V_{h+1}^\star)\big)(s_h^k, a_h^k) + \zeta_h^k,$$

where $\zeta_h^k = c''H^2S\iota/N$. Since $V_{h+1}^{\pi_k} \leq V_{h+1}^\star$, the term $P_h(V_{h+1}^k - V_{h+1}^\star)$ is at most $P_h(V_{h+1}^k - V_{h+1}^{\pi_k})$. Define the martingale difference $\xi_h^k = \big(P_h(V_{h+1}^k - V_{h+1}^{\pi_k})\big)(s_h^k, a_h^k) - \delta_{h+1}^k$. Then

$$\delta_h^k \leq \Big(1 + \frac{1}{H}\Big)\big(\delta_{h+1}^k + \xi_h^k\big) + \zeta_h^k + 2b_h^k.$$

With the conventions $b(0) = H$ and $\zeta = 0$ for unvisited pairs, this also holds when $N = 0$, since then $\delta_h^k \leq H$ and $\delta_{h+1}^k + \xi_h^k \geq 0$.

Unrolling. Since $\delta_{H+1}^k = 0$ and $(1+1/H)^h \le e$,

$$\delta_1^k \le \sum_{h=1}^{H} \Big(1+\frac{1}{H}\Big)^h \xi_h^k + e\sum_{h=1}^{H}\big(\zeta_h^k + 2b_h^k\big), \qquad \mathrm{Regret}(K) \le \sum_{k=1}^{K}\sum_{h=1}^{H}\Big(1+\frac{1}{H}\Big)^h \xi_h^k + e\sum_{k=1}^{K}\sum_{h=1}^{H}\big(\zeta_h^k + 2b_h^k\big).$$

Bonus terms. Grouping visits by (h,s,a), with $N_h(s,a) = N_h^{K+1}(s,a)$ and $\sum_{h,s,a} N_h(s,a) = T$, Cauchy-Schwarz gives

$$\begin{aligned}\sum_{k,h} b_h^k &\le H^2SA + c\sqrt{H^2\iota}\sum_{h,s,a}\sum_{i=1}^{N_h(s,a)}\frac{1}{\sqrt{i}} \le H^2SA + 2c\sqrt{H^2\iota}\sum_{h,s,a}\sqrt{N_h(s,a)} \\ &\le H^2SA + 2c\sqrt{H^2\iota}\sqrt{HSA\textstyle\sum_{h,s,a} N_h(s,a)} = H^2SA + 2c\sqrt{H^3SAT\iota},\end{aligned}$$

where H^2SA accounts for the first visits ($b(0) = H$).
Second-order terms. Similarly $\sum_{k,h}\zeta_h^k \le c''H^2S\iota\sum_{h,s,a}\sum_{i=1}^{N_h(s,a)}\frac{1}{i} \le c''H^2S\iota\cdot HSA(1+\log T) \le c'''H^3S^2A\iota^2$.
Martingale terms. Let $\mathcal{F}_{k,h}$ contain all randomness up to step h of episode k, ordered $\mathcal{F}_{1,1} \subseteq \cdots \subseteq \mathcal{F}_{1,H} \subseteq \mathcal{F}_{2,1} \subseteq \cdots$. The function $V_{h+1}^k - V_{h+1}^{\pi_k}$ is determined by the first $k-1$ episodes, and only s_{h+1}^k is random given $\mathcal{F}_{k,h}$, so $\mathbb{E}[\xi_h^k \mid \mathcal{F}_{k,h}] = 0$; also $|\xi_h^k| \le 2H$. By Azuma-Hoeffding, with probability at least $1-\delta$, $\sum_{k,h}(1+1/H)^h\xi_h^k \le cH\sqrt{T\iota}$, a lower-order term.
Adding the three bounds proves the theorem. □

From regret to sample complexity. As in the conversion lemma for bandits (Lemma 8.1), dividing by $K = T/H$,

$$V_1^\star(s_1) - \frac{1}{K}\sum_{k=1}^{K} V_1^{\pi_k}(s_1) \le c\Big(\sqrt{\frac{H^5SA\iota}{T}} + \frac{H^4S^2A\iota^2}{T}\Big),$$

so playing a uniformly random one of $\pi_1,\dots,\pi_K$ is ϵ-optimal in expectation once

$$T \ge c\Big(\underbrace{\frac{H^5SA\iota}{\epsilon^2}}_{\text{leading term}} + \underbrace{\frac{H^4S^2A\iota^2}{\epsilon}}_{\text{burn-in term}}\Big).$$

The leading term $\widetilde{O}(H^5SA/\epsilon^2)$ is the same as with a simulator. The burn-in term means that UCB-VI pays about S^2A samples at the start; whether this is necessary is open.

The combination lock revisited. In the combination lock (Figure 3) every untried action starts with the optimistic value H, so every untried path looks as good as possible. Once a path is tried and returns no reward, its estimated values drop quickly, and the greedy policy is pushed toward untried actions. Optimism therefore finds the rewarding path in a polynomial number of episodes, where random exploration needed $\Omega(2^H)$.

9.2 Q-learning with UCB

Q-learning can also be made optimistic.

Algorithm 9.2: Q-learning with UCB

$Q_h(s,a) \leftarrow H$ and $N_h(s,a) \leftarrow 0$ for all (s,a,h)
for episode $k = 1, \dots, K$ **do**
  receive s_1
  for $h = 1, \dots, H$ **do**
    take $a_h = \arg\max_a Q_h(s_h, a)$ and observe s_{h+1}
    $t = N_h(s_h, a_h) \leftarrow N_h(s_h, a_h) + 1$; $\quad b_t \leftarrow c\sqrt{H^3\iota/t}$
    $Q_h(s_h, a_h) \leftarrow (1-\alpha_t)Q_h(s_h, a_h) + \alpha_t\left[r_h(s_h, a_h) + V_{h+1}(s_{h+1}) + b_t\right]$
    $V_h(s_h) \leftarrow \min\{H, \max_a Q_h(s_h, a)\}$

The learning rate depends on t, the number of visits to (s_h, a_h) so far. The algorithm never forms an empirical model $\hat{P}$ and stores no data: it is model-free. Choosing $\alpha_t = \frac{H+1}{H+t} = O(H/t)$, which puts more weight on later updates (Lemma 7.1), is crucial: it keeps the regret from depending exponentially on H.

Theorem 9.3 [Q-learning with UCB]. *With $\alpha_t = \frac{H+1}{H+t}$ and $\iota = \log(HSAT/\delta)$, with probability at least $1-\delta$, $\mathrm{Regret}(K) \le c\sqrt{H^4SAT\iota}$, with no lower-order term.*

The proof has two parts: first, thanks to the bonus, $Q_h^k \ge Q_h^\star$ with high probability, together with an upper bound on $Q_h^k - Q_h^\star$ in terms of the weighted errors at the next step (as in the analysis of Q-learning with a simulator); second, a regret analysis that unrolls these bounds over steps using the properties of the weights in Lemma 7.1.

9.3 Summary

Algorithm	Regret	Space
E^3, R-max	$\mathrm{poly}(H,S,A)\,T^{1-\alpha}$	
UCRL (confidence sets on transitions)	at least $\sqrt{S^2AT}$	
UCB-VI	$\widetilde{O}(\sqrt{H^3SAT}) + S^2A$ terms	$\min\{S^2AH, T\}$
UCB-VI (Bernstein)	$\widetilde{O}(\sqrt{H^2SAT}) + S^2A$ terms	
Q-learning with UCB	$\widetilde{O}(\sqrt{H^4SAT})$	$\min\{SAH, T\}$
Q-learning (Bernstein)	$\widetilde{O}(\sqrt{H^3SAT}) + S^2A$ terms	
Lower bound	$\Omega(\sqrt{H^2SAT})$	

- Running value iteration in every episode makes UCB-VI cost $\widetilde{O}(TS^2A)$ time; redoing it only in $\log T$ of the episodes reduces the time to $T + HS^2A\log T$. Q-learning with UCB takes $\widetilde{O}(T)$ time.
- The Bernstein versions of UCB-VI and Q-learning estimate the empirical variance to build their

upper bounds, which is more involved.

- UCB-VI with Bernstein bonuses matches the lower bound in the leading term. Q-learning with UCB is the only bound so far with no S^2A lower-order term. Achieving the minimax-optimal regret without paying an extra S^2A in a burn-in phase is open.

Simulator versus exploration. With a simulator the agent can query any (s, a); the minimax sample complexity is H^4SA/ϵ^2, and the guarantee is uniform, $\|V^{\hat{\pi}} - V^\star\|_\infty \le \epsilon$, obtained by learning a model $\hat{P}$ close to P at every (s, a) and planning in it. In the exploration setting the agent must start from the initial state and can only choose actions, yet the minimax sample complexity is essentially the same, H^4SA/ϵ^2 up to a lower-order term. The guarantee, however, is only at the initial state, $V_1^\star(s_1) - V_1^{\hat{\pi}}(s_1) \le \epsilon$, and the mechanism differs: the agent cannot learn P at every (s, a), since some states are reached with tiny probability (say 10^{-100}), but such states barely affect $V_1^\pi(s_1)$, so not learning them does not prevent finding a near-optimal policy.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 9 (Spring 2024); written from the Spring 2020 Lecture 9 and Lecture 10 notes.

Lecture 10: Lower Bounds for Multi-Arm Bandits

UCB guarantees, with probability at least $1-\delta$, $\text{Regret}(T) \le c\sqrt{AT\log(AT/\delta)}$ (Lecture 8), better than ϵ-greedy and explore-then-commit. Can any algorithm do better? This unit proves that it cannot, beyond logarithmic factors.

Theorem 10.1 [Regret lower bound]. *For any algorithm $\mathcal{A}$, randomized or deterministic, and any $T \ge A \ge 2$, there is a multi-armed bandit instance with*

$$\mathbb{E}_{\mathcal{A}}[\text{Regret}(T)] \ge c\sqrt{AT},$$

where the expectation is over the randomness of the rewards and of the algorithm.

An upper bound says that *some* algorithm does well on *every* instance; it captures the power of an algorithm. A lower bound says that *every* algorithm does poorly on *some* instance; it is independent of the algorithm and captures the information-theoretic limit of the problem. In a bandit, the limiting factor is the noise: a pull reveals a random reward, not its mean.

10.1 Intuition with two arms

Let arm 1 give reward 1 with probability $\frac{1}{2}+\epsilon$ and arm 2 with probability $\frac{1}{2}$, for $\epsilon \ll 1$. Arm 1 is better, but with small ϵ many samples are needed to tell. After n pulls of each arm, the central limit theorem gives $\hat{r}_1 \approx \frac{1}{2}+\epsilon+\mathcal{N}(0,\frac{1}{4n})$ and $\hat{r}_2 \approx \frac{1}{2}+\mathcal{N}(0,\frac{1}{4n})$, with noise of standard deviation $1/(2\sqrt{n})$. If $1/(2\sqrt{n}) \approx \epsilon$, then with constant probability $\hat{r}_1 < \hat{r}_2$. Conversely, Hoeffding gives $|\hat{r}_j - r_j| \le c\sqrt{\iota/n}$ with $\iota = \log(A/\delta)$, and to rank the arms correctly we need $c\sqrt{\iota/n} \le \epsilon/2$, that is $n \ge c'\iota/\epsilon^2$. Taking $\epsilon = 1/\sqrt{T}$, an algorithm cannot tell the arms apart within T rounds; guessing, it pays ϵ per round on the wrong arm, so $\text{Regret}(T) \approx T\epsilon = \sqrt{T}$. This argument concerns one particular algorithm and instance; the formal proof must handle every algorithm.

10.2 Reduction to deterministic algorithms

No single deterministic instance is hard: if one arm is always optimal, the algorithm that always pulls it has zero regret, and among the A such algorithms one is right. Instead, we construct a *distribution $\mathcal{D}$* over hard instances, so that an algorithm that does well on one instance does poorly on

others (Yao's principle). Writing $\mathrm{Alg}_{\mathrm{rand}}$ and $\mathrm{Alg}_{\mathrm{det}}$ for randomized and deterministic algorithms,

$$\begin{aligned}\min_{\mathcal{A}\in\mathrm{Alg}_{\mathrm{rand}}} \max_{\mathcal{P}} \mathbb{E}_{\mathcal{A}}[\mathrm{Regret}_{\mathcal{P}}(\mathcal{A},T)] &\geq \min_{\mathcal{A}\in\mathrm{Alg}_{\mathrm{rand}}} \mathbb{E}_{\mathcal{P}\sim\mathcal{D}}\mathbb{E}_{\mathcal{A}}[\mathrm{Regret}_{\mathcal{P}}(\mathcal{A},T)] \\ &= \min_{\mathcal{A}\in\mathrm{Alg}_{\mathrm{rand}}} \mathbb{E}_{\mathcal{A}}\mathbb{E}_{\mathcal{P}\sim\mathcal{D}}[\mathrm{Regret}_{\mathcal{P}}(\mathcal{A},T)] \\ &\geq \min_{\mathcal{A}\in\mathrm{Alg}_{\mathrm{det}}} \mathbb{E}_{\mathcal{P}\sim\mathcal{D}}[\mathrm{Regret}_{\mathcal{P}}(\mathcal{A},T)],\end{aligned}$$

since a randomized algorithm is a distribution over deterministic ones, and the best expected regret against a fixed distribution of instances is attained by a deterministic algorithm. It therefore suffices to lower bound the expected regret of every deterministic algorithm under $\mathcal{D}$.

10.3 Proof of the lower bound

Definition: *The hard instances.* Choose a special arm a uniformly at random from $[A]$. The special arm gives reward 1 with probability $\frac{1}{2}+\epsilon$ and 0 otherwise; every other arm gives reward 1 with probability $\frac{1}{2}$.

Fix a deterministic algorithm. Write $\mathbb{E}_a$ for expectations over the rewards when a is the special arm, $\mathbb{E}_0$ when there is no special arm (all arms are fair coins), and $\mathbb{E}_{\mathcal{D}} = \frac{1}{A}\sum_{a=1}^{A}\mathbb{E}_a$. Let $R_{1:T} = (R_1,\dots,R_T)$ be the observed rewards. Since the algorithm is deterministic, a_t is a function of $R_{1:t-1}$, and the number of pulls n_a of arm a is a function of $R_{1:T}$. Each pull of a non-special arm loses ϵ, so

$$\mathbb{E}_{\mathcal{D}}[\mathrm{Regret}(T)] = \frac{1}{A}\sum_{a=1}^{A}\mathbb{E}_a\big[\epsilon(T-n_a)\big] = \epsilon\Big(T - \frac{1}{A}\sum_{a=1}^{A}\mathbb{E}_a[n_a]\Big).$$

A good algorithm pulls the special arm often; we show it cannot, by comparing with the instance without a special arm. Since $0 \leq n_a \leq T$,

$$\begin{aligned}\mathbb{E}_a[n_a] - \mathbb{E}_0[n_a] &= \sum_{R_{1:T}} n_a(R_{1:T})\big[P_a(R_{1:T}) - P_0(R_{1:T})\big] \\ &\leq T\sum_{R_{1:T}} \max\{P_a(R_{1:T}) - P_0(R_{1:T}), 0\} = T\,\|P_a - P_0\|_{TV},\end{aligned}$$

and by Pinsker's inequality $\|P_a - P_0\|_{TV} \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(P_0\|P_a)}$. The KL divergence is easier to handle because of the chain rule:

$$\begin{aligned}\mathrm{KL}(P_0\|P_a) &= \sum_{t=1}^{T}\sum_{R_{1:t}} P_0(R_{1:t})\log\frac{P_0(R_t \mid R_{1:t-1})}{P_a(R_t \mid R_{1:t-1})} = \sum_{t=1}^{T}\sum_{R_{1:t-1}:\,a_t=a} P_0(R_{1:t-1})\;\mathrm{KL}\big(\mathrm{Bern}(\tfrac{1}{2})\big\|\mathrm{Bern}(\tfrac{1}{2}+\epsilon)\big) \\ &= \sum_{t=1}^{T} P_0(a_t = a)\cdot\frac{1}{2}\log\frac{1}{1-4\epsilon^2} = \frac{\mathbb{E}_0[n_a]}{2}\log\frac{1}{1-4\epsilon^2}.\end{aligned}$$

Only rounds that pull arm a contribute, because the two instances differ only on arm a, and $\frac{1}{2}\log\frac{1/2}{1/2-\epsilon} + \frac{1}{2}\log\frac{1/2}{1/2+\epsilon} = \frac{1}{2}\log\frac{1}{1-4\epsilon^2}$. Every algorithm satisfies $\sum_a \mathbb{E}_0[n_a] = T$, so by Cauchy-Schwarz,

$$\sum_{a=1}^{A} \mathbb{E}_a[n_a] \le T + T\sum_{a=1}^{A}\sqrt{\frac{\mathbb{E}_0[n_a]}{4}\log\frac{1}{1-4\epsilon^2}} \le T + T\sqrt{\frac{AT}{4}\log\frac{1}{1-4\epsilon^2}}.$$

Using $\log x \le x - 1$, that is $\log\frac{1}{1-4\epsilon^2} \le \frac{4\epsilon^2}{1-4\epsilon^2}$,

$$\mathbb{E}_{\mathcal{D}}[\mathrm{Regret}(T)] \ge \epsilon T\Big(1 - \frac{1}{A} - \sqrt{\frac{T}{A}\cdot\frac{\epsilon^2}{1-4\epsilon^2}}\Big).$$

Choose $\epsilon = \frac{1}{4}\min\{\sqrt{A/T}, 1\}$. For $A \ge 2$, $1 - \frac{1}{A} \ge \frac{1}{2}$, and $\frac{T}{A}\cdot\frac{\epsilon^2}{1-4\epsilon^2} \le \frac{1}{16}\cdot\frac{4}{3} = \frac{1}{12}$, so

$$\mathbb{E}_{\mathcal{D}}[\mathrm{Regret}(T)] \ge \epsilon T\Big(\frac{1}{2} - \frac{1}{\sqrt{12}}\Big) \ge \frac{\min\{\sqrt{AT}, T\}}{24}.$$

With $T \ge A$ this is $\sqrt{AT}/24$, proving Theorem 10.1. The derivation did not use $T \ge A$: when $A \ge T$ the bound is $\Omega(T)$, which is tight, since regret is trivially at most T.

10.4 Lower bound for best-arm identification

The same technique bounds the sample complexity of finding a good arm.

Theorem 10.2 [Sample complexity lower bound]. *For any algorithm that pulls arms adaptively for a total of T times and then outputs a (possibly random) arm ψ, there is a bandit instance with*

$$r_{i^\star} - \mathbb{E}_{\mathcal{A}} r_\psi \ge c\sqrt{\frac{A}{T}}.$$

Hence finding an ϵ-optimal arm requires $T \ge c'A/\epsilon^2$ samples.

A low-regret algorithm can be turned into a best-arm identification algorithm by outputting one of the T arms it pulled, chosen uniformly at random (Lemma 8.1). So Theorem 10.2 immediately implies Theorem 10.1.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 10 (Spring 2024); written from the Spring 2020 Lecture 11 and Lecture 12 notes.

Lecture 11: Lower Bounds for MDPs

This unit extends the bandit lower bound of Lecture 10 to MDPs. Recall that UCB-VI (Lecture 9) satisfies, with probability at least $1-\delta$,

$$\mathrm{Regret}(K) = \sum_{k=1}^{K} \left[V_1^\star(s_1^k) - V_1^{\pi_k}(s_1^k) \right] \le c\sqrt{H^3 SAT\,\iota} + \text{lower-order terms},$$

where $T = KH$ and $\iota = \log(HSAT/\delta)$. The proof was given for a fixed initial state, and it goes through unchanged when the initial states $\{s_1^k\}_{k=1}^K$ are chosen adversarially. Replacing Hoeffding by Bernstein bonuses improves the leading term:

	Non-stationary ($P_1, \dots, P_H$ may differ)	Stationary ($P_h = P$)
UCB-VI (Hoeffding)	$\widetilde{O}(\sqrt{H^3SAT})$	
UCB-VI (Bernstein)	$\widetilde{O}(\sqrt{H^2SAT})$	$\widetilde{O}(\sqrt{HSAT})$ (modified)
Lower bound	$\Omega(\sqrt{H^2SAT})$	$\Omega(\sqrt{HSAT})$

Proving the non-stationary lower bound is a bit more involved. This unit proves the stationary one, which a modified version of UCB-VI with Bernstein bonuses matches in the leading term, so it is sharp.

Theorem 11.1 [Regret lower bound for MDPs]. *For any algorithm $\mathcal{A}$, possibly randomized, there is a stationary MDP with initial states $\{s_1^k\}_{k=1}^K$ such that*

$$\mathbb{E}_{\mathcal{A}}[\mathrm{Regret}(K)] = \mathbb{E}_{\mathcal{A}} \sum_{k=1}^{K} \left[V_1^\star(s_1^k) - V_1^{\pi_k}(s_1^k) \right] \ge c\sqrt{HSAT}.$$

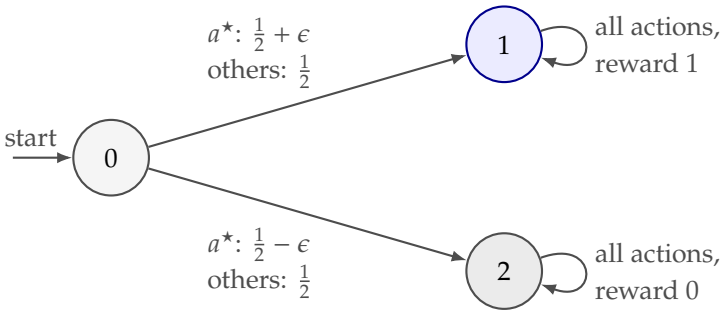


Figure 4: The three-state hard instance. Only the first action matters; it is a bandit whose rewards are scaled by $H-1$.

Proof. Three states. First take $S = 3$ and show $\mathbb{E}[\text{Regret}(K)] \geq c\sqrt{HAT}$. Consider the distribution over hard instances in Figure 4, with a special action $a^\star$ drawn uniformly from the A actions. Every episode starts in state 0. From state 0, the special action moves to state 1 with probability $\frac{1}{2} + \epsilon$ and to state 2 otherwise; every other action moves to states 1 and 2 with probability $\frac{1}{2}$ each. States 1 and 2 are absorbing under every action, with reward 1 per step in state 1 and 0 in state 2. The transitions do not depend on the step, so the MDP is stationary.

The choice of action in states 1 and 2 does not matter, so only the first action matters, and all information about $a^\star$ is revealed at the first step; nothing more is learned later in the episode. Landing in state 1 yields cumulative reward $H - 1$ and landing in state 2 yields 0. So each episode is one round of the bandit of Lecture 10, with a Bernoulli$(\frac{1}{2} + \epsilon)$ special arm and Bernoulli$(\frac{1}{2})$ other arms, and the only difference is that the regret is scaled by $H - 1$. Theorem 10.1 gives, for $H \geq 2$,

$$\mathbb{E}_{\mathcal{A}}[\text{Regret}(K)] \geq (H-1)\, c\sqrt{AK} \geq c'\sqrt{HAT},$$

since $H\sqrt{AK} = \sqrt{HAT}$.

General S. Make $S/3$ copies of this instance, each with its own independently drawn special action, and start precisely $3K/S$ of the episodes at state 0 of each copy. The copies share no information, so each contributes the three-state bound with $3K/S$ episodes:

$$\mathbb{E}_{\mathcal{A}}[\text{Regret}(K)] \geq \frac{S}{3} \cdot c(H-1)\sqrt{A \cdot \frac{3K}{S}} \geq c''\sqrt{HSAT}. \qquad \square$$

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 11 (Spring 2024); written from the Spring 2020 Lecture 12 notes.

Lecture 12: Offline RL

In offline RL the learner cannot interact with the environment at all. It receives a dataset $\mathcal{D} = \{(s_h^\tau, a_h^\tau, r_h^\tau)\}_{\tau\in[K],h\in[H]}$ of K trajectories collected beforehand by an experimenter in an episodic MDP, and must output a policy $\hat{\pi}$ with small suboptimality

$$\mathrm{SubOpt}(\hat{\pi}; s) = V_1^\star(s) - V_1^{\hat{\pi}}(s).$$

Optimism, the principle behind online exploration, is the wrong idea here: the learner cannot collect more data to correct an optimistic mistake. The right principle is its opposite, *pessimism*.

12.1 Setting

Write $(\mathbb{B}_h V)(s,a) = r_h(s,a) + (P_h V)(s,a)$ for the Bellman operator applied to a function $V : \mathcal{S} \to \mathbb{R}$. At step h of trajectory τ the experimenter takes a_h^τ in s_h^τ, receives $r_h^\tau = r_h(s_h^\tau, a_h^\tau)$, and observes $s_{h+1}^\tau \sim P_h(\cdot \mid s_h^\tau, a_h^\tau)$; the actions a_h^τ may be chosen arbitrarily.

Definition: The dataset is compliant with the MDP if, for every h and τ, conditionally on (s_h^τ, a_h^τ) the pair (r_h^τ, s_{h+1}^τ) is generated by the MDP's reward and transition, independently of the data $\{(s_h^j, a_h^j, r_h^j, s_{h+1}^j)\}_{j<\tau}$ of the earlier trajectories. We assume that the data are compliant.

Compliance holds when the experimenter follows a fixed behavior policy, but also when the actions are chosen adaptively or even adversarially, for example by an online algorithm that improves its policy between trajectories, so the trajectories may be dependent. Crucially, nothing requires the data to explore the state and action spaces well.

12.2 What causes suboptimality

Consider any algorithm that builds estimates $\hat{Q}_h$ and $\hat{V}_h$ from $\mathcal{D}$, with $\hat{V}_h(s) = \langle \hat{Q}_h(s,\cdot), \hat{\pi}_h(\cdot \mid s)\rangle$ for its policy $\hat{\pi}$, and define the model evaluation error

$$e_h(s,a) = (\mathbb{B}_h \hat{V}_{h+1})(s,a) - \hat{Q}_h(s,a),$$

the error in estimating the Bellman backup from the data.

Lemma 12.1 [Decomposition of suboptimality]. *For any such $\hat{\pi}$ and any s,*

$$\mathrm{SubOpt}(\hat{\pi}; s) = \underbrace{-\sum_{h=1}^{H} \mathbb{E}_{\hat{\pi}}\left[e_h(s_h, a_h) \mid s_1 = s\right]}_{\text{(i) spurious correlation}} + \underbrace{\sum_{h=1}^{H} \mathbb{E}_{\pi^\star}\left[e_h(s_h, a_h) \mid s_1 = s\right]}_{\text{(ii) intrinsic uncertainty}}$$
$$+ \underbrace{\sum_{h=1}^{H} \mathbb{E}_{\pi^\star}\left[\langle \hat{Q}_h(s_h, \cdot), \pi^\star_h(\cdot \mid s_h) - \hat{\pi}_h(\cdot \mid s_h)\rangle \,\middle|\, s_1 = s\right]}_{\text{(iii) optimization error}},$$

where $\mathbb{E}_{\hat{\pi}}$ and $\mathbb{E}_{\pi^\star}$ are over trajectories of $\hat{\pi}$ and $\pi^\star$ in the true MDP, with e_h and $\hat{Q}_h$ held fixed.

Proof. For any policies π, π' and any $\hat{Q}$ with $\hat{V}_h(s) = \langle \hat{Q}_h(s, \cdot), \pi_h(\cdot \mid s)\rangle$,

$$\hat{V}_h(s) - V_h^{\pi'}(s) = \langle \hat{Q}_h(s, \cdot), \pi_h(\cdot \mid s) - \pi'_h(\cdot \mid s)\rangle + \langle \hat{Q}_h(s, \cdot) - Q_h^{\pi'}(s, \cdot), \pi'_h(\cdot \mid s)\rangle,$$

and $\hat{Q}_h - Q_h^{\pi'} = (\hat{Q}_h - \mathbb{B}_h \hat{V}_{h+1}) + P_h(\hat{V}_{h+1} - V_{h+1}^{\pi'})$. Recursing over h gives $\hat{V}_1(s) - V_1^{\pi'}(s) = \sum_h \mathbb{E}_{\pi'}[\langle \hat{Q}_h(s_h, \cdot), \pi_h - \pi'_h\rangle] - \sum_h \mathbb{E}_{\pi'}[e_h(s_h, a_h)]$. Apply this with $\pi = \hat{\pi}$ and $\pi' = \pi^\star$, then with $\pi = \pi' = \hat{\pi}$, and add the two results in $\mathrm{SubOpt}(\hat{\pi}; s) = \big(V_1^{\pi^\star}(s) - \hat{V}_1(s)\big) + \big(\hat{V}_1(s) - V_1^{\hat{\pi}}(s)\big)$. □

Term (iii) is nonpositive whenever $\hat{\pi}$ is greedy with respect to $\hat{Q}$. Term (ii) involves only $\pi^\star$, which is intrinsic to the MDP and does not depend on the data. Term (i) is the dangerous one: $\hat{\pi}$ and e_h both depend on $\mathcal{D}$, and this spurious correlation can be large even when every e_h has mean zero.

Example: *Spurious correlation in a bandit.* Take one state, $H = 1$, and rewards $r(a) = \mu(a) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$. With $N(a)$ samples of action a, the sample average satisfies $\hat{\mu}(a) \sim \mathcal{N}(\mu(a), 1/N(a))$, and the greedy policy $\hat{\pi}$ picks $\arg\max_a \hat{\mu}(a)$. Here $e(a) = \mu(a) - \hat{\mu}(a)$ has mean zero, so term (ii) has mean zero, but term (i) equals $\hat{\mu}(a^\natural) - \mu(a^\natural)$ for the chosen action $a^\natural$. If all $\mu(a) = 0$, it is the maximum of independent $\mathcal{N}(0, 1/N(a))$ variables, which is large when some $N(a)$ is small (say $N(a) = 1$). The greedy policy may choose an action only because it was sampled rarely and its estimate happens to be inflated. Datasets rarely cover every action uniformly, and with $H > 1$ uniform coverage of all states and actions is even harder to ensure; earlier analyses assumed it through concentrability coefficients, which often fail in practice.

12.3 Pessimistic value iteration

The fix is to subtract an uncertainty penalty, the online exploration bonus with its sign flipped. Suppose the algorithm builds an estimate $\hat{\mathbb{B}}_h \hat{V}_{h+1}$ of $\mathbb{B}_h \hat{V}_{h+1}$ from the data.

Definition 12.2 [Uncertainty quantifier]. Functions $\{\Gamma_h\}_{h=1}^H$ form a δ-uncertainty quantifier if

the event

$$\mathcal{E} = \left\{ \left| (\hat{\mathbb{B}}_h \hat{V}_{h+1})(s,a) - (\mathbb{B}_h \hat{V}_{h+1})(s,a) \right| \le \Gamma_h(s,a) \text{ for all } (s,a,h) \right\}$$

has probability at least $1-\delta$ over the data.

Algorithm 12.1: Pessimistic value iteration (PEVI)

Input: dataset $\mathcal{D} = \{(s_h^\tau, a_h^\tau, r_h^\tau)\}_{\tau,h}$

$\hat{V}_{H+1} \leftarrow 0$

for $h = H, H-1, \dots, 1$ **do**

  construct $(\hat{\mathbb{B}}_h \hat{V}_{h+1})(\cdot,\cdot)$ and $\Gamma_h(\cdot,\cdot)$ from $\mathcal{D}$ *// estimation and uncertainty*

  $Q_h(\cdot,\cdot) \leftarrow (\hat{\mathbb{B}}_h \hat{V}_{h+1})(\cdot,\cdot) - \Gamma_h(\cdot,\cdot)$ *// pessimism*

  $\hat{Q}_h(\cdot,\cdot) \leftarrow \min\{Q_h(\cdot,\cdot), H-h+1\}^+$ *// truncation*

  $\hat{\pi}_h(\cdot \mid \cdot) \leftarrow \arg\max_{\pi_h} \langle \hat{Q}_h(\cdot,\cdot), \pi_h(\cdot \mid \cdot) \rangle$

  $\hat{V}_h(\cdot) \leftarrow \langle \hat{Q}_h(\cdot,\cdot), \hat{\pi}_h(\cdot \mid \cdot) \rangle$

return $\hat{\pi} = \{\hat{\pi}_h\}_{h=1}^H$

Theorem 12.3 [Suboptimality of PEVI]. *If $\{\Gamma_h\}$ is a δ-uncertainty quantifier, then on $\mathcal{E}$, which has probability at least $1-\delta$, for every s,*

$$\mathrm{SubOpt}(\hat{\pi}; s) \le 2 \sum_{h=1}^{H} \mathbb{E}_{\pi^\star} \left[\Gamma_h(s_h, a_h) \,\middle|\, s_1 = s \right].$$

Lemma 12.4 [Pessimism]. *On $\mathcal{E}$, $0 \le e_h(s,a) \le 2\Gamma_h(s,a)$ for all (s,a,h).*

Proof. If $Q_h(s,a) < 0$, then $\hat{Q}_h(s,a) = 0$ and $e_h(s,a) = (\mathbb{B}_h \hat{V}_{h+1})(s,a) \ge 0$, since rewards and $\hat{V}_{h+1}$ are nonnegative; moreover $(\mathbb{B}_h \hat{V}_{h+1})(s,a) \le (\hat{\mathbb{B}}_h \hat{V}_{h+1})(s,a) + \Gamma_h(s,a) < 2\Gamma_h(s,a)$. If $Q_h(s,a) \ge 0$, then $\hat{Q}_h(s,a) \le Q_h(s,a)$, so $e_h(s,a) \ge (\mathbb{B}_h \hat{V}_{h+1})(s,a) - (\hat{\mathbb{B}}_h \hat{V}_{h+1})(s,a) + \Gamma_h(s,a) \ge 0$ on $\mathcal{E}$. For the upper bound, either the truncation at $H-h+1$ is active, and then $e_h \le 0$ because $\mathbb{B}_h \hat{V}_{h+1} \le 1 + (H-h)$, or $\hat{Q}_h = Q_h$, and then $e_h = (\mathbb{B}_h \hat{V}_{h+1}) - (\hat{\mathbb{B}}_h \hat{V}_{h+1}) + \Gamma_h \le 2\Gamma_h$ on $\mathcal{E}$. □

The nonnegativity of e_h makes $\hat{Q}$ pessimistic: by induction on h, $\hat{Q}_h \le Q_h^\star$ everywhere.

Proof of Theorem 12.3. In Lemma 12.1, term (iii) is nonpositive because $\hat{\pi}$ is greedy with respect to $\hat{Q}$, term (i) is nonpositive because $e_h \ge 0$, and term (ii) is at most $2\sum_h \mathbb{E}_{\pi^\star}[\Gamma_h(s_h,a_h)]$ because $e_h \le 2\Gamma_h$ (Lemma 12.4). □

Pessimism thus removes the spurious correlation entirely; what remains is the intrinsic uncertainty, measured only along the trajectory of $\pi^\star$.

12.4 Linear and tabular MDPs

In a LINEAR MDP (developed in Lecture 13) there is a known feature map $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$ with $P_h(s' \mid s, a) = \langle \phi(s,a), \mu_h(s') \rangle$ and $r_h(s,a) = \langle \phi(s,a), \theta_h \rangle$, normalized so that $\|\phi(s,a)\| \le 1$ and $\max\{\|\mu_h(\mathcal{S})\|, \|\theta_h\|\} \le \sqrt{d}$. PEVI estimates the backup by ridge regression and penalizes by the elliptical uncertainty:

$$\Lambda_h = \sum_{\tau=1}^{K} \phi(s_h^\tau, a_h^\tau)\phi(s_h^\tau, a_h^\tau)^\top + \lambda I, \qquad \hat{w}_h = \Lambda_h^{-1} \sum_{\tau=1}^{K} \phi(s_h^\tau, a_h^\tau)\big(r_h^\tau + \hat{V}_{h+1}(s_{h+1}^\tau)\big),$$

$$(\hat{\mathbb{B}}_h \hat{V}_{h+1})(s,a) = \phi(s,a)^\top \hat{w}_h, \qquad \Gamma_h(s,a) = \beta\big(\phi(s,a)^\top \Lambda_h^{-1} \phi(s,a)\big)^{1/2}.$$

Theorem 12.5 [PEVI FOR LINEAR MDPs]. *For compliant data from a linear MDP, with $\lambda = 1$ and $\beta = c\, dH\sqrt{\zeta}$, $\zeta = \log(2dHK/\delta)$, the penalties Γ_h form a δ-uncertainty quantifier. Hence with probability at least $1 - \delta$, for every s,*

$$\mathrm{SubOpt}(\hat{\pi}; s) \le 2\beta \sum_{h=1}^{H} \mathbb{E}_{\pi^\star}\left[\big(\phi(s_h, a_h)^\top \Lambda_h^{-1} \phi(s_h, a_h)\big)^{1/2} \,\middle|\, s_1 = s \right],$$

where Λ_h depends on the data but is held fixed in the expectation.

The guarantee is assumption-free in the sense that it requires only compliance: no uniform coverage and no closeness to a behavior policy. It is also minimax optimal: an information-theoretic lower bound shows that no algorithm can improve it beyond the factor β and constants.

The tabular case. With $\phi(s,a) = e_{(s,a)} \in \mathbb{R}^{SA}$, the matrix $\Lambda_h = \mathrm{diag}\big(N_h(s,a) + \lambda\big)$ is diagonal, where $N_h(s,a)$ counts the visits to (s,a) at step h in $\mathcal{D}$. If the transitions are deterministic and $\pi^\star$ follows the trajectory $(s_h^\star, a_h^\star)_{h=1}^H$, the bound reads

$$\mathrm{SubOpt}(\hat{\pi}; s_1) \le 2\beta \sum_{h=1}^{H} \big(N_h(s_h^\star, a_h^\star) + \lambda\big)^{-1/2}.$$

It depends only on how well the data cover the trajectory of the optimal policy, not on coverage of all states and actions; how often other pairs appear does not matter (an ORACLE PROPERTY, since $\pi^\star$ is unknown). In particular PEVI can outperform the behavior policy that generated the data: a pair that is visited often but lies off the optimal trajectory can make the behavior policy poor without affecting PEVI. More generally, if the data cover the optimal policy's feature directions, in the sense that $\Lambda_h \ge I + c^\dagger K\, \mathbb{E}_{\pi^\star}[\phi(s_h,a_h)\phi(s_h,a_h)^\top]$ for a constant $c^\dagger > 0$, then the suboptimality is $O\big(d^{3/2}H^2\sqrt{\zeta/K}\big)$, which decays at the rate $K^{-1/2}$.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 12 (Spring 2024); written from Jin, Yang, and Wang, Is Pessimism Provably Efficient for Offline RL? (ICML 2021).

Lecture 13: RL in Large State Space

13.1 Why function approximation

In the tabular setting the sample complexity is $O\big(\text{poly}(H)\, SA/\epsilon^2\big)$, linear in the number of states S. In many applications this is unaffordable: in an Atari game the number of states is exponential in the number of pixels. Worse, the same state is essentially never seen twice; a robot's joints are never in exactly the same configuration. The learner must *generalize*: use what it observed at visited states to predict the behavior of the dynamics at unseen states, relying on similarities between states.

Function approximation achieves this by learning from a restricted family $\mathcal{F}$ of functions parametrized by θ. One can approximate value functions, or policies, or both (actor-critic methods). Instead of learning an entire table, the learner fits θ, and the hope is a sample complexity that depends on the dimension of the parametrization rather than on S. There are three challenges:

- *generalization*: observed states are sparse, and we must generalize to unseen ones;
- *misspecification*: the optimal policy or $Q^\star$ may not belong to $\mathcal{F}$, which introduces bias;
- *exploration*: random exploration can already need exponentially many samples in tabular MDPs, and extending UCB to function approximation is itself challenging.

The next units approximate value functions, without misspecification, and address generalization and exploration, starting with linear function approximation.

13.2 Linear function approximation

We are given a feature map $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$ and use

$$Q(s, a; w) = w^\top \phi(s, a).$$

The feature map is always assumed known (it may come from domain knowledge), and the task is to learn w. This covers the tabular setting by taking $\phi(s, a) = e_{(s,a)} \in \mathbb{R}^{SA}$. In general $Q^\star$ need not be linear in ϕ; we first study the well-specified case, in which it is, and the following model guarantees it.

Definition 13.1 [Linear MDP]. An MDP $(\mathcal{S}, \mathcal{A}, r, P, H)$, where $\mathcal{S}$ may be infinite and $\mathcal{A}$ is finite, is a linear MDP with feature map $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$ if for every $h \in [H]$ there are a vector $\theta_h \in \mathbb{R}^d$ and a vector μ_h of d (signed) measures over $\mathcal{S}$ such that

$$r_h(s, a) = \theta_h^\top \phi(s, a) \qquad \text{and} \qquad P_h(\cdot \mid s, a) = \langle \phi(s, a), \mu_h(\cdot) \rangle.$$

Both the reward and the transition are linear in the features. The transition can still have infinitely many degrees of freedom, since the measures μ_h are unknown.

Example: *Tabular MDPs.* With $\phi(s,a) = e_{(s,a)}$, choose θ_h and μ_h with $e_{(s,a)}^\top \theta_h = r_h(s,a)$ and $e_{(s,a)}^\top \mu_h(\cdot) = P_h(\cdot \mid s,a)$. Every tabular MDP is a linear MDP with this feature map.

Example: *Simplex features.* Suppose every $\phi(s,a)$ lies in the d-dimensional probability simplex Δ_d. For any $\theta_h \in \mathbb{R}^d$ and any probability measures $e_i^\top \mu_h(\cdot)$, the reward r_h and transition P_h are well defined: $P_h(\cdot \mid s,a)$ is a mixture of the d measures with weights $\phi(s,a)$. The number of states may be infinite.

The linear quadratic regulator cannot be framed as a linear MDP. The point of the definition is that every value function is linear.

Proposition 13.2 [Linear Q-functions]. *In a linear MDP, for every policy π there are weights $\{w_h^\pi\}_{h\in[H]}$ such that $Q_h^\pi(s,a) = \langle \phi(s,a), w_h^\pi \rangle$ for all (s,a,h).*

Proof. By the Bellman equation and linearity of expectation,

$$Q_h^\pi(s,a) = r_h(s,a) + (P_h V_{h+1}^\pi)(s,a) = \langle \phi(s,a), \theta_h \rangle + \left\langle \phi(s,a), \int V_{h+1}^\pi(s')\, d\mu_h(s') \right\rangle,$$

so $w_h^\pi = \theta_h + \int V_{h+1}^\pi(s')\, d\mu_h(s')$. □

The model is not free, however: it constrains the geometry of the features.

Proposition 13.3 [Feature constraints]. *In a linear MDP, for all (s,a,h) and every measurable $B \subseteq \mathcal{S}$,*

$$\phi(s,a)^\top \mu_h(\mathcal{S}) = 1 \qquad \text{and} \qquad \phi(s,a)^\top \mu_h(B) \geq 0.$$

Proof. $P_h(\cdot \mid s,a) = \phi(s,a)^\top \mu_h(\cdot)$ must be a probability measure: it has total mass $P_h(\mathcal{S} \mid s,a) = 1$ and $P_h(B \mid s,a) \geq 0$. □

By the equality constraint, all features lie in the affine hyperplane $\{x \in \mathbb{R}^d : x^\top \mu_h(\mathcal{S}) = 1\}$. For example, the unit ℓ_2 ball cannot be the feature set of a linear MDP. (If $\mu_h(\mathcal{S})$ is the all-ones vector, the hyperplane is $\{x : \sum_i x_i = 1\}$, as for simplex features.)

When P and r are known, a linear MDP is solved by value iteration as before. Lecture 14 learns it from samples.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 13 (Spring 2024); written from the Spring 2020 Lecture 17 notes.

Lecture 14: Least-Squares Value Iteration

This unit learns a linear MDP (Definition 13.1) from samples with a simulator, using least squares in place of the empirical transition matrix.

14.1 Least squares as an approximate Bellman backup

In the tabular setting, given samples $\{(s_i, a_i, r_i, s_i')\}_{i=1}^N$ at step h, value iteration computes $\hat{Q}_h(s,a) = r_h(s,a) + (\hat{P}_h \hat{V}_{h+1})(s,a)$ with the empirical average $\hat{P}(s' \mid s,a) = N(s,a,s')/N(s,a)$. In a linear MDP with infinitely many states, $N(s,a) = 0$ for most pairs and this estimator is useless. The linear structure helps instead: parametrize $\hat{Q}_h(s,a) = \hat{w}_h^\top \phi(s,a)$ and fit

$$\hat{w}_h = \arg\min_{w \in \mathbb{R}^d} \sum_{i=1}^N \big(w^\top \phi_i - r_i - \hat{V}_{h+1}(s_i')\big)^2, \qquad \phi_i = \phi(s_i, a_i).$$

This approximation of the Bellman backup by regression is least-squares value iteration (LSVI). Setting the gradient to zero, $\sum_i \phi_i \big(\phi_i^\top \hat{w}_h - r_i - \hat{V}_{h+1}(s_i')\big) = 0$, so

$$\hat{w}_h = \Lambda^{-1} \sum_{i=1}^N \phi_i \big(r_i + \hat{V}_{h+1}(s_i')\big), \qquad \Lambda = \sum_{i=1}^N \phi_i \phi_i^\top.$$

If Λ is not invertible, replace it by $\Lambda + \lambda I$ for some $\lambda > 0$, which corresponds to adding the ridge penalty $\lambda \|w\|_2^2$ to the objective and guarantees a unique solution.

14.2 Core sets

With a simulator we may query any (s, a, h), but with infinitely many states, which pairs should we query? Since the reward and transition are linear in ϕ, only d coordinates must be learned. Ideally we would query pairs with $\phi(s_i, a_i) \approx e_i$; querying such a pair is like querying the ith coordinate of θ_h and of the transition.

Definition 14.1 [Core set]. A core set is a set $\mathcal{K} = \{(s_i, a_i)\}_{i=1}^d$ whose features $\psi_i = \phi(s_i, a_i)$ form a matrix $\Psi = (\psi_1, \ldots, \psi_d)$ with

- Ψ of full rank, and
- $\|\Psi^{-1} \phi(s,a)\|_1 \le L$ for all (s,a).

So every feature vector is a combination of the core features with small weights: $\phi(s,a) = \sum_i \tilde{w}_i \psi_i =$

$\Psi\tilde{w}$ with $\tilde{w} = \Psi^{-1}\phi(s,a)$ and $\|\tilde{w}\|_1 \le L$. If the feature set $\{\phi(s,a)\}$ spans $\mathbb{R}^d$, a core set can be constructed by an algorithm (omitted here). For simplex features, the pairs with $\phi(s_i,a_i) = e_i$ form a core set with $\Psi = I$ and $L = \max\|\phi(s,a)\|_1 = 1$.

Algorithm 14.1: LSVI with a core set

Input: core set $\mathcal{K}$, number of samples n

for $h \in [H]$ and $(s,a) \in \mathcal{K}$ **do**
  query (s,a,h) n times, obtaining $s'_1,\dots,s'_n$, and add $\{(s,a,s'_i)\}_{i=1}^n$ to $\mathcal{D}_h$
$\hat{V}_{H+1} \leftarrow 0$
for $h = H,\dots,1$ **do**
  $\hat{w}_h \leftarrow \arg\min_w \sum_{(s,a,s')\in\mathcal{D}_h} \big(w^\top\phi(s,a) - r_h(s,a) - \hat{V}_{h+1}(s')\big)^2$
  $\hat{Q}_h(s,a) \leftarrow \hat{w}_h^\top\phi(s,a)$ and $\hat{V}_h(s) \leftarrow \max_a \hat{Q}_h(s,a)$
return $\hat{\pi}_h(s) = \arg\max_a \hat{Q}_h(s,a)$

The last two assignments define functions (evaluated when needed), not updates over all states.

Theorem 14.2 [LSVI with a core set]. *Let $\iota = \log(Hd/\delta)$. If $n \ge cL^2H^4\iota/\epsilon^2$, then with probability at least $1-\delta$ the output $\hat{\pi}$ is ϵ-optimal. The total number of samples is $nHd = cH^5dL^2\iota/\epsilon^2$, independent of the number of states.*

14.3 Analysis

Since every query is at a core pair, $\Lambda = \sum_i \phi_i\phi_i^\top = n\Psi\Psi^\top$ is invertible, and LSVI is value iteration with a signed "empirical" transition. Using $r_h(s_i,a_i) = \phi_i^\top\theta_h$,

$$\hat{Q}_h(s,a) = \phi(s,a)^\top\Lambda^{-1}\sum_i \phi_i\phi_i^\top\theta_h + \phi(s,a)^\top\Lambda^{-1}\sum_i \phi_i\hat{V}_{h+1}(s'_i) = r_h(s,a) + (\hat{P}_h\hat{V}_{h+1})(s,a),$$

where

$$\hat{P}_h(\cdot \mid s,a) = \sum_i \phi(s,a)^\top\Lambda^{-1}\phi_i\,\delta(\cdot,s'_i)$$

puts mass $\phi(s,a)^\top\Lambda^{-1}\phi_i$ on the sampled next state s'_i. These masses may be negative, which does not matter.

Lemma 14.3 [Error decomposition].

$$V_1^\star(s_1) - \hat{V}_1(s_1) \le \mathbb{E}_{\pi^\star}\Big[\sum_{h=1}^H \big((P_h - \hat{P}_h)\hat{V}_{h+1}\big)(s_h,a_h)\Big],$$

$$\hat{V}_1(s_1) - V_1^{\hat{\pi}}(s_1) = \mathbb{E}_{\hat{\pi}}\Big[\sum_{h=1}^H \big((\hat{P}_h - P_h)\hat{V}_{h+1}\big)(s_h,a_h)\Big].$$

Proof. As in the simulation lemma (Lecture 6), $Q_h^\star - \hat{Q}_h = P_h V_{h+1}^\star - \hat{P}_h \hat{V}_{h+1} = (P_h - \hat{P}_h)\hat{V}_{h+1} + P_h(V_{h+1}^\star - \hat{V}_{h+1})$, and since $\hat{V}_h(s) = \max_a \hat{Q}_h(s,a) \geq \hat{Q}_h(s, \pi_h^\star(s))$,

$$\begin{aligned} V_h^\star(s) - \hat{V}_h(s) &\leq Q_h^\star(s, \pi_h^\star(s)) - \hat{Q}_h(s, \pi_h^\star(s)) \\ &= \mathbb{E}_{\pi^\star}\left[((P_h - \hat{P}_h)\hat{V}_{h+1})(s_h, a_h) + (V_{h+1}^\star - \hat{V}_{h+1})(s_{h+1}) \,\middle|\, s_h = s\right]. \end{aligned}$$

Unrolling over h gives the first claim. For the second, $\hat{V}_h(s) = \hat{Q}_h(s, \hat{\pi}_h(s))$ and $V_h^{\hat{\pi}}(s) = Q_h^{\hat{\pi}}(s, \hat{\pi}_h(s))$, so the same computation holds with equality along $\hat{\pi}$. □

Proof of Theorem 14.2. Fix h and write the error at any (s,a). Since $P_h(\cdot \mid s_i, a_i) = \phi_i^\top \mu_h(\cdot)$ and $\Lambda^{-1}\sum_i \phi_i \phi_i^\top = I$,

$$\begin{aligned} \big((P_h - \hat{P}_h)\hat{V}_{h+1}\big)(s,a) &= \phi(s,a)^\top \int \hat{V}_{h+1}(s')\, d\mu_h(s') - \phi(s,a)^\top \Lambda^{-1} \sum_i \phi_i \hat{V}_{h+1}(s_i') \\ &= \phi(s,a)^\top \Lambda^{-1} \sum_i \phi_i \Big[\mathbb{E}\big[\hat{V}_{h+1}(s') \,\big|\, s_i, a_i\big] - \hat{V}_{h+1}(s_i')\Big]. \end{aligned}$$

The data at step h consist of n samples at each core pair, so $\Lambda = n\Psi\Psi^\top$ and, grouping by core pair,

$$\big((P_h - \hat{P}_h)\hat{V}_{h+1}\big)(s,a) = \phi(s,a)^\top (\Psi\Psi^\top)^{-1} \sum_{j=1}^{d} \psi_j \varepsilon_j = \phi(s,a)^\top (\Psi\Psi^\top)^{-1}\Psi\varepsilon = (\Psi^{-1}\phi(s,a))^\top \varepsilon,$$

where $\varepsilon_j = \mathbb{E}[\hat{V}_{h+1}(s') \mid \psi_j] - \frac{1}{n}\sum_{l=1}^n \hat{V}_{h+1}(s'_{j,l})$ is the error of an average of n next-state values at core pair j. The function $\hat{V}_{h+1}$ is computed from the data at steps $h+1, \dots, H$ only, so it is independent of the samples at step h, and Hoeffding with a union bound over $j \in [d]$ and $h \in [H]$ gives $|\varepsilon_j| \leq cH\sqrt{\iota/n}$. By Hölder's inequality and the core-set property,

$$\big|\big((P_h - \hat{P}_h)\hat{V}_{h+1}\big)(s,a)\big| \leq \|\Psi^{-1}\phi(s,a)\|_1 \|\varepsilon\|_\infty \leq cLH\sqrt{\frac{\iota}{n}}.$$

Lemma 14.3 then gives $V_1^\star(s_1) - \hat{V}_1(s_1) \leq cLH^2\sqrt{\iota/n} \leq \epsilon/2$ and likewise $\hat{V}_1(s_1) - V_1^{\hat{\pi}}(s_1) \leq \epsilon/2$ once $n \geq c' L^2 H^4 \iota/\epsilon^2$. Adding the two proves the theorem. □

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 14 (Spring 2024); written from the Spring 2020 Lecture 17 and Lecture 18 notes.

Lecture 15: Exploration in Large State Space

In the exploration setting there is no simulator, so the learner cannot query a core set: it must generalize from whatever (non-core) data its own policies produce, and it needs a scheme that makes this data diverse. This unit adds optimism to least-squares value iteration, first through local bonuses (LSVI-UCB) and then through global optimization (ELEANOR).

15.1 LSVI-UCB

The agent plays K episodes in a linear MDP with regret $\mathrm{Regret}(K) = \sum_k [V_1^\star(s_1^k) - V_1^{\pi_k}(s_1^k)]$ and $T = KH$. The features are normalized so that $\|\phi(s,a)\| \le 1$ for all (s,a), and $\|\mu_h(\mathcal{S})\| \le \sqrt{d}$ and $\|\theta_h\| \le \sqrt{d}$ for all h. Write $\phi_h^\tau = \phi(s_h^\tau, a_h^\tau)$ for the features visited in episode τ.

Algorithm 15.1: LSVI-UCB

for episode $k = 1, \dots, K$ **do**
  receive the initial state s_1^k; set $Q_{H+1} \leftarrow 0$
  for $h = H, \dots, 1$ **do**
    $\Lambda_h \leftarrow \sum_{\tau=1}^{k-1} \phi_h^\tau (\phi_h^\tau)^\top + \lambda I$
    $w_h \leftarrow \Lambda_h^{-1} \sum_{\tau=1}^{k-1} \phi_h^\tau \big[r_h(s_h^\tau, a_h^\tau) + \max_a Q_{h+1}(s_{h+1}^\tau, a) \big]$
    $Q_h(\cdot,\cdot) \leftarrow \min\big\{ H,\ w_h^\top \phi(\cdot,\cdot) + \beta \sqrt{\phi(\cdot,\cdot)^\top \Lambda_h^{-1} \phi(\cdot,\cdot)} \big\}$
  for $h = 1, \dots, H$ **do**
    take $a_h^k = \arg\max_a Q_h(s_h^k, a)$ (ties broken arbitrarily), receive $r_h(s_h^k, a_h^k)$, observe s_{h+1}^k

The update for w_h is the solution of the regularized least-squares problem of Lecture 14 on all past data. The term $\beta\sqrt{\phi^\top \Lambda_h^{-1} \phi}$ is a bonus quantifying the uncertainty in the direction ϕ. In the tabular case, $\phi(s,a) \in \{e_{(s,a)}\}$ and $\Lambda_h = \sum_{(s,a)} N^k(s,a) e_{(s,a)} e_{(s,a)}^\top + \lambda I$ is diagonal, so $\sqrt{\phi(s,a)^\top \Lambda_h^{-1} \phi(s,a)} \approx \sqrt{1/N^k(s,a)}$, the Hoeffding bonus of UCB-VI up to logarithmic factors. In general $\phi^\top \Lambda_h^{-1} \phi \approx 1/\tilde{n}(\phi)$, where $\tilde{n}(\phi)$ is the effective number of samples in the direction ϕ: it measures how much the observed data reveal about a feature direction, even one never observed exactly.

Theorem 15.1 [Regret of LSVI-UCB]. *There is an absolute constant c such that, with $\lambda = 1$, $\beta = c\, dH\sqrt{\iota}$, and $\iota = \log(2dT/\delta)$, with probability at least $1 - \delta$,*

$$\mathrm{Regret}(K) \le O\big(\sqrt{d^3 H^3 T \iota^2}\big).$$

The bound is polynomial in H and independent of S and A, replacing the SA of the tabular minimax rate $\tilde{\Theta}(\sqrt{H^2 SAT})$ by d^3. By the usual conversion, with a fixed initial state an ϵ-optimal policy is

found with constant probability from $\widetilde{O}(d^3H^4/\epsilon^2)$ samples, by running $K = \widetilde{O}(d^3H^3/\epsilon^2)$ episodes and outputting the greedy policy of a uniformly random episode.

Why the regression approximates the Bellman backup. Ignoring the regularization, the least-squares solution is $\phi^\top w_h = \phi^\top \Lambda_h^{-1} \sum_\tau \phi_h^\tau [r_h^\tau + V_{h+1}(s_{h+1}^\tau)]$, with $V_{h+1} = \max_a Q_{h+1}(\cdot, a)$. The reward part recovers r_h, since $r_h = \phi^\top \theta_h$. The rest is $\hat{P}_h V_{h+1}$ for the signed empirical measure $\hat{P}_h(\cdot \mid s, a) = \phi(s,a)^\top \Lambda_h^{-1} \sum_\tau \phi_h^\tau \delta(\cdot, s_{h+1}^\tau)$, and it is compared with $P_h V_{h+1}$ through the bridge $\bar{P}_h(\cdot \mid s, a) = \phi(s,a)^\top \Lambda_h^{-1} \sum_\tau \phi_h^\tau P_h(\cdot \mid s_h^\tau, a_h^\tau)$.

1. $\hat{P}_h V_{h+1} \approx \bar{P}_h V_{h+1}$ by concentration. The difference $\phi^\top \Lambda_h^{-1} \sum_\tau \phi_h^\tau [V_{h+1}(s_{h+1}^\tau) - (P_h V_{h+1})(s_h^\tau, a_h^\tau)]$ is a self-normalized sum of zero-mean terms in $[-H, H]$. Because V_{h+1} is computed from the same data, the bound must hold uniformly over the class of functions $\min\{\max_a \phi(\cdot, a)^\top w + \beta\sqrt{\phi(\cdot, a)^\top \Lambda^{-1} \phi(\cdot, a)}, H\}$ with bounded parameters, which has a small covering number. This gives $|(\hat{P}_h - \bar{P}_h)V_{h+1}(s,a)| \le \widetilde{O}(dH)\sqrt{\phi(s,a)^\top \Lambda_h^{-1} \phi(s,a)}$.
2. $\bar{P}_h V_{h+1} \approx P_h V_{h+1}$ by linearity: since $P_h(\cdot \mid s_h^\tau, a_h^\tau) = (\phi_h^\tau)^\top \mu_h(\cdot)$,

$$\bar{P}_h(\cdot \mid s, a) = \phi(s,a)^\top \Lambda_h^{-1} \sum_\tau \phi_h^\tau (\phi_h^\tau)^\top \mu_h(\cdot) \approx \phi(s,a)^\top \mu_h(\cdot) = P_h(\cdot \mid s, a).$$

No finite amount of data can make $\hat{P}_h$ close to P_h in total variation, since μ_h has infinitely many degrees of freedom; the algorithm needs $\hat{P}_h V \approx P_h V$ only for V in a small class, which is why it can be viewed as model-free.

Proof sketch of Theorem 15.1. Combining the two steps with the regularization terms gives the key estimate: for any policy π, on a high-probability event,

$$\left|\phi(s,a)^\top w_h^k - Q_h^\pi(s,a) - \left(P_h(V_{h+1}^k - V_{h+1}^\pi)\right)(s,a)\right| \le \beta\sqrt{\phi(s,a)^\top (\Lambda_h^k)^{-1} \phi(s,a)}.$$

With $\pi = \pi^\star$ and backward induction on h this gives optimism, $Q_h^k \ge Q_h^\star$, as for UCB-VI. With $\pi = \pi_k$ it gives the recursion

$$\delta_h^k \le \delta_{h+1}^k + \zeta_{h+1}^k + 2\beta\sqrt{(\phi_h^k)^\top (\Lambda_h^k)^{-1} \phi_h^k}, \qquad \delta_h^k = (V_h^k - V_h^{\pi_k})(s_h^k),$$

where $\zeta_{h+1}^k = \mathbb{E}[\delta_{h+1}^k \mid s_h^k, a_h^k] - \delta_{h+1}^k$ is a martingale difference bounded by $2H$. Hence $\mathrm{Regret}(K) \le \sum_k \delta_1^k \le \sum_{k,h} \zeta_h^k + 2\beta \sum_{k,h} \sqrt{(\phi_h^k)^\top (\Lambda_h^k)^{-1} \phi_h^k}$. Azuma-Hoeffding bounds the first sum by $2H\sqrt{T\iota}$. For the second, the ELLIPTICAL POTENTIAL LEMMA (using $\lambda = 1$ and $\|\phi\| \le 1$) gives

$$\sum_{k=1}^K (\phi_h^k)^\top (\Lambda_h^k)^{-1} \phi_h^k \le 2 \log \frac{\det \Lambda_h^{K+1}}{\det \Lambda_h^1} \le 2d \log\left(1 + \frac{K}{\lambda}\right) \le 2d\iota,$$

so by Cauchy-Schwarz the bonus sum is at most $2\beta \cdot H\sqrt{K \cdot 2d\iota} = O\big(\sqrt{d^3H^4K\iota^2}\big) = O\big(\sqrt{d^3H^3T\iota^2}\big)$.

15.2 Global optimism

LSVI-UCB relies on the linear MDP structure: the bonus makes Q_h nonlinear, and this is harmless only because the low-rank transition maps any next-step value function back to a linear one. A weaker and more general condition asks only that the Bellman backup of *linear* functions be nearly linear. Let $\phi_h : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^{d_h}$ be features for step h, $\mathcal{B}_h = \{\theta \in \mathbb{R}^{d_h} : |\phi_h(s,a)^\top \theta| \le D \text{ for all } (s,a)\}$ the parameters giving bounded values, and $(\mathcal{T}_h Q)(s,a) = r_h(s,a) + \mathbb{E}_{s' \sim P_h(\cdot|s,a)} \max_{a'} Q(s',a')$ the Bellman optimality operator.

Definition 15.2 [Inherent Bellman error]. The inherent Bellman error $\mathcal{I}$ is the maximum over $h \in [H]$ of

$$\sup_{\theta_{h+1} \in \mathcal{B}_{h+1}} \inf_{\theta_h \in \mathcal{B}_h} \sup_{(s,a)} \big|\phi_h(s,a)^\top \theta_h - \big(\mathcal{T}_h(\phi_{h+1}^\top \theta_{h+1})\big)(s,a)\big|.$$

With $\mathcal{I} = 0$ the backup maps linear functions to linear functions. Every linear MDP has $\mathcal{I} = 0$, but not conversely, so low inherent Bellman error is strictly more general. In this setting the value function used for the backup must stay linear: adding bonuses state by state would leave the linear class, and the error could then propagate exponentially. The ELEANOR algorithm therefore keeps everything linear and is optimistic only at the initial state, through a global optimization over all steps. In episode k it solves

$$\max_{\bar\theta_1,\dots,\bar\theta_H} \max_a \phi_1(s_1^k, a)^\top \bar\theta_1 \quad \text{subject to} \quad \hat\theta_h = \Sigma_{hk}^{-1} \sum_{i<k} \phi_{hi} \big[r_{hi} + \max_a \phi_{h+1}(s_{h+1,i}, a)^\top \bar\theta_{h+1} \big],$$

$$\bar\theta_h = \hat\theta_h + \xi_h, \qquad \|\xi_h\|_{\Sigma_{hk}} \le \sqrt{\alpha_{hk}}, \qquad \bar\theta_h \in \mathcal{B}_h,$$

where $\Sigma_{hk} = \sum_{i<k} \phi_{hi}\phi_{hi}^\top + \lambda I$ is the empirical covariance of the features ϕ_{hi} observed at step h, the radius α_{hk} accounts for noise, regularization, and misspecification, and then plays the greedy policy of $\bar\theta$. The least-squares solutions are perturbed within confidence ellipsoids, but the perturbations are chosen jointly to maximize the value at the initial state, rather than state by state.

Theorem 15.3 [Regret of ELEANOR]. *With high probability, ELEANOR has regret $\widetilde{O}\big(\sum_{h=1}^H d_h\sqrt{K} + \sum_{h=1}^H \sqrt{d_h}\,\mathcal{I}K\big)$, and a matching lower bound shows that this is unimprovable beyond constants and logarithmic factors.*

The misspecification is amplified only by $\sqrt{d_h}$, even in the online setting. The global optimization is statistically optimal but not known to be computationally efficient.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 15 (Spring 2024); written from the Spring 2020 Lecture 19 notes and the LSVI-UCB and ELEANOR papers.

Lecture 16: General Function Approximation

Beyond linear features, the Q-functions may be approximated by general function classes $\mathcal{F}_h$, one for each step, with infinitely many states but finitely many actions. The question is what structure makes reinforcement learning sample-efficient; computational efficiency is set aside.

16.1 Assumptions and the Bellman error

Define the Bellman optimality operator on functions $f : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$,

$$(\mathcal{T}_h f)(s,a) = r_h(s,a) + \mathbb{E}_{s' \sim P_h(\cdot|s,a)}\Big[\max_{a'} f(s',a')\Big],$$

and write $V_f(s) = \max_a f(s,a)$.

Assumption 16.1 (Realizability)**.** $Q_h^\star \in \mathcal{F}_h$ for all h.

Assumption 16.2 (Completeness)**.** $\mathcal{T}_h f \in \mathcal{F}_h$ for all $f \in \mathcal{F}_{h+1}$ and all h.

Completeness together with $0 \in \mathcal{F}_{H+1}$ implies realizability, by the Bellman optimality equation; conversely, realizability gives $Q_{H+1}^\star = 0 \in \mathcal{F}_{H+1}$. Both assumptions hold for linear MDPs with linear classes. To separate the statistical question from exploration, first assume the data are representative.

Assumption 16.3 (Representative data)**.** For each h we are given data $\mathcal{D}_h = \{(s_h^{(i)}, a_h^{(i)}, r^{(i)}, s_{h+1}^{(i)})\}_{i=1}^N$ with $(s_h^{(i)}, a_h^{(i)}) \overset{\text{i.i.d.}}{\sim} \mu_h$, and there is a constant C such that for every policy π and every (s,a),

$$\frac{P_h^\pi(s,a)}{\mu_h(s,a)} \le C,$$

where $P_h^\pi(s,a)$ is the probability of being at (s,a) at step h under π (from the fixed initial state).

In a tabular MDP with a simulator, querying all pairs uniformly gives $\mu_h(s,a) = 1/(SA)$ and $P_h^\pi(s,a) \le 1$, so $C = SA$. With general function approximation this assumption is needed by the analysis but may be very hard to satisfy.

The natural empirical loss is

$$L_{\mathcal{D},h}(f_h, f_{h+1}) = \frac{1}{N} \sum_{(s,a,r,s') \in \mathcal{D}_h} \Big(f_h(s,a) - r - \max_{a'} f_{h+1}(s',a')\Big)^2.$$

Its expectation is not the Bellman error. Adding and subtracting $(P_h V_{f_{h+1}})(s,a)$, the cross term vanishes because $(P_h V_{f_{h+1}})(s,a) - V_{f_{h+1}}(s')$ has mean zero given (s,a), so

$$\mathbb{E} L_{\mathcal{D},h}(f_h, f_{h+1}) = \|f_h - \mathcal{T}_h f_{h+1}\|_{2,\mu_h}^2 + \mathbb{E}_{(s,a)\sim\mu_h}\Big[\mathrm{Var}\big(V_{f_{h+1}}(s') \,\big|\, s,a\big)\Big],$$

where $\|g\|_{2,\mu}^2 = \mathbb{E}_{(s,a)\sim\mu}[g(s,a)^2]$ and the first term is the Bellman error of f_h with respect to f_{h+1}. The extra variance term depends only on f_{h+1}.

16.2 Fitted Q-iteration

Algorithm 16.1: Fitted Q-iteration (FQI)

Input: function classes $\{\mathcal{F}_h\}$, data $\{\mathcal{D}_h\}$

$\hat{f}_{H+1} \leftarrow 0$

for $h = H, \dots, 1$ **do**

  $\hat{f}_h \leftarrow \arg\min_{f\in\mathcal{F}_h} L_{\mathcal{D},h}(f, \hat{f}_{h+1})$

return $\hat{\pi}_h(s) = \arg\max_a \hat{f}_h(s,a)$

LSVI is the special case of linear classes. In general the minimizer may not be efficiently computable.

The infinite-sample limit. As $N \to \infty$, $L_{\mathcal{D},h}(f, \hat{f}_{h+1}) \to \|f - \mathcal{T}_h \hat{f}_{h+1}\|_{2,\mu_h}^2 +$ (a term independent of f), so $\hat{f}_h = \arg\min_{f\in\mathcal{F}_h} \|f - \mathcal{T}_h \hat{f}_{h+1}\|_{2,\mu_h}^2$. By completeness $\mathcal{T}_h \hat{f}_{h+1} \in \mathcal{F}_h$, so it is a global minimizer, and it is the unique one because $\mu_h(s,a) > 0$ everywhere by Assumption 16.3. Hence $\hat{f}_h = \mathcal{T}_h \hat{f}_{h+1}$ for all h, and since $\hat{f}_{H+1} = 0 = Q^\star_{H+1}$, the Bellman optimality equation gives $\hat{f}_h = Q^\star_h$ for all h. The theorem quantifies how many samples approximate this limit.

Theorem 16.4 [FQI]. *Suppose $|f_h(s,a)| \le B$ for all h, all $f_h \in \mathcal{F}_h$, and all (s,a), and let $|\mathcal{F}| = \max_h |\mathcal{F}_h|$. Under Assumptions 16.1–16.3, with probability at least $1-\delta$, FQI outputs an ϵ-optimal policy using*

$$N = O\Big(\frac{H^8 B^4 C^2}{\epsilon^4} \log \frac{|\mathcal{F}|}{\delta}\Big)$$

samples per step.

The dependence on the class is through $\log|\mathcal{F}|$, not $|\mathcal{F}|$. For a tabular MDP with $\mathcal{F}_h = [0,H]^{SA}$, an ϵ-discretization $\tilde{\mathcal{F}}$ has $|\tilde{\mathcal{F}}| = (H/\epsilon)^{SA}$, so $\log|\tilde{\mathcal{F}}| = \widetilde{O}(SA)$; for a linear MDP, $\log|\tilde{\mathcal{F}}| = \widetilde{O}(d)$. A refined concentration analysis improves the bound to $O\big(H^4 B^2 C \log(|\mathcal{F}|/\delta)/\epsilon^2\big)$.

For a policy π, write $\|g\|_{p,\pi}$ for the L_p norm under the state-action distribution P_h^π at the relevant step h.

Lemma 16.5 [Error propagation]. *For any policy π and any functions $f_h \in \mathcal{F}_h$ with $f_{H+1} = 0$,*

$$\|Q^\star_h - f_h\|_{2,\pi} \le \sqrt{C} \sum_{h'=h}^{H} \|f_{h'} - \mathcal{T}_{h'} f_{h'+1}\|_{2,\mu_{h'}}.$$

Proof. Backward induction; at $h = H+1$ both sides vanish. For step h, by the triangle inequality

and $Q_h^\star = \mathcal{T}_h Q_{h+1}^\star$,

$$\|Q_h^\star - f_h\|_{2,\pi} \le \|f_h - \mathcal{T}_h f_{h+1}\|_{2,\pi} + \underbrace{\|\mathcal{T}_h f_{h+1} - \mathcal{T}_h Q_{h+1}^\star\|_{2,\pi}}_{A}.$$

For the second term, Jensen's inequality and $|V_f(s) - V^\star(s)| \le \max_a |(f - Q^\star)(s,a)|$ give

$$\begin{aligned} A^2 &= \mathbb{E}_\pi\Big[\big(\mathbb{E}\big[(V_{f_{h+1}} - V_{h+1}^\star)(s_{h+1}) \,\big|\, s_h, a_h\big]\big)^2\Big] \\ &\le \mathbb{E}_{s_{h+1}\sim P_{h+1}^\pi}\Big[\max_a (f_{h+1} - Q_{h+1}^\star)^2(s_{h+1}, a)\Big] = \|f_{h+1} - Q_{h+1}^\star\|_{2,\tilde\pi}^2, \end{aligned}$$

where $\tilde\pi$ follows π for steps $1, \dots, h$ and at step $h+1$ takes $\arg\max_a |(f_{h+1} - Q_{h+1}^\star)(s_{h+1}, a)|$. For the first term, a change of measure using Assumption 16.3,

$$\|f_h - \mathcal{T}_h f_{h+1}\|_{2,\pi} = \Big(\sum_{s,a} \frac{P_h^\pi(s,a)}{\mu_h(s,a)} \mu_h(s,a)\big[(f_h - \mathcal{T}_h f_{h+1})(s,a)\big]^2\Big)^{1/2} \le \sqrt{C}\,\|f_h - \mathcal{T}_h f_{h+1}\|_{2,\mu_h}.$$

Applying the induction hypothesis to $\tilde\pi$ (it holds for every policy) finishes the proof. □

Lemma 16.6 [Uniform concentration of the loss]. *With probability at least $1-\delta$, for all h, $f_h \in \mathcal{F}_h$, and $f_{h+1} \in \mathcal{F}_{h+1}$,*

$$\big|L_{\mathcal{D},h}(f_h, f_{h+1}) - \mathbb{E}L_{\mathcal{D},h}(f_h, f_{h+1})\big| \le O\Big(B^2\sqrt{\frac{1}{N}\log\frac{|\mathcal{F}_h||\mathcal{F}_{h+1}|H}{\delta}}\Big).$$

This is Hoeffding's inequality for the bounded squared losses and a union bound over the pairs of functions and the steps.

Proof of Theorem 16.4. Performance difference. For any policy $\hat\pi$,

$$V_1^\star(s_1) - V_1^{\hat\pi}(s_1) = \sum_{h=1}^H \mathbb{E}_{\hat\pi}\big[Q_h^\star(s_h, \pi_h^\star(s_h)) - Q_h^\star(s_h, \hat\pi_h(s_h))\big],$$

by telescoping $V_h^\star(s_h) - V_h^{\hat\pi}(s_h) = \big[V_h^\star(s_h) - Q_h^\star(s_h, \hat\pi_h(s_h))\big] + \mathbb{E}\big[(V_{h+1}^\star - V_{h+1}^{\hat\pi})(s_{h+1})\big]$. Since $\hat\pi$ is greedy for $\hat f$, $\hat f_h(s, \hat\pi_h(s)) \ge \hat f_h(s, \pi_h^\star(s))$, so each term is at most

$$\big[Q_h^\star - \hat f_h\big](s_h, \pi_h^\star(s_h)) + \big[\hat f_h - Q_h^\star\big](s_h, \hat\pi_h(s_h)).$$

Taking expectations, the two terms are L_1 errors under the policy $\tilde\pi$ that follows $\hat\pi$ before step h

and $\pi^\star$ at step h, and under $\hat{\pi}$; bounding L_1 by L_2 norms and applying Lemma 16.5 to each,

$$V_1^\star(s_1) - V_1^{\hat{\pi}}(s_1) \le \sum_h \Big(\|Q_h^\star - \hat{f}_h\|_{2,\tilde{\pi}} + \|Q_h^\star - \hat{f}_h\|_{2,\hat{\pi}} \Big) \le 2\sqrt{C} H \sum_{h=1}^{H} \|\hat{f}_h - \mathcal{T}_h \hat{f}_{h+1}\|_{2,\mu_h}.$$

Bellman errors. By the decomposition of $\mathbb{E}L_{\mathcal{D},h}$, the variance terms cancel in

$$\begin{aligned} \|\hat{f}_h - \mathcal{T}_h \hat{f}_{h+1}\|_{2,\mu_h}^2 &= \mathbb{E}L_{\mathcal{D},h}(\hat{f}_h, \hat{f}_{h+1}) - \mathbb{E}L_{\mathcal{D},h}(\mathcal{T}_h \hat{f}_{h+1}, \hat{f}_{h+1}) \\ &\le L_{\mathcal{D},h}(\hat{f}_h, \hat{f}_{h+1}) - L_{\mathcal{D},h}(\mathcal{T}_h \hat{f}_{h+1}, \hat{f}_{h+1}) + 2B^2 \sqrt{\frac{1}{N} \log \frac{|\mathcal{F}|^2 H}{\delta}}, \end{aligned}$$

using Lemma 16.6 twice (note $\mathcal{T}_h \hat{f}_{h+1} \in \mathcal{F}_h$ by completeness). The empirical difference is nonpositive because $\hat{f}_h$ minimizes $L_{\mathcal{D},h}(\cdot, \hat{f}_{h+1})$ over $\mathcal{F}_h$. Hence

$$\begin{aligned} \|\hat{f}_h - \mathcal{T}_h \hat{f}_{h+1}\|_{2,\mu_h} &\le O\Big(B \Big(\frac{1}{N} \log \frac{|\mathcal{F}|}{\delta} \Big)^{1/4} \Big), \\ V_1^\star(s_1) - V_1^{\hat{\pi}}(s_1) &\le O\Big(\sqrt{C} H^2 B \Big(\frac{1}{N} \log \frac{|\mathcal{F}|}{\delta} \Big)^{1/4} \Big) \le \epsilon \end{aligned}$$

once $N = O\big(C^2 H^8 B^4 \log(|\mathcal{F}|/\delta)/\epsilon^4\big)$. □

The proof relies on representative data, a constant C that bounds every policy's state-action distribution relative to μ_h. In general, how to obtain such data, and whether it exists with small C, is itself a challenging problem; Lecture 17 removes this assumption by exploring.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 16 (Spring 2024); written from the Spring 2020 Lecture 19 and Lecture 20 notes.

Lecture 17: Exploration in General Function Approximation

Fitted Q-iteration needs representative data (Lecture 16), and whether such data exist, or how to collect them, is unclear. Can a learner explore with a general function class and still find a near-optimal policy with few samples? Statistically yes, ignoring computation, for problems with a suitable low-dimensional structure. This unit introduces two such structures, Bellman rank and Bellman Eluder dimension, and the GOLF algorithm.

Throughout, $\mathcal{F} = \mathcal{F}_1 \times \cdots \times \mathcal{F}_H$ with $f = (f_1, \ldots, f_H) \in \mathcal{F}$, $f_{H+1} = 0$, and π_f is the greedy policy of f, $\pi_{f,h}(s) = \arg\max_a f_h(s, a)$.

17.1 Bellman rank

Definition 17.1 [Average Bellman error]. For $f \in \mathcal{F}$, a roll-in policy π, and a step h,

$$\mathcal{E}_h(f, \pi) = \mathbb{E}_{s_h \sim \pi,\, a_h \sim \pi_f}\left[f_h(s_h, a_h) - (\mathcal{T}_h f_{h+1})(s_h, a_h)\right],$$

where s_h is generated by running π for steps $1, \ldots, h-1$ and a_h is chosen by the greedy policy π_f.

Definition 17.2 [Bellman rank]. Let $\mathcal{E}_h \in \mathbb{R}^{|\mathcal{F}| \times |\mathcal{F}|}$ be the matrix with (f, f') entry $\mathcal{E}_h(f, \pi_{f'})$. The Bellman rank of an MDP M with class $\mathcal{F}$ is $\max_{h \in [H]} \operatorname{rank}(\mathcal{E}_h)$.

Small Bellman rank says that the matrix of average Bellman errors, indexed by the candidate function and the roll-in policy, is low rank.

Theorem 17.3 [Bellman rank suffices]. *If $Q^\star \in \mathcal{F}$ and the Bellman rank is at most M, then an algorithm (OLIVE) finds an ϵ-optimal policy with a number of samples polynomial in M, H, A, $\log|\mathcal{F}|$, and $1/\epsilon$, quadratic in M.*

OLIVE is based on hypothesis elimination: it plays the greedy policy of the most optimistic surviving function, estimates that function's average Bellman error, and eliminates all functions with a large average Bellman error under the current roll-in policy. It is not computationally efficient; but whenever the Bellman rank is small, a near-optimal policy can be found statistically with few samples.

Proposition 17.4 [Tabular MDPs]. *For a tabular MDP and any class $\mathcal{F}$, the Bellman rank is at most S.*

Proof.

$$\mathcal{E}_h(f, \pi_{f'}) = \sum_{s_h} P_h^{\pi_{f'}}(s_h)\, \mathbb{E}_{a_h \sim \pi_f(\cdot|s_h)}\big[(f_h - \mathcal{T}_h f_{h+1})(s_h, a_h)\big] = \langle \nu_h(f'), \xi_h(f)\rangle,$$

an inner product in $\mathbb{R}^S$ of a vector depending only on f' (the state distribution of $\pi_{f'}$) and one depending only on f. So $\mathcal{E}_h$ factors through $\mathbb{R}^S$ and has rank at most S. □

Proposition 17.5 [Linear MDPs]. *For a linear MDP with features in $\mathbb{R}^d$ and any class $\mathcal{F}$, the Bellman rank is at most d.*

Proof. Let $\Delta_h(f, s, a) = f_h(s, a) - (\mathcal{T}_h f_{h+1})(s, a)$. Conditioning on the previous step and using $P_{h-1}(s_h \mid s_{h-1}, a_{h-1}) = \langle \phi(s_{h-1}, a_{h-1}), \mu_{h-1}(s_h)\rangle$,

$$\mathcal{E}_h(f, \pi_{f'}) = \left\langle \mathbb{E}_{(s_{h-1}, a_{h-1}) \sim \pi_{f'}} \phi(s_{h-1}, a_{h-1}),\ \int \mathbb{E}_{a_h \sim \pi_f}\big[\Delta_h(f, s_h, a_h)\big]\, d\mu_{h-1}(s_h) \right\rangle,$$

an inner product in $\mathbb{R}^d$ of a vector depending only on f' and one depending only on f. □

17.2 Bellman Eluder dimension

The GOLF analysis uses a related variant, the Q-type Bellman rank, in which the action at step h also follows the roll-in policy: the Bellman rank is the smallest d such that there are maps $\phi_h, \psi_h : \mathcal{F} \to \mathbb{R}^d$ with

$$\mathbb{E}_{\pi_{f'}}\big[(f_h - \mathcal{T}_h f_{h+1})(s_h, a_h)\big] = \langle \phi_h(f), \psi_h(f')\rangle, \qquad \|\phi_h(f)\|_2 \|\psi_h(f')\|_2 \le \zeta,$$

for a normalization parameter ζ. A more general measure replaces the linear-algebraic notion of independence by an approximate, distributional one.

Definition 17.6 [Distributional Eluder dimension]. Let $\mathcal{G}$ be a class of functions on $\mathcal{X}$ and Π a family of probability measures on $\mathcal{X}$. A measure ν is ϵ-independent of $\mu_1, \dots, \mu_n$ with respect to $\mathcal{G}$ if some $g \in \mathcal{G}$ has $\sqrt{\sum_{i=1}^n (\mathbb{E}_{\mu_i}[g])^2} \le \epsilon$ but $|\mathbb{E}_\nu[g]| > \epsilon$. The distributional Eluder dimension $\dim_{DE}(\mathcal{G}, \Pi, \epsilon)$ is the length of the longest sequence $\rho_1, \dots, \rho_n \in \Pi$ such that, for some $\epsilon' \ge \epsilon$, each ρ_i is ϵ'-independent of $\rho_1, \dots, \rho_{i-1}$.

Just as a vector space has dimension d exactly when the longest sequence of vectors, each linearly independent of its predecessors, has length d, the Eluder dimension measures how long a sequence can keep revealing something new, allowing nonlinear and approximate independence. Taking point masses $\Pi = \{\delta_x\}$ recovers the original Eluder dimension of a class.

Definition 17.7 [Bellman Eluder dimension]. Let $(I - \mathcal{T}_h)\mathcal{F} = \{f_h - \mathcal{T}_h f_{h+1} : f \in \mathcal{F}\}$ be the Bellman residuals at step h, and $\Pi = \{\Pi_h\}_{h=1}^H$ families of distributions on $\mathcal{S} \times \mathcal{A}$. The Bellman Eluder (BE)

dimension is

$$\dim_{BE}(\mathcal{F}, \Pi, \epsilon) = \max_{h \in [H]} \dim_{DE}\big((I - \mathcal{T}_h)\mathcal{F}, \Pi_h, \epsilon\big).$$

Two families are used: $\mathcal{D}_{\mathcal{F}}$, where $\mathcal{D}_{\mathcal{F},h}$ contains the state-action distributions at step h of the greedy policies π_f, $f \in \mathcal{F}$; and $\mathcal{D}_{\Delta}$, where $\mathcal{D}_{\Delta,h}$ contains the point masses $\delta_{(s,a)}$. A problem has low BE dimension if $\min_{\Pi \in \{\mathcal{D}_{\mathcal{F}}, \mathcal{D}_{\Delta}\}} \dim_{BE}(\mathcal{F}, \Pi, \epsilon)$ is small. This class contains the two most general previously known tractable classes, which include tabular and linear MDPs and many others:

- low Bellman rank: Bellman rank d with normalization ζ gives $\dim_{BE}(\mathcal{F}, \mathcal{D}_{\mathcal{F}}, \epsilon) \le O\big(1 + d \log(1 + \zeta/\epsilon)\big)$;
- low Eluder dimension: under completeness, $\dim_{BE}(\mathcal{F}, \mathcal{D}_{\Delta}, \epsilon) \le \max_h \dim_E(\mathcal{F}_h, \epsilon)$;

and the inclusion is strict: for every m there are problems with BE dimension at most 5 whose Eluder dimension and Bellman rank are both at least m.

17.3 GOLF

Algorithm 17.1: GOLF: global optimism based on local fitting

Input: classes $\mathcal{F}$ and $\mathcal{G}$, number of episodes K, confidence parameter β

$\mathcal{D}_1, \dots, \mathcal{D}_H \leftarrow \emptyset$; $\quad \mathcal{B}^0 \leftarrow \mathcal{F}$

for episode $k = 1, \dots, K$ **do**

  $f^k \leftarrow \arg\max_{f \in \mathcal{B}^{k-1}} f_1(s_1, \pi_f(s_1))$; $\quad \pi^k \leftarrow \pi_{f^k}$ *// optimistic planning*

  play π^k for one episode, collecting $(s_1, a_1, r_1, \dots, s_H, a_H, r_H, s_{H+1})$

  $\mathcal{D}_h \leftarrow \mathcal{D}_h \cup \{(s_h, a_h, r_h, s_{h+1})\}$ for all h

  $\mathcal{B}^k \leftarrow \big\{f \in \mathcal{F} : L_{\mathcal{D}_h}(f_h, f_{h+1}) \le \inf_{g \in \mathcal{G}_h} L_{\mathcal{D}_h}(g, f_{h+1}) + \beta \text{ for all } h\big\}$

return π^{out} drawn uniformly from $\{\pi^k\}_{k=1}^K$

Here $L_{\mathcal{D}_h}(\xi_h, \zeta_{h+1}) = \sum_{(s,a,r,s') \in \mathcal{D}_h} \big[\xi_h(s,a) - r - \max_{a'} \zeta_{h+1}(s', a')\big]^2$ is the squared loss of Lecture 16, a proxy for the squared Bellman error. GOLF is an optimistic version of FQI: FQI sets $f_h \leftarrow \arg\min_{\varphi \in \mathcal{F}_h} L_{\mathcal{D}_h}(\varphi, f_{h+1})$, while GOLF keeps every function whose loss is within β of the best loss over an auxiliary class $\mathcal{G}_h$, and among these plays the most optimistic one. It extends ELEANOR (Lecture 15) from linear to arbitrary classes. The planning step is in general not computationally efficient.

Assumption 17.8 (Generalized completeness)**.** $\mathcal{T}_h \mathcal{F}_{h+1} \subseteq \mathcal{G}_h$ for all h. With $\mathcal{G} = \mathcal{F}$ this is the completeness of Lecture 16.

Theorem 17.9 [Regret of GOLF]**.** *Under realizability and generalized completeness, there is an absolute constant c such that, with $\beta = c \log[\mathcal{N}_{\mathcal{F} \cup \mathcal{G}}(1/K) \cdot KH/\delta]$, with probability at least $1 - \delta$, for all $k \in [K]$,*

$$\sum_{t=1}^{k} \big[V_1^\star(s_1) - V_1^{\pi^t}(s_1)\big] \le O\big(H\sqrt{dk\beta}\big), \qquad d = \min_{\Pi \in \{\mathcal{D}_\Delta, \mathcal{D}_{\mathcal{F}}\}} \dim_{BE}(\mathcal{F}, \Pi, 1/\sqrt{K}),$$

where $\mathcal{N}_{\mathcal{F}\cup\mathcal{G}}(\epsilon)$ is the ϵ-covering number in the metric $\max_h \|f_h - g_h\|_\infty$.

So the regret is $\widetilde{O}\big(H\sqrt{dK\log \mathcal{N}_{\mathcal{F}\cup\mathcal{G}}}\big)$, independent of the number of states, and by online-to-batch conversion GOLF finds an $O(\epsilon)$-optimal policy (with probability at least $\frac{1}{2}$) after $K = \widetilde{O}\big(H^2 d \log(\mathcal{N}_{\mathcal{F}\cup\mathcal{G}})/\epsilon^2\big)$ episodes, linear in both the BE dimension and the log covering number. For linear classes of dimension d_{lin} this gives regret $\widetilde{O}(Hd_{\text{lin}}\sqrt{K})$, the best known rate; for low Bellman rank, the sample complexity is linear rather than quadratic in the rank. In comparison, OLIVE needs $\widetilde{O}\big(H^3 d^2 \log(\mathcal{N}_{\mathcal{F}})/\epsilon^2\big)$ episodes with d the BE dimension for $\mathcal{D}_{\mathcal{F}}$, and has no sublinear regret, but it does not need completeness.

Proof sketch. There are three steps (here with $\Pi = \mathcal{D}_{\mathcal{F}}$).

1. *Optimism.* By martingale concentration and the construction of the confidence set, $Q^\star \in \mathcal{B}^k$ for all k with high probability, so the planning step gives $V_1^\star(s_1) \le \max_a f_1^k(s_1, a)$. Then, by the policy loss decomposition,

$$\text{Regret}(K) \le \sum_{k=1}^{K}\Big(\max_a f_1^k(s_1,a) - V_1^{\pi^k}(s_1)\Big) = \sum_{h=1}^{H}\sum_{k=1}^{K}\mathbb{E}_{\pi^k}\big[(f_h^k - \mathcal{T}_h f_{h+1}^k)(s_h,a_h)\big].$$

   The equality telescopes: for f and its greedy policy π_f, with $a_h = \pi_f(s_h)$,

$$f_h(s_h,a_h) - V_h^{\pi_f}(s_h) = (f_h - \mathcal{T}_h f_{h+1})(s_h,a_h) + \mathbb{E}\big[f_{h+1}(s_{h+1},\pi_f(s_{h+1})) - V_{h+1}^{\pi_f}(s_{h+1})\big].$$

2. *Sharpness of the confidence set.* The data in $\mathcal{D}_h$ come from playing $\pi^1,\dots,\pi^{k-1}$ once each, so martingale concentration and completeness turn the loss constraint into

$$\sum_{i=1}^{k-1}\Big(\mathbb{E}_{\pi^i}\big[(f_h^k - \mathcal{T}_h f_{h+1}^k)(s_h,a_h)\big]\Big)^2 \le O(\beta) \qquad \text{for all } (k,h).$$

3. *A pigeonhole principle for Eluder dimension.* Step 1 needs the Bellman errors of f^k under its own policy π^k; step 2 controls them only under earlier policies, and squared. The BE dimension bridges the two: for functions φ_k with $|\varphi_k| \le 1$ and measures $\mu_k \in \Pi$, if $\sum_{i<k}(\mathbb{E}_{\mu_i}[\varphi_k])^2 \le \beta$ for all k, then $\sum_{i\le k}|\mathbb{E}_{\mu_i}[\varphi_i]| \le O\big(\sqrt{\dim_{DE}(\Phi,\Pi,1/k)\,\beta k}\big)$. Applied with $\varphi_k = f_h^k - \mathcal{T}_h f_{h+1}^k$ and μ_k the distribution of π^k at step h, and summed over h, this gives the theorem.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 17 (Spring 2024); written from the Spring 2020 Lecture 20 notes and the GOLF paper (Jin, Liu, and Miryoosefi, 2021).

Lecture 18: Multiagent Reinforcement Learning

Many applications, such as Go and Starcraft, combine several agents with sequential decision making. The agents may cooperate or compete. Classical game theory (normal-form and extensive-form games) does not handle sequential games with long horizons efficiently, and single-agent RL looks for the best policy in a fixed environment, whereas in multiagent RL the opponents are not fixed and can adapt. Multiagent RL combines the two fields and has its own challenges. The goal is again a theoretical foundation with sample and computational efficiency; AlphaGo Zero, for example, trained on at least 10^7 games for at least a month. This unit introduces the two-player zero-sum setting, its solution concepts, and the direct combinations of game theory with single-agent RL.

18.1 Matrix games

As for single-agent RL, start without transitions. In rock-paper-scissors both players simultaneously choose $a, b \in \{\text{rock}, \text{paper}, \text{scissors}\}$, and the row player's utility is

$$R = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix},$$

with rows (and columns) ordered rock, paper, scissors; the column player receives $-R(a, b)$. This is a zero-sum game: the utilities of the two players sum to zero.

Definition 18.1 [Nash equilibrium]. A pair of strategies is a Nash equilibrium if neither player gains by changing only its own strategy while the other keeps playing its equilibrium strategy.

A pure strategy plays a single deterministic action, and a pure Nash equilibrium need not exist. If $(a^\star, b^\star)$ were one in rock-paper-scissors, then

$$-1 = \max_a \min_b R(a, b) \geq \min_b R(a^\star, b) \geq R(a^\star, b^\star) \geq \max_a R(a, b^\star) \geq \min_b \max_a R(a, b) = 1,$$

a contradiction. Intuitively, whatever one player commits to, the other beats it, and this creates a loop with no equilibrium. In general $\min_b \max_a R(a, b) \geq \max_a \min_b R(a, b)$, and the existence of a Nash equilibrium forces equality. A mixed strategy is a distribution over actions; with $\mu, \nu \in \Delta_{\mathcal{A}}$ the utility becomes the bilinear function

$$f(\mu, \nu) = \sum_{a,b} \mu(a) R(a, b) \nu(b) = \mu^\top R \nu.$$

Theorem 18.2 [von Neumann's minimax theorem]. *Let U, V be convex compact sets and $f : U \times V \to \mathbb{R}$ continuous, with $f(\cdot, v)$ concave for every v and $f(\mu, \cdot)$ convex for every μ. Then*

$$\max_{\mu \in U} \min_{v \in V} f(\mu, v) = \min_{v \in V} \max_{\mu \in U} f(\mu, v),$$

and there is at least one Nash equilibrium $(\mu^\star, v^\star)$, which attains both sides: $f(\mu^\star, v^\star)$ equals the common value.

Simplices are convex and compact, so every matrix game has a mixed Nash equilibrium; in rock-paper-scissors, both players play uniformly at random. For convex-concave f an equilibrium can be found by projected gradient descent-ascent, $x_{t+1} = x_t + \eta\, \partial_x f(x_t, y_t)$ and $y_{t+1} = y_t - \eta\, \partial_y f(x_t, y_t)$ (each followed by a projection), or by mirror descent-ascent.

18.2 Markov games

Definition 18.3 [Two-player zero-sum Markov game]. A two-player zero-sum Markov game, or stochastic game, written $\mathrm{MG}(\mathcal{S}, \mathcal{A}, \mathcal{B}, P, r, H)$, has a set of states $\mathcal{S}$, action sets $\mathcal{A}$ and $\mathcal{B}$ of the max-player and the min-player, transitions $P_h(s_{h+1} \mid s_h, a_h, b_h)$ that depend on both actions, rewards $r_h(s, a, b) \in [0, 1]$ for the max-player (a loss for the min-player), and horizon H. Each state s carries the matrix game $r_h(s, \cdot, \cdot)$.

Algorithm 18.1: Markov game interaction

```
the environment samples the initial state s_1
for step h = 1, ..., H do
    the two agents take their actions (a_h, b_h) simultaneously
    they receive the rewards ±r_h(s_h, a_h, b_h)
    the environment transitions to s_{h+1} ~ P_h(· | s_h, a_h, b_h)
```

We focus on fully observable tabular games: states and joint actions are revealed to both agents, and S, $A = |\mathcal{A}|$, $B = |\mathcal{B}|$ are finite and small. Normal-form (matrix) games have no state and no transition; extensive-form games have tree-structured transitions.

A general policy of the max-player maps histories to distributions, $\mu_h : (\mathcal{S} \times \mathcal{A} \times \mathcal{B})^{h-1} \times \mathcal{S} \to \Delta_{\mathcal{A}}$; a Markov policy uses only the current state, $\mu_h : \mathcal{S} \to \Delta_{\mathcal{A}}$, and the min-player's policies v are defined symmetrically. The agents act simultaneously, each based on the state, not on the other's current action. For a pair of Markov policies (μ, v),

$$V_h^{\mu,v}(s) = \mathbb{E}_{\mu,v}\Big[\sum_{h'=h}^{H} r_{h'}(s_{h'}, a_{h'}, b_{h'}) \,\Big|\, s_h = s\Big],$$

$$Q_h^{\mu,v}(s, a, b) = \mathbb{E}_{\mu,v}\Big[\sum_{h'=h}^{H} r_{h'}(s_{h'}, a_{h'}, b_{h'}) \,\Big|\, s_h = s, a_h = a, b_h = b\Big],$$

and $V^{\mu,\nu} = V_1^{\mu,\nu}(s_1)$ is the max-player's expected total reward.

18.3 Solution concepts

What makes a policy good: beating the world champion by a large margin, or not losing to anyone?

Definition 18.4 [Best response]. A best response to the min-player's policy ν is $\mathrm{BR}(\nu) \in \arg\max_\mu V^{\mu,\nu}$, the policy that best exploits ν, with value $V^{\dagger,\nu} = \max_\mu V^{\mu,\nu}$. Symmetrically, the min-player's best response to μ is $\mathrm{BR}(\mu) \in \arg\min_\nu V^{\mu,\nu}$, with value $V^{\mu,\dagger} = \min_\nu V^{\mu,\nu}$.

A best response is good against a fixed opponent but can be bad against others. A pair $(\mu^\star, \nu^\star)$ is a Nash equilibrium if no player has an incentive to deviate:

$$V^{\mu,\nu^\star} \le V^{\mu^\star,\nu^\star} \le V^{\mu^\star,\nu} \qquad \text{for all } \mu, \nu.$$

In two-player zero-sum Markov games the minimax theorem holds,

$$\max_\mu \min_\nu V^{\mu,\nu} = \min_\nu \max_\mu V^{\mu,\nu},$$

but not because of von Neumann's theorem, since $V^{\mu,\nu}$ is not convex-concave in the policies; it is proved by dynamic programming (Lecture 19). A Nash policy is optimal when always facing best responses: it may not win by a large margin, but no one beats it.

Definition: The objective is an ϵ-approximate Nash equilibrium $(\hat\mu, \hat\nu)$,

$$\max_\mu V^{\mu,\hat\nu} - \min_\nu V^{\hat\mu,\nu} = V^{\dagger,\hat\nu} - V^{\hat\mu,\dagger} \le \epsilon,$$

found with a number of samples mildly dependent on $S, A, B, H, 1/\epsilon$. The left side, the duality gap, is the sum of the suboptimality of each player against best responses, $\left[V^{\dagger,\hat\nu} - \min_\nu V^{\dagger,\nu}\right] + \left[\max_\mu V^{\mu,\dagger} - V^{\hat\mu,\dagger}\right]$, because the two optimal values coincide.

Three challenges stand out:

- the policy space is huge: an ϵ-discretization of the Markov policies alone has $\Omega((1/\epsilon)^{HSA})$ elements in the tabular setting;
- a Nash equilibrium policy is Markov, but a best response to a non-Markov policy may not be;
- Markov games do not allow efficient *no-regret* learning: there is no efficient algorithm guaranteeing $\max_\mu \sum_{t=1}^T V^{\mu,\nu^t} - \sum_{t=1}^T V^{\mu^t,\nu^t} \le \mathrm{poly}(H,S,A,B)\, T^{1-\alpha}$ against an arbitrary opponent (Lecture 19).

18.4 Direct combinations of game theory and single-agent RL

The key observation is that, against a *fixed* opponent, computing a best response is a single-agent RL problem. Combining Nash-finding algorithms that use a best-response oracle (self-play, fictitious play, double oracle) with single-agent RL algorithms (value iteration, Q-learning, DQN, PPO) gives methods commonly used in practice. Write μ^k, ν^k for the players' policies at iteration k.

Self-play. $\mu^{k+1} = \mathrm{BR}(\nu^k)$ and $\nu^{k+1} = \mathrm{BR}(\mu^{k+1})$. It does not converge to a Nash equilibrium even in rock-paper-scissors, and averaging the iterates does not help.

Fictitious play. Each player best responds to the opponent's *average* policy: $\mu^{k+1} = \mathrm{BR}\big(\frac{1}{k}(\nu^1 + \cdots + \nu^k)\big)$ and $\nu^{k+1} = \mathrm{BR}\big(\frac{1}{k+1}(\mu^1 + \cdots + \mu^{k+1})\big)$, which makes more sense in rock-paper-scissors. Fictitious play does converge to a Nash equilibrium, but slowly: the classical convergence proof needs $(1/\epsilon)^{\Omega(A)}$ iterations for an ϵ-Nash equilibrium of a normal-form game with A actions; the conjecture that $O(1/\epsilon^2)$ iterations suffice was refuted, and $(1/\epsilon)^{\Omega(A)}$ is the true worst case.

Double oracle. Let $M_k \in \mathbb{R}^{k\times k}$ be the reward matrix of the subgame whose row actions are $\mu^1, \ldots, \mu^k$ and column actions are $\nu^1, \ldots, \nu^k$, with entries $(M_k)_{ij} = V^{\mu^i, \nu^j}$. At iteration k, compute a Nash equilibrium (p, q) of M_k and add best responses to the equilibrium mixtures: $\mu^{k+1} = \mathrm{BR}\big(\sum_j q_j \nu^j\big)$ and $\nu^{k+1} = \mathrm{BR}\big(\sum_i p_i \mu^i\big)$. The weights are better informed than the uniform average of fictitious play. For a normal-form game with A actions, double oracle finds a Nash equilibrium in $O(A)$ iterations, since by then M_A is the full game matrix; but converting a Markov game into a normal form gives $A' = (1/\epsilon)^{HSA}$ actions, the size of the policy space.

Drawbacks. These algorithms use single-agent RL as a black box and do not exploit the structure of Markov games; the normal-form conversion has $(1/\epsilon)^{HSA}$ actions; and computing a best response is *not* an easy single-agent problem, because when the min-player deploys a fixed non-Markov policy, the game is not an MDP from the max-player's perspective, and existing single-agent results do not apply.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 18 (Spring 2024); written from the Spring 2022 multiagent RL slides and the Spring 2020 Lecture 22 notes.

Lecture 19: Two-Player Zero-Sum Games

This unit works with the two-player zero-sum Markov game $\mathrm{MG}(\mathcal{S}, \mathcal{A}, \mathcal{B}, P, r, H)$ of Lecture 18, with max-player policy μ and min-player policy ν. It solves the planning problem by dynamic programming, adds optimism for exploration, and then asks for decentralized algorithms whose sample complexity scales with $\max\{A, B\}$ rather than AB.

19.1 Bellman equations and planning

For Markov policies (μ, ν) the Bellman equations are

$$Q_h^{\mu,\nu}(s,a,b) = r_h(s,a,b) + (P_h V_{h+1}^{\mu,\nu})(s,a,b), \qquad V_h^{\mu,\nu}(s) = \sum_{a,b} \mu_h(a \mid s)\, Q_h^{\mu,\nu}(s,a,b)\, \nu_h(b \mid s).$$

For fixed strategies everything is as in an MDP; the interesting part is the optimal strategies. For a fixed μ, the game is an MDP for the min-player, so a best response $\nu^\dagger(\mu)$ exists that attains $\min_\nu V_h^{\mu,\nu}(s)$ at every state simultaneously and does not depend on the state; write $V_h^{\mu,\dagger} = V_h^{\mu,\nu^\dagger(\mu)}$. It is computed by dynamic programming:

$$Q_h^{\mu,\dagger}(s,a,b) = r_h(s,a,b) + (P_h V_{h+1}^{\mu,\dagger})(s,a,b), \qquad V_h^{\mu,\dagger}(s) = \min_{\nu_h} \sum_{a,b} \mu_h(a \mid s) Q_h^{\mu,\dagger}(s,a,b) \nu_h(b \mid s).$$

For the equilibrium, define the Nash value of a matrix $Q \in \mathbb{R}^{A \times B}$ as $\max_{\mu \in \Delta_{\mathcal{A}}} \min_{\nu \in \Delta_{\mathcal{B}}} \mu^\top Q \nu$, and the Bellman optimality equation

$$Q_h^\star(s,a,b) = r_h(s,a,b) + (P_h V_{h+1}^\star)(s,a,b), \qquad V_h^\star(s) = \max_{\mu \in \Delta_{\mathcal{A}}} \min_{\nu \in \Delta_{\mathcal{B}}} \sum_{a,b} \mu(a) \nu(b) Q_h^\star(s,a,b),$$

with $V_{H+1}^\star = 0$; by von Neumann's theorem at each state, the max and min can be exchanged.

Algorithm 19.1: Nash value iteration

```
V*_{H+1}(s) ← 0 for all s
for h = H, ..., 1 do
    Q*_h(s,a,b) ← r_h(s,a,b) + E_{s'~P_h(·|s,a,b)} V*_{h+1}(s') for all (s,a,b)
    for each s do
        (μ*_h(· | s), ν*_h(· | s)) ← a Nash equilibrium of the matrix game Q*_h(s,·,·)
        V*_h(s) ← μ*_h(· | s)^⊤ Q*_h(s,·,·) ν*_h(· | s)
```

Each matrix game is solved by linear programming, so Nash value iteration computes a Nash equilibrium of the Markov game in $\mathrm{poly}(H, S, A, B)$ time.

Proposition 19.1 [Nash equilibria are Markov]. *The Markov policies $(\mu^\star, \nu^\star)$ computed by Nash value*

iteration form a Nash equilibrium with value $V^\star$: $V_h^{\mu,\nu^\star}(s) \le V_h^\star(s) \le V_h^{\mu^\star,\nu}(s)$ for all policies μ, ν and all (s,h). Consequently $\max_\mu \min_\nu V_h^{\mu,\nu}(s) = V_h^\star(s) = \min_\nu \max_\mu V_h^{\mu,\nu}(s)$.

Proof. Against the fixed Markov policy $\nu^\star$ the max-player faces an MDP, so it suffices to consider Markov μ. By backward induction, if $V_{h+1}^{\mu,\nu^\star} \le V_{h+1}^\star$, then

$$V_h^{\mu,\nu^\star}(s) = \sum_{a,b} \mu_h(a \mid s)\nu_h^\star(b \mid s)\big[r_h + P_h V_{h+1}^{\mu,\nu^\star}\big](s,a,b) \le \mu_h(\cdot \mid s)^\top Q_h^\star(s,\cdot,\cdot)\,\nu_h^\star(\cdot \mid s) \le V_h^\star(s),$$

because $\nu_h^\star(\cdot \mid s)$ is an equilibrium strategy of the matrix game $Q_h^\star(s,\cdot,\cdot)$, against which no strategy of the max-player gets more than its value. The other inequality is symmetric. □

With a simulator (query any (s,a,b)), one can sample all triples uniformly and plug sample averages for P and r into variants of Nash value iteration or Nash Q-learning.

19.2 Exploration: optimistic Nash value iteration

When the agents cannot choose the state they must trade off exploration and exploitation. As in MDPs, ϵ-greedy needs exponentially many samples in the worst case, while optimism works. Optimistic Nash value iteration maintains an upper bound $\overline{Q}$ (optimistic for the max-player) and a lower bound $\underline{Q}$ (optimistic for the min-player), with the empirical model $\hat{P}_h(s' \mid s,a,b) = N(s,a,b,s')/N(s,a,b)$ and a bonus β.

Algorithm 19.2: Optimistic Nash value iteration

for episode $k = 1, \dots, K$ **do**
  for $h = H, \dots, 1$ **do**
    for each (s,a,b) **do**
      $\overline{Q}_h(s,a,b) \leftarrow r_h(s,a,b) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot \mid s,a,b)} \overline{V}_{h+1}(s') + \beta$
      $\underline{Q}_h(s,a,b) \leftarrow r_h(s,a,b) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot \mid s,a,b)} \underline{V}_{h+1}(s') - \beta$
    for each s **do**
      $\pi_h(\cdot,\cdot \mid s) \leftarrow \mathrm{CCE}\big(\overline{Q}_h(s,\cdot,\cdot), \underline{Q}_h(s,\cdot,\cdot)\big)$
      $\overline{V}_h(s) \leftarrow \langle \pi_h(\cdot,\cdot \mid s), \overline{Q}_h(s,\cdot,\cdot)\rangle$; $\underline{V}_h(s) \leftarrow \langle \pi_h(\cdot,\cdot \mid s), \underline{Q}_h(s,\cdot,\cdot)\rangle$
  execute π, collect samples, and update $\hat{P}$

This is a multiagent version of UCB-VI. Its main techniques are: sample averages for the transitions; the optimistic pair $(\overline{Q}, \underline{Q})$, with a bonus chosen differently from the single-agent case to get a sharp guarantee; and, at each state, a COARSE CORRELATED EQUILIBRIUM (CCE) of the general-sum matrix game $(\overline{Q}, \underline{Q})$ instead of a Nash equilibrium, since Nash equilibria of general-sum games are PPAD-hard to compute. Here a CCE is a joint distribution π over action pairs (a,b) from which neither player gains by switching to a fixed action: $\mathbb{E}_\pi \overline{Q}(a,b) \ge \max_{a'} \mathbb{E}_\pi \overline{Q}(a',b)$ and $\mathbb{E}_\pi \underline{Q}(a,b) \le \min_{b'} \mathbb{E}_\pi \underline{Q}(a,b')$; it can be computed by linear programming (Lecture 20 treats these equilibria in general).

Theorem 19.2 [Optimistic Nash value iteration**].** *With high probability, optimistic Nash value iteration finds an ϵ-approximate Nash equilibrium within $\widetilde{O}(H^3SAB/\epsilon^2)$ episodes. The information-theoretic lower bound is $\Omega(H^3S\max\{A,B\}/\epsilon^2)$.*

19.3 Decentralized learning

Optimistic Nash value iteration is *centralized*: at every step a central solver computes a CCE of $(\overline{Q}, \underline{Q})$ for both players. In a decentralized algorithm each agent runs the same algorithm on its own observations, as if it were alone. Decentralized algorithms are easier to implement, more versatile (agnostic to the other agents' actions), and faster, with less communication. A second challenge is that most single-agent methods estimate $Q^\star$, which in a Markov game has $\Omega(HAB)$ entries and so needs at least that many samples, while the lower bound scales with $\max\{A,B\}$. Is there a decentralized algorithm with sample complexity $O(\max\{A,B\})$?

Normal-form games. In matrix games the answer is yes: each agent independently runs a no-regret algorithm for adversarial bandits, such as EXP3, which guarantees $\sum_{t=1}^T \langle \mu_t, \ell_t\rangle - \min_a \sum_{t=1}^T \langle e_a, \ell_t\rangle \le \mathrm{poly}(A)\,T^{1-\alpha}$ for its losses ℓ_t, using only bandit feedback. In a two-player zero-sum game the averaged strategies $\bar\mu \times \bar\nu$, where $\bar\mu = \frac{1}{T}\sum_t \mu_t$ and $\bar\nu = \frac{1}{T}\sum_t \nu_t$, form an approximate Nash equilibrium: with $f(\mu,\nu) = \mu^\top R \nu$,

$$\max_\mu f(\mu,\bar\nu) = \max_\mu \frac{1}{T}\sum_t f(\mu,\nu_t) \le \frac{1}{T}\sum_t f(\mu_t,\nu_t) + \frac{\mathrm{Regret}_1}{T},$$
$$\min_\nu f(\bar\mu,\nu) = \min_\nu \frac{1}{T}\sum_t f(\mu_t,\nu) \ge \frac{1}{T}\sum_t f(\mu_t,\nu_t) - \frac{\mathrm{Regret}_2}{T},$$

so the duality gap is at most $(\mathrm{Regret}_1 + \mathrm{Regret}_2)/T \to 0$. The sample complexity scales with $\widetilde{O}(A+B)$.

Markov games. Simply running no-regret algorithms in Markov games is impossible, since Markov games do not allow efficient no-regret learning:

$$\max_\mu \sum_{t=1}^T V_1^{\mu,\nu^t} - \sum_{t=1}^T V_1^{\mu^t,\nu^t} \le \mathrm{poly}(H,S,A,B)\,T^{1-\alpha}$$

cannot be guaranteed efficiently. Computationally, a polynomial-time no-regret algorithm for Markov games would give a polynomial-time algorithm for learning parity with noise; statistically, no-regret learning in Markov games is at least as hard as learning the best Markov policy in a POMDP.

19.4 V-learning

Algorithm 19.3: V-learning (run by one agent)

for episode $k = 1, \dots, K$ **do**
  receive s_1
  for $h = 1, \dots, H$ **do**
    take $a_h \sim \pi_h(\cdot \mid s_h)$, observe the reward r_h and the next state s_{h+1}
    $t = N_h(s_h) \leftarrow N_h(s_h) + 1$
    $V_h(s_h) \leftarrow (1 - \alpha_t) V_h(s_h) + \alpha_t \big(r_h + V_{h+1}(s_{h+1}) + \beta_t\big)$
    $\pi_h(\cdot \mid s_h) \leftarrow \textsc{AdvBanditUpdate}\big(a_h,\ r_h + V_{h+1}(s_{h+1})\big)$ on the (s_h, h)th adversarial bandit

V-learning updates V incrementally rather than Q, with the same learning rate $\alpha_t = \frac{H+1}{H+t}$ as Q-learning. It is a single-agent algorithm: it treats each pair (s, h) as an adversarial bandit and uses any bandit algorithm with a weighted regret guarantee,

$$\sum_{t=1}^{T} \gamma_T^t \langle \mu_t, \ell_t \rangle - \min_a \sum_{t=1}^{T} \gamma_T^t \langle e_a, \ell_t \rangle \le \mathrm{poly}(A)\, T^{1-\alpha},$$

as a black box, where γ_T^t are the weights of Lecture 7. It has a no-regret guarantee at each state for the losses it is fed, but it is not a no-regret algorithm for the Markov game.

Theorem 19.3 [V-learning, zero-sum]. *If both agents run V-learning independently with the FTRL bandit subroutine, then in a two-player zero-sum Markov game they find an ϵ-approximate Nash equilibrium within $\widetilde{O}(H^5 S \max\{A, B\}/\epsilon^2)$ episodes.*

V-learning is decentralized and achieves the optimal $\max\{A, B\}$ dependence; a sharp dependence on H remains open.

Algorithm	Training	Main estimand	Sample complexity
Nash value iteration (optimistic)	centralized	$P_h(s' \mid s, a, b)$	$\widetilde{O}(H^3 SAB/\epsilon^2)$
Nash Q-learning	centralized	$Q_h^\star(s, a, b)$	$\widetilde{O}(H^5 SAB/\epsilon^2)$
V-learning	decentralized	$V_h^\star(s)$	$\widetilde{O}(H^5 S \max\{A, B\}/\epsilon^2)$
Lower bound			$\Omega(H^3 S \max\{A, B\}/\epsilon^2)$

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 19 (Spring 2024); written from the Spring 2022 multiagent RL slides and the Spring 2020 Lecture 22 notes.

Lecture 20: Multiplayer General-Sum Games

With more than two players, or with rewards that are not opposed, the players may cooperate as well as compete. This unit defines general-sum Markov games and their equilibria, explains why the natural extension of Nash value iteration suffers from the *curse of multiagents*, and shows how V-learning breaks it.

20.1 General-sum Markov games

Definition 20.1 [Markov game]. A (general-sum) Markov game $\mathrm{MG}(\mathcal{S}, \{\mathcal{A}_i\}_{i=1}^m, P, \{r_i\}_{i=1}^m, H)$ has m players, a set of states $\mathcal{S}$, an action set $\mathcal{A}_i$ with A_i actions for each player i, transitions $P_h(s_{h+1} \mid s_h, \mathbf{a}_h)$ that depend on the joint action $\mathbf{a}_h = (a_h^{(1)}, \ldots, a_h^{(m)})$, a reward $r_{i,h}(s_h, \mathbf{a}_h) \in [0,1]$ for each player i, and horizon H.

The relation among the players' rewards is arbitrary, so both cooperation and competition are covered. A general policy of player i maps histories to distributions, $\pi_{i,h} : (\mathcal{S} \times \prod_j \mathcal{A}_j)^{h-1} \times \mathcal{S} \to \Delta_{\mathcal{A}_i}$, and a Markov policy uses only the current state. A joint policy $\pi = \pi_1 \odot \cdots \odot \pi_m$ may correlate the players through a shared random seed; a product policy $\pi = \pi_1 \times \cdots \times \pi_m$ uses independent randomness for each player. Player i's value of a joint policy is

$$V_i^\pi = \mathbb{E}_\pi \Big[\sum_{h=1}^H r_{i,h}(s_h, \mathbf{a}_h) \Big],$$

from the fixed initial state s_1. Write π_{-i} for the joint policy of the players other than i, and $V_i^{\dagger,\pi_{-i}} = \max_{\pi_i'} V_i^{\pi_i' \times \pi_{-i}}$ for the value of player i's best response, a policy independent of the randomness in π_{-i}.

20.2 Equilibria

Definition 20.2 [Nash, CE, CCE]. Let $\epsilon \geq 0$.

- A product policy π is an ϵ-approximate Nash equilibrium if $\max_{i \in [m]} \big(V_i^{\dagger,\pi_{-i}} - V_i^\pi\big) \leq \epsilon$: no player gains more than ϵ by changing only its own policy.
- A joint policy π is an ϵ-approximate coarse correlated equilibrium (CCE) if $\max_i \big(V_i^{\dagger,\pi_{-i}} - V_i^\pi\big) \leq \epsilon$: no player gains by deviating to an independent policy, one that does not see the actions sampled for it from π.
- A joint policy π is an ϵ-approximate correlated equilibrium (CE) if $\max_i \max_{\phi_i} \big(V_i^{(\phi_i \diamond \pi_i) \odot \pi_{-i}} -$

$V_i^{\pi}) \le \epsilon$, where a STRATEGY MODIFICATION ϕ_i maps the history and the action recommended to player i to a new action: no player gains even if it can observe its sampled action before deviating.

The only difference between Nash and CCE is that a Nash equilibrium must be a product policy. A strategy modification can use the shared randomness, so it is more powerful than a best response, and

$$\text{Nash} \subseteq \text{CE} \subseteq \text{CCE},$$

in normal-form games and in Markov games alike. Even for normal-form games, computing a Nash equilibrium of a general-sum game is PPAD-hard, while CEs and CCEs can be computed by linear programming.

20.3 Optimistic Nash value iteration and the curse of multiagents

Optimistic Nash value iteration (Lecture 19) extends to general-sum games by maintaining an upper bound $\overline{Q}_{i,h}(s,\cdot)$ for every player and replacing the CCE subroutine by an equilibrium of the general-sum matrix game at each state,

$$\pi_h(\cdot \mid s) \leftarrow \textsc{Equilibrium}\big(\overline{Q}_{1,h}(s,\cdot), \dots, \overline{Q}_{m,h}(s,\cdot)\big), \qquad \textsc{Equilibrium} \in \{\text{Nash}, \text{CE}, \text{CCE}\}.$$

Theorem 20.3 [OPTIMISTIC NASH VALUE ITERATION, GENERAL SUM**].** *With high probability, optimistic Nash value iteration finds an ϵ-approximate {Nash, CE, CCE} of a general-sum Markov game within $\widetilde{O}\big(H^4 S \prod_{i=1}^m A_i/\epsilon^2\big)$ episodes.*

(For Nash, the per-state equilibrium computation is the PPAD-hard step.) The sample complexity scales with $\prod_i A_i \approx A^m$, the size of the joint action space, which grows exponentially in the number of players; even the Q-table $Q(s,\mathbf{a})$ has $\Omega(S\prod_i A_i)$ entries. This is the CURSE OF MULTIAGENTS. Can the sample complexity be polynomial in m?

Normal-form games. If each agent independently runs a no-regret algorithm for adversarial bandits, the averaged joint play $\mathbb{E}_{t\sim\text{Unif}([T])}\big(\mu_t^{(1)} \times \cdots \times \mu_t^{(m)}\big)$ is an approximate CCE, with sample complexity $\widetilde{O}(\max_i A_i)$. If each agent instead runs a no-*swap*-regret algorithm, guaranteeing

$$\sum_{t=1}^{T} \langle \mu_t, \ell_t\rangle - \min_{\psi\in\Psi} \sum_{t=1}^{T} \langle \psi \diamond \mu_t, \ell_t\rangle \le \text{poly}(A)\, T^{1-\alpha},$$

where Ψ is the set of all maps $\psi : \mathcal{A} \to \mathcal{A}$ and $\psi \diamond \mu$ moves the probability of each action a to $\psi(a)$, the averaged play is an approximate CE, with sample complexity $\widetilde{O}((\max_i A_i)^2)$.

20.4 V-learning for general-sum games

V-learning (Lecture 19) carries this idea to Markov games. Each player runs it independently, on its own actions only, with three ingredients made precise below: a normalized loss, a weighted-regret bandit subroutine, and a correlated output policy. Player j maintains $V_h(s)$, initialized to $H+1-h$, a counter $N_h(s)$, and a policy $\pi_h(\cdot \mid s)$, initialized to uniform. At step h of episode k, with $t = N_h(s_h)$ after incrementing,

$$\tilde{V}_h(s_h) \leftarrow (1-\alpha_t)\tilde{V}_h(s_h) + \alpha_t\big(r_h + V_{h+1}(s_{h+1}) + \beta_t\big), \qquad V_h(s_h) \leftarrow \min\{H+1-h, \tilde{V}_h(s_h)\},$$

and the (s_h, h)th bandit receives the action a_h with loss $\big(H - r_h - V_{h+1}(s_{h+1})\big)/H \in [0,1]$. Since only V-values are kept, each player stores $O(HS)$ values rather than a Q-table over joint actions.

Weighted regret. With the weights $\gamma_t^i = \alpha_i \prod_{j=i+1}^{t}(1-\alpha_j)$ of Lecture 7, the subroutine must guarantee, with probability at least $1-\delta$,

$$\max_{\theta \in \Delta_{\mathcal{A}}} \sum_{i=1}^{t} \gamma_t^i \big[\langle \theta_i, \ell_i\rangle - \langle \theta, \ell_i\rangle\big] \le \xi(A, t, \log(1/\delta)),$$

with $\sum_{t' \le t} \xi(A, t', \log(1/\delta)) \le \Xi(A, t, \log(1/\delta))$ concave in t. Here θ_i is the bandit's distribution and ℓ_i its loss vector at its ith round, of which it observes only the noisy loss $\tilde{\ell}_i(a_i)$ of the action played. Follow-the-regularized-leader with weights w_i chosen so that $w_i/w_j = \gamma_t^i/\gamma_t^j$ for every t, and importance-weighted loss estimates,

$$\theta_{t+1}(a) \propto \exp\Big(-\frac{\eta_t}{w_t}\sum_{i \le t} w_i \hat{\ell}_i(a)\Big), \qquad \hat{\ell}_i(a) = \frac{\tilde{\ell}_i(a_i)\mathbf{1}[a_i = a]}{\theta_i(a) + \kappa_i},$$

with step sizes η_t and exploration parameters κ_t both equal to $\sqrt{H\log A/(At)}$, achieves $\xi \le O\big(\sqrt{HA\log(A/\delta)/t}\big)$ and $\Xi \le O\big(\sqrt{HAt\log(A/\delta)}\big)$.

Output policy. The output $\hat{\pi}$ is not Markov. To execute it, sample k uniformly from $[K]$; at each step h, observe s_h, let $t = N_h^k(s_h)$ be the number of earlier visits to s_h at step h before episode k, sample $i \in [t]$ with probability γ_t^i, set k to the episode k^i of the ith such visit, and play $a_h \sim \pi_h^k(\cdot \mid s_h)$ with this new k. This mixture mirrors the incremental update: without the bonus, $V_1^K(s_1)$ is a stochastic estimate of the value of $\hat{\pi}$. For a CCE or CE, all players use a common random seed when executing their output policies, so that they sample the same indices; training is fully decentralized, and only this seed is communicated. The resulting correlated policy $\hat{\pi} = \hat{\pi}_1 \odot \cdots \odot \hat{\pi}_m$ is stored with $O(HSK\sum_j A_j)$ space, linear in m, instead of the $\Omega(HS\prod_j A_j)$ needed to write down a general correlated policy.

Theorem 20.4 [V-learning, general sum]. *Let $A = \max_j A_j$ and $\iota = \log(mHSAK/\delta)$. With the bonuses chosen so that $\sum_{i=1}^{t}\gamma_t^i\beta_{j,i} = \Theta\big(H\xi(A_j, t, \iota) + \sqrt{H^3\iota/t}\big)$, with probability at least $1-\delta$ the correlated output*

policy satisfies

$$\max_{j\in[m]}\left(V_j^{\dagger,\hat{\pi}_{-j}} - V_j^{\hat{\pi}}\right) \le O\left(\frac{H^2 S}{K}\,\Xi(A, K/S, \iota) + \sqrt{\frac{H^5 S\iota}{K}}\right).$$

With FTRL and $\beta_{j,t} = c\sqrt{H^3 A_j \iota/t}$ this is $O\left(\sqrt{H^5 SA\iota/K}\right)$, so V-learning finds an ϵ-CCE within $\widetilde{O}(H^5SA/\epsilon^2)$ episodes. With a weighted no-swap-regret subroutine (FTRL for swap regret) it finds an ϵ-CE within $\widetilde{O}(H^5SA^2/\epsilon^2)$ episodes.

The sample complexity depends on $\max_j A_j$, not on $\prod_j A_j$: V-learning breaks the curse of multiagents. In the single-agent case the same bound gives $\widetilde{O}(H^5SA/\epsilon^2)$, a factor H^2 above the lower bound; one factor of H comes from the incremental update, as for Q-learning.

Proof idea. Three estimates combine. *Optimism:* with high probability $V_{j,h}^k(s) \ge V_{j,h}^{\dagger,\hat{\pi}^k_{-j,h}}(s)$, where $\hat{\pi}_h^k$ is the output policy started at step h and episode k: by backward induction, the weighted-regret guarantee says that player j's bandit at (s,h) did nearly as well, on the weighted average, as any fixed action, so the incremental estimate plus the bonus upper bounds the value of the best response. *Pessimism:* a lower estimate $\underline{V}_{j,h}^k$, built for the analysis by the same update with the bonus subtracted, satisfies $\underline{V}_{j,h}^k(s) \le V_{j,h}^{\hat{\pi}_h^k}(s)$, by martingale concentration. *The gap:* the CCE gap of $\hat{\pi}$ is at most $\frac{1}{K}\sum_k \max_j(V_{j,1}^k - \underline{V}_{j,1}^k)(s_1)$, and the difference $\delta_h^k = \max_j(V_{j,h}^k - \underline{V}_{j,h}^k)(s_h^k)$ satisfies the Q-learning-style recursion

$$\delta_h^k \le \gamma_{n_h^k}^0 H + \sum_{i=1}^{n_h^k} \gamma_{n_h^k}^i \delta_{h+1}^{k^i} + O\left(H\xi(A, n_h^k, \iota) + \sqrt{H^3\iota/n_h^k}\right),$$

where n_h^k is the visit count and $\gamma_t^0 = \prod_{j\le t}(1-\alpha_j)$. Summing over k and using $\sum_{t\ge i}\gamma_t^i \le 1 + 1/H$ (Lemma 7.1) gives $\sum_k \delta_h^k \le SH + (1+\frac{1}{H})\sum_k \delta_{h+1}^k + \sum_k O(\cdots)$, which unrolls over h to the theorem.

Objective	Two-player zero-sum	Multiplayer general-sum
Nash	$\widetilde{O}(H^5SA/\epsilon^2)$	PPAD-complete
CCE	$\widetilde{O}(H^5SA/\epsilon^2)$	$\widetilde{O}(H^5SA/\epsilon^2)$
CE	$\widetilde{O}(H^5SA^2/\epsilon^2)$	$\widetilde{O}(H^5SA^2/\epsilon^2)$

The table summarizes the sample complexity of V-learning, with $A = \max_i A_i$.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 20 (Spring 2024); written from the Spring 2022 multiagent RL slides and Jin, Liu, Wang, and Yu, V-Learning (2021).

Lecture 21: Partially Observable Reinforcement Learning I

So far the agents observed the state: MDPs and Markov games are fully observable models. Many applications are partially observable: in poker a player does not know the opponents' hands, and in Starcraft the fog of war hides part of the map. This unit introduces the general model, partially observable MDPs, its belief-state view, and a still more general representation based only on observable quantities.

21.1 Partially observable MDPs

Definition 21.1 [POMDP]. A partially observable Markov decision process

$$\text{POMDP}(\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathbb{T}, \mathbb{O}, r, H),$$

with initial state distribution μ_1, has a set of hidden states $\mathcal{S}$, a set of actions $\mathcal{A}$, a set of observations $\mathcal{O}$, transition probabilities $\mathbb{T}(s' \mid s, a)$, observation (emission) probabilities $\mathbb{O}(o \mid s)$, a reward function $r(o)$ of the observation, and episode length H.

For simplicity the model is stationary ($\mathbb{T}$ and $\mathbb{O}$ do not change with h), and the reward is part of the observation. At each step the hidden state s_h emits an observation $o_h \sim \mathbb{O}(\cdot \mid s_h)$, the agent acts, and the hidden state moves to $s_{h+1} \sim \mathbb{T}(\cdot \mid s_h, a_h)$ (Figure 5). A POMDP is Markovian in the hidden states but not in the observations, and the hidden states are in general unknown.

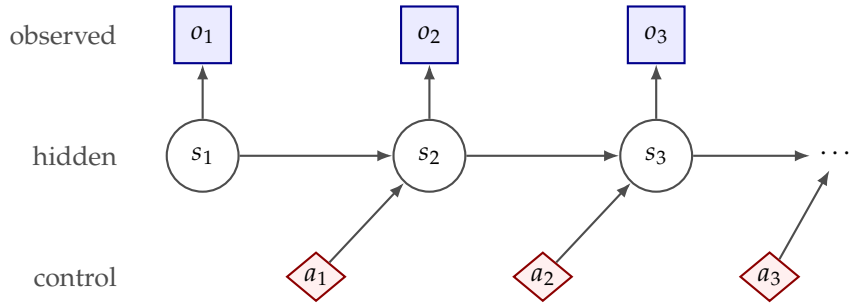


Figure 5: A POMDP. Hidden states form a controlled Markov chain; the agent sees only the emitted observations and its own actions.

	fully observable	partially observable
uncontrolled	Markov chain	hidden Markov model (HMM)
controlled	MDP	POMDP

The agent's information at step h is the history $\tau_h = (o_1, a_1, o_2, a_2, \ldots, o_{h-1}, a_{h-1}, o_h) \in (\mathcal{O} \times \mathcal{A})^{h-1} \times \mathcal{O}$,

and a policy maps histories to actions, $\pi_h : \tau_h \mapsto \Delta_{\mathcal{A}}$. Policies that depend only on the current observation are no longer sufficient; decisions must be based on the entire history. Values are functions of the history rather than of a current state:

$$V_h^\pi(\tau_h) = \mathbb{E}_\pi\Big[\sum_{h'=h+1}^{H} r(o_{h'}) \;\Big|\; \tau_h\Big], \qquad Q_h^\pi(\tau_h, a_h) = \mathbb{E}_\pi\Big[\sum_{h'=h+1}^{H} r(o_{h'}) \;\Big|\; \tau_h, a_h\Big].$$

By definition they satisfy the Bellman equations

$$V_h^\pi(\tau_h) = \sum_{a\in\mathcal{A}} \pi_h(a \mid \tau_h) Q_h^\pi(\tau_h, a), \qquad Q_h^\pi(\tau_h, a_h) = \mathbb{E}_{o_{h+1}\sim\mathbb{P}(\cdot\mid\tau_h,a_h)}\big[r(o_{h+1}) + V_{h+1}^\pi(\tau_h a_h o_{h+1})\big],$$

where $\mathbb{P}(o_{h+1} \mid \tau_h, a_h) = \mathbb{P}(\tau_h a_h o_{h+1})/\mathbb{P}(\tau_h a_h)$ depends on $\mathbb{T}$ and $\mathbb{O}$. The optimal policy is the deterministic $\pi_h^\star(\tau_h) = \arg\max_a Q_h^\star(\tau_h, a)$, with the Bellman optimality equation

$$V_h^\star(\tau_h) = \max_{a\in\mathcal{A}} Q_h^\star(\tau_h, a), \qquad Q_h^\star(\tau_h, a_h) = \mathbb{E}_{o_{h+1}\sim\mathbb{P}(\cdot\mid\tau_h,a_h)}\big[r(o_{h+1}) + V_{h+1}^\star(\tau_h a_h o_{h+1})\big].$$

In general τ_h takes about $(|\mathcal{O}||\mathcal{A}|)^h$ values, so representing $V^\star$ needs exponential space, and planning in a POMDP cannot in general be done computationally efficiently.

21.2 Belief states

A history determines a distribution over the current hidden state, the BELIEF STATE $b_h \in \Delta_{\mathcal{S}}$, $b_h(s) = \mathbb{P}(s_h = s \mid \tau_h)$. The belief is a sufficient statistic for the future: two histories that generate the same belief behave identically from then on, so the history-based view has redundancy. Beliefs update recursively,

$$b_{h+1}(s_{h+1}) = \frac{\mathbb{P}(s_{h+1}, o_{h+1} \mid a_h, b_h)}{\mathbb{P}(o_{h+1} \mid a_h, b_h)} \propto \mathbb{O}(o_{h+1} \mid s_{h+1}) \sum_s \mathbb{T}(s_{h+1} \mid s, a_h) b_h(s),$$

so $b_{h+1} = f(b_h, a_h, o_{h+1})$. Hence a POMDP is equivalent to the BELIEF-STATE MDP, whose states are beliefs, and policies $\pi_h : \Delta_{\mathcal{S}} \to \Delta_{\mathcal{A}}$ that depend only on this sufficient statistic suffice. The values and Bellman equations become

$$V_h^\pi(b_h) = \sum_a \pi_h(a \mid b_h) Q_h^\pi(b_h, a), \qquad Q_h^\pi(b_h, a_h) = \mathbb{E}_{o_{h+1}\sim\mathbb{P}(\cdot\mid b_h,a_h)}\big[r(o_{h+1}) + V_{h+1}^\pi(f(b_h, a_h, o_{h+1}))\big],$$

with a maximum over actions for the optimal values. But beliefs lie in a continuous space and $V^\star(b)$ is in general not linear in b; it can be shown to be piecewise linear, with a number of pieces exponential in H, so planning remains computationally intractable in general.

When $\mathbb{T}$ and $\mathbb{O}$ are unknown and must be estimated from data, the problem is harder still. It is related to learning hidden Markov models (the uncontrolled version), where open problems remain, such as the weakest assumptions under which HMMs are learnable and their sharp sample

complexity.

21.3 Predictive state representations

In a POMDP we observe only $(o_1, a_1, o_2, a_2, \dots)$; the hidden states are latent variables we assume exist. In practice interpretable hidden states may or may not exist, and they need be neither unique nor intrinsic. A more general view works with observable quantities only. Write histories and futures as action-observation sequences.

Definition 21.2 [Tests and the system-dynamics matrix]. A test of length k is a sequence $t = a^1 o^1 \cdots a^k o^k$, and $p(t) = \mathbb{P}(o^1, \dots, o^k \mid a^1, \dots, a^k)$ is the probability of seeing its observations when its actions are taken. The system-dynamics vector lists $p(t)$ for all tests in lexicographic order; it is infinite-dimensional, and knowing it means knowing everything about the dynamical system. For a history $h = a_1 o_1 \cdots a_\ell o_\ell$ and a test t,

$$p(t \mid h) = \mathbb{P}\big(o_{\ell+1} = o^1, \dots, o_{\ell+k} = o^k \,\big|\, h,\ a_{\ell+1} = a^1, \dots, a_{\ell+k} = a^k\big) = \frac{p(ht)}{p(h)},$$

and the system-dynamics matrix has rows indexed by histories (starting with the empty history $\emptyset$), columns indexed by tests, and entries $p(t \mid h)$. It is computed from the system-dynamics vector, and the structure of the system is easier to see in this matrix form.

Proposition 21.3 [Rank of a POMDP]. *For a POMDP with S hidden states, the system-dynamics matrix has rank at most S.*

Proof. Let $b[h] \in \Delta_{\mathcal{S}}$ be the belief over the current hidden state after history h, and let $u_t(s)$ be the probability of the observations of t when its actions are taken from hidden state s. Then $p(t \mid h) = \langle b[h], u_t \rangle$, an inner product in $\mathbb{R}^S$, so the matrix factors through $\mathbb{R}^S$. □

The converse fails: there are dynamical systems whose system-dynamics matrix has rank 3 but which cannot be represented by any POMDP with finitely many hidden states. So POMDPs with finitely many hidden states form a strict subset of the dynamical systems with finite-rank system-dynamics matrices, which raises the question of how to represent the larger class.

Definition 21.4 [Predictive state representation]. A set of core tests $Q = \{q_1, \dots, q_k\}$ is a set of tests whose columns of the system-dynamics matrix have full column rank k equal to the rank of the matrix. The vector

$$\psi(h) = \big(p(q_1 \mid h), \dots, p(q_k \mid h)\big) \in \mathbb{R}^k$$

is the predictive state representation (PSR) of the history h.

Every column is then a linear combination of the core columns: for every test t there is a vector m_t,

not depending on h, with

$$p(t \mid h) = m_t^\top \psi(h) \quad \text{for all histories } h.$$

So $\psi(h)$ is a sufficient statistic, playing the role of the belief state. A linear PSR model specifies the entire dynamical system through the parameters $\psi(\emptyset)$, the PSR of the empty history, and the linear weights $\{m_{ao}\}_{a\in\mathcal{A},o\in\mathcal{O}}$ and $\{m_{aoq}\}_{a\in\mathcal{A},o\in\mathcal{O},q\in\mathcal{Q}}$ that predict the one-step tests ao and the extended core tests aoq.

Proposition 21.5 [PSR PARAMETERS DETERMINE THE SYSTEM]. *Let $M_{ao} = [m_{aoq_1}, \dots, m_{aoq_k}] \in \mathbb{R}^{k\times k}$. For every test $t = a^1o^1 \cdots a^no^n$,*

$$m_t = M_{a^1o^1} \cdots M_{a^{n-1}o^{n-1}}\, m_{a^no^n},$$

and therefore $p(t) = m_t^\top \psi(\emptyset)$ is determined by the parameters.

Proof. First, the PSR updates linearly: for each core test,

$$p(q_i \mid hao) = \frac{p(aoq_i \mid h)}{p(ao \mid h)} = \frac{m_{aoq_i}^\top \psi(h)}{m_{ao}^\top \psi(h)}, \qquad \text{so} \qquad \psi(hao) = \frac{M_{ao}^\top \psi(h)}{m_{ao}^\top \psi(h)}.$$

Next, for any test t, $p(aot \mid h) = p(t \mid hao)\, p(ao \mid h)$ for every history h, that is,

$$m_{aot}^\top \psi(h) = m_t^\top \psi(hao) \cdot m_{ao}^\top \psi(h) = m_t^\top M_{ao}^\top \psi(h).$$

The vectors $\psi(h)$ over all histories span $\mathbb{R}^k$, because the core columns have rank k, so $m_{aot} = M_{ao} m_t$. Induction on the length of the test gives the formula. □

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 21 (Spring 2024); written from the Spring 2020 Lecture 23 notes.

Lecture 22: Partially Observable Reinforcement Learning II

This unit asks when partially observable RL is tractable. Learning a POMDP is hard in the worst case, both computationally and statistically, but the hard instances have uninformative observations. Ruling them out with a weakly revealing condition makes a simple algorithm, optimism combined with maximum likelihood estimation, sample-efficient.

22.1 Setting

Consider an episodic, tabular, time-inhomogeneous POMDP with S hidden states, A actions, O observations, horizon H, and initial distribution $\mu_1 \in \Delta_{\mathcal{S}}$. Its transitions are $S \times S$ matrices $\mathbb{T}_{h,a}$, with $\mathbb{T}_{h,a}(\cdot \mid s)$ the next-state distribution, and its emissions are $O \times S$ matrices $\mathbb{O}_h$, with $\mathbb{O}_h(\cdot \mid s)$ the observation distribution in state s; the rewards $r_h(o) \in [0,1]$ are known functions of the observation. In each step the agent observes $o_h \sim \mathbb{O}_h(\cdot \mid s_h)$, receives $r_h(o_h)$, and takes a_h, and then $s_{h+1} \sim \mathbb{T}_{h,a_h}(\cdot \mid s_h)$. With trajectories $\tau_h = (o_1, a_1, \dots, o_h, a_h)$, a policy maps $(\mathcal{O} \times \mathcal{A})^{h-1} \times \mathcal{O}$ to $\Delta_{\mathcal{A}}$, and $V^\pi = \mathbb{E}_\pi\left[\sum_{h=1}^H r_h(o_h)\right]$. Optimal policies are in general history-dependent. The goal is an ϵ-optimal policy, $V^\pi \geq V^\star - \epsilon$, from a number of samples polynomial in all parameters, or low regret $\mathrm{Regret}(K) = \sum_{k=1}^K (V^\star - V^{\pi_k})$.

22.2 Why POMDPs are hard

Computationally, even with a known model, planning in a POMDP is intractable in general (Lecture 21). *Statistically*, learning can require exponentially many samples.

Proposition 22.1 [Statistical hardness]. *There is a class of POMDPs with two states per step and horizon H, whose observations reveal no information about the states until the end, such that any algorithm needs $A^{\Omega(H)}$ samples to learn an $O(1)$-optimal policy with probability at least $\frac{1}{2}$.*

The hard instance is a combination lock with unobserved states (Figure 6). There are states $s_{h,\mathrm{g}}$ and $s_{h,\mathrm{b}}$ for each step, and the episode starts in $s_{1,\mathrm{g}}$. For a special action sequence $a_1^\star, \dots, a_{H-1}^\star$, the next state is $s_{h+1,\mathrm{g}}$ only if the current state is $s_{h,\mathrm{g}}$ and the action is $a_h^\star$; otherwise it is $s_{h+1,\mathrm{b}}$. The reward is 1 only in $s_{H,\mathrm{g}}$. The emissions $\mathbb{O}_h(\cdot \mid s_{h,\mathrm{g}})$ and $\mathbb{O}_h(\cdot \mid s_{h,\mathrm{b}})$ coincide for every step except the last, so for $H-1$ steps the agent effectively observes nothing. The optimal policy plays $a_h^\star$ and earns 1, but the agent has no option except trying out action sequences, and needs $A^{\Omega(H)}$ episodes to find the right one with constant probability.

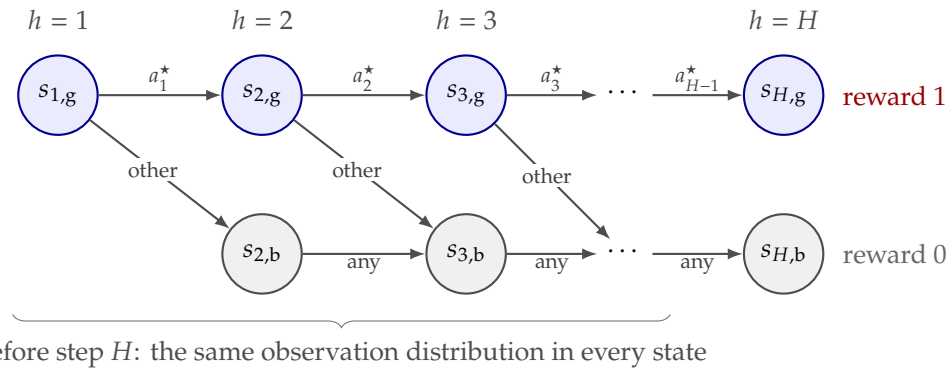


Figure 6: The combination lock with unobserved states. The observations carry no information about the hidden state before step H.

22.3 The weakly revealing condition

The hardness comes from trivial observations. More generally, if two mixtures of states with disjoint supports induce the same observation distribution, the observations cannot tell them apart and learning is hard. Identify a mixture with a probability vector $\nu \in \Delta_{\mathcal{S}}$, so that $\mathbb{O}_h \nu$ is its observation distribution.

Proposition 22.2 [Distinguishable mixtures]. *$\mathbb{O}_h$ has rank S if and only if $\mathbb{O}_h \nu_1 \neq \mathbb{O}_h \nu_2$ for all mixtures ν_1, ν_2 with disjoint supports.*

Proof. If $\mathbb{O}_h$ has rank S, it is injective, and disjoint-support mixtures are distinct vectors. Conversely, if the rank is less than S, take $x \neq 0$ with $\mathbb{O}_h x = 0$. The columns of $\mathbb{O}_h$ are distributions, so $\mathbf{1}^\top x = \mathbf{1}^\top \mathbb{O}_h x = 0$, and the positive and negative parts x^+, x^- have disjoint supports and equal mass $c > 0$. Then $\nu_1 = x^+/c$ and $\nu_2 = x^-/c$ have disjoint supports and $\mathbb{O}_h \nu_1 = \mathbb{O}_h \nu_2$. □

The weakly revealing condition is a robust version of full rank.

Assumption 22.3 (α-weakly revealing)**.** There is $\alpha > 0$ with $\min_h \sigma_S(\mathbb{O}_h) \geq \alpha$, where σ_S is the Sth largest singular value.

The observations then contain enough information to distinguish any two mixtures of states, given enough samples. The condition is "weakly" revealing in contrast to rich-observation (block) MDPs, where the state can be decoded from a single observation and the problem is effectively fully observable. Since $\mathbb{O}_h$ is $O \times S$, the assumption requires $S \leq O$: it covers the *undercomplete* case. In the *overcomplete* case $S > O$, a single observation cannot distinguish all mixtures, so one inspects m consecutive steps. The m-step emission-action matrix $\mathbb{M}_h \in \mathbb{R}^{(A^{m-1}O^m) \times S}$ has entries

$$[\mathbb{M}_h]_{(\mathbf{a},\mathbf{o}),s} = \mathbb{P}\big(o_{h:h+m-1} = \mathbf{o} \;\big|\; s_h = s,\ a_{h:h+m-2} = \mathbf{a}\big),$$

and the m-step α-weakly revealing condition asks $\min_h \sigma_S(\mathbb{M}_h) \geq \alpha$. When $O^m \geq S$, it holds if at every step some action sequence $\mathbf{a}$ makes the m-step observations α-revealing, $\sigma_S(\mathbb{M}_{h,\mathbf{a}}) \geq \alpha$.

22.4 Optimistic maximum likelihood estimation

Take a model-based approach. Write $\theta = (\mathbb{T}, \mathbb{O}, \mu_1)$ for the parameters of a POMDP, Θ for all POMDPs with S states, A actions, and O observations, $V^\pi(\theta)$ for the value of π in model θ, and $\mathbb{P}^\pi_\theta(\tau)$ for the probability of trajectory τ under π. Maximum likelihood estimation (MLE) picks the model that maximizes the total log-likelihood $\sum_{(\pi,\tau)\in\mathcal{D}} \log \mathbb{P}^\pi_\theta(\tau)$ of the policy-trajectory pairs collected so far. OMLE keeps every weakly revealing model whose likelihood is close to the maximum, and plans optimistically over them.

Algorithm 22.1: Optimistic MLE (OMLE)

$\mathcal{B}^1 \leftarrow \{\hat\theta \in \Theta : \min_h \sigma_S(\mathbb{O}_h(\hat\theta)) \geq \alpha\}$; $\quad \mathcal{D} \leftarrow \emptyset$
for $k = 1, \dots, K$ **do**
  $(\theta^k, \pi^k) \leftarrow \arg\max_{\hat\theta \in \mathcal{B}^k,\, \pi} V^\pi(\hat\theta)$ *// optimistic planning*
  execute π^k to collect a trajectory τ^k, and add (π^k, τ^k) to $\mathcal{D}$
  $\mathcal{B}^{k+1} \leftarrow \left\{\hat\theta \in \Theta : \sum_{(\pi,\tau)\in\mathcal{D}} \log \mathbb{P}^\pi_{\hat\theta}(\tau) \geq \max_{\theta'\in\Theta} \sum_{(\pi,\tau)\in\mathcal{D}} \log \mathbb{P}^\pi_{\theta'}(\tau) - \beta\right\} \cap \mathcal{B}^1$

With $\beta = 0$ the confidence set collapses to the maximum likelihood estimates. The algorithm is much simpler than earlier spectral methods, but, like all known algorithms for large classes of POMDPs, it is computationally inefficient: the planning step searches over the confidence set.

Theorem 22.4 [Regret of OMLE]. *There is an absolute constant c such that with $\beta = c\big(H(S^2A + SO)\log(SAOHK) + \log(K/\delta)\big)$, for every POMDP satisfying Assumption 22.3, with probability at least $1-\delta$,*

$$\mathrm{Regret}(k) \leq \mathrm{poly}\big(S, A, O, H, \alpha^{-1}, \log(K/\delta)\big) \cdot \sqrt{k} \qquad \text{for all } k \in [K].$$

Consequently, after $K \geq \mathrm{poly}(S, A, O, H, \alpha^{-1}, \log(1/(\epsilon\delta)))/\epsilon^2$ episodes, the uniform mixture of $\pi^1, \dots, \pi^K$ is ϵ-optimal.

The $\sqrt{k}$ growth and the $1/\epsilon^2$ rate are optimal. The polynomial dependence on $1/\alpha$ is also necessary: for every $\alpha \in (0, \frac{1}{2})$ there is an undercomplete α-weakly revealing POMDP with $S, O = O(1)$ on which any algorithm needs $\Omega(\min\{1/(\alpha H), A^{H-1}\})$ samples to learn a $\frac{1}{2}$-optimal policy. In the overcomplete case, a multi-step variant requires the m-step condition in $\mathcal{B}^1$ and, for each optimistic policy π^k and each $h \in \{0, \dots, H-m\}$, executes π^k for the first h steps and then uniformly random actions, to learn the dynamics reachable by π^k. It finds an ϵ-optimal policy from $\mathrm{poly}(S, A^m, O, H, \alpha^{-1}, \log(1/(\epsilon\delta)))/\epsilon^2$ samples, and the A^m dependence is unavoidable: some POMDPs with $S, H, O = O(m)$ satisfying the m-step condition with $\alpha \geq 1$ need $\Omega(A^{m-1})$ samples.

Proof overview. The analysis uses OBSERVABLE OPERATORS, an equivalent parametrization of the POMDP by

$$\mathbf{B}_h(o,a) = \mathbb{O}_{h+1}\mathbb{T}_{h,a}\,\mathrm{diag}\big(\mathbb{O}_h(o \mid \cdot)\big)\mathbb{O}_h^\dagger, \qquad \mathbf{b}_0 = \mathbb{O}_1\mu_1,$$

where $\mathbb{O}_h^\dagger$ is the pseudoinverse. The probability of a trajectory is a product of operators, $\mathbb{P}_\theta^\pi(\tau_h) = \pi(\tau_h)\cdot \mathbf{e}_{o_h}^\top \mathbf{B}_{h-1}(o_{h-1},a_{h-1})\cdots\mathbf{B}_1(o_1,a_1)\mathbf{b}_0$, with $\pi(\tau_h)$ the product of the policy's action probabilities, and this linear structure allows tools from matrix analysis.

1. *Optimism and operator errors.* With the choice of β, the true model $\theta^\star$ lies in every confidence set with high probability, so $V^\star \le V^{\pi^k}(\theta^k)$ and

$$\sum_{t=1}^k \big(V^\star - V^{\pi^t}(\theta^\star)\big) \le \sum_{t=1}^k \big(V^{\pi^t}(\theta^t) - V^{\pi^t}(\theta^\star)\big) \le H\sum_{t=1}^k\sum_{\tau_H}\big|\mathbb{P}_{\theta^t}^{\pi^t}(\tau_H) - \mathbb{P}_{\theta^\star}^{\pi^t}(\tau_H)\big|,$$

   since each trajectory's total reward is at most H. Using the operator representation, this is at most $\frac{H\sqrt{S}}{\alpha}$ times the errors of the operators $\mathbf{B}_h^t$ of θ^t, projected onto belief vectors $\mathbf{b}(\tau_{h-1})$ (the operator products applied to $\mathbf{b}_0$) and weighted by π^t. Only directions covered by these weighted beliefs matter, so the operators need not be recovered at hard-to-reach states.

2. *What MLE guarantees.* Classical MLE analysis gives, with high probability,

$$\sum_{t<k}\big\|\mathbb{P}_{\theta^k}^{\pi^t}(\tau_h=\cdot) - \mathbb{P}_{\theta^\star}^{\pi^t}(\tau_h=\cdot)\big\|_1^2 = O(\beta) \qquad \text{for all } (k,h).$$

   So the current estimate predicts the behavior of the past policies. Rewritten through the operators, the past-policy-weighted operator errors of θ^k are $O(\sqrt{S\beta k}/\alpha)$.

3. *Bridging the two.* Step 1 needs the errors of θ^t under its own policy π^t, while step 2 controls θ^k under earlier policies. An ℓ_1-norm eluder argument, playing the role of the elliptical potential lemma, closes the gap: if $\sum_{t<k}|w_k^\top x_t| \le \sqrt{k}$ for all k, then $\sum_{t\le k}|w_t^\top x_t| = \widetilde{O}(\sqrt{\zeta k})$ with ζ depending only on the dimension.

Source. Chi Jin, *ECE524: Foundations of Reinforcement Learning*, Lecture 22 (Spring 2024); written from the OMLE paper (Liu, Chung, Szepesvári, and Jin, 2022).